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Are economists amoral?

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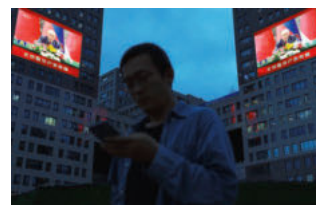
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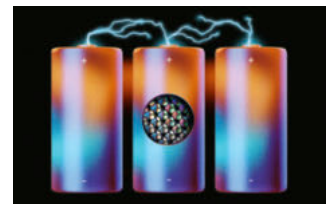
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to take part in "a severe contest between intelligence, which presses forward, and an unworthy, timid ignorance obstructing our progress."

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The world this week Politics



America's Justice Department charged Raúl Castro with the "murder" of four Americans of Cuban descent. Mr Castro succeeded his brother Fidel as Cuba's leader before stepping down in 2021. The indictments relate to the shooting down of two small aircraft, operated by a Cuban-exiles group, in 1996 by Cuban military jets, which killed the four men. Mr Castro was Cuba's defence minister at the time. The charges intensify America's campaign against the Cuban regime and threat of an invasion. Speaking in Spanish, Marco Rubio, America's secretary of state, released a video to Cubans telling them that America was offering "to build a better future" for the island.

Venezuela deported Alex Saab, an ally of Nicolás Maduro, the deposed president, to America. Mr Saab, often referred to as Mr Maduro's "bag man", appeared in court in Miami where a charge of money-laundering was presented. He has been arrested before, but prosecutors may now lean on Mr Saab to flesh out their case against Mr Maduro, who was snatched by American forces in January.

More anti-government demonstrations took place in Bolivia, amid widening opposition to austerity measures proposed by Rodrigo Paz, the centrist president. Bolivian officials in part blame Evo Morales, a former leftist president, for stirring up trouble. Mr Paz's election to office last year ended two decades of socialist rule in the country. America's State Department compared the protests to an attempted "coup d'état".

Indonesia announced a proposal to centralise the export of commodities such as palm oil and coal through a new state-operated export company. Exporters of natural resources will also have to place their export earnings in state-owned banks, in order to stabilise the falling rupiah. Indonesia's central bank raised interest rates for the first time in two years, lifting them by a larger-than-expected half a percentage point.

Trade flows

Vladimir Putin visited China, where he signed several deals on trade and co-operation with Xi Jinping. The leaders discussed a new pipeline that would provide China with Russian natural gas. Meanwhile, China confirmed it will buy 200 Boeing jets following Donald Trump's recent trip to Beijing. Although the order is smaller than had been expected it represents a step forward; at the height of the trade war China paused deliveries of Boeing jets.



The WHO declared an outbreak of Ebola to be a public-health emergency of international concern. The virus kills up to half of those who catch it. The epidemic in eastern Democratic Republic of Congo is thought to have killed at least 139 people so far, with hundreds more infected. The numbers will probably rise. The response to the outbreak is being hampered by war, aid cuts and the lack of a vaccine for the strain that is causing it.

America and Nigeria claimed that a top Islamic State commander and 175 other militants

were killed in air strikes in north-eastern Nigeria, part of a series of collaborative operations aimed at tackling the country's overlapping security crises. America has stepped up support for Nigeria's counter-terrorism efforts, but there is no evidence that this is reducing the violence.

At least four people were killed in protests over high fuel prices in Kenya. In response the government cut diesel prices and entered negotiations with transport unions to resolve a strike by bus and minibus drivers. The war in Iran has driven up prices in Kenya, which, like much of east Africa, depends on the Gulf for oil supplies.

Mr Trump threatened to restart strikes on Iran, warning the Islamic regime that the "clock is ticking". The American president then said that he had called off an attack at the request of Qatar, Saudi Arabia and the United Arab Emirates. He claimed that "serious negotiations" were under way. Meanwhile, Iran announced the establishment of the Persian Gulf Strait Authority, a new entity tasked with "managing the passage and transit through the Strait of Hormuz".

The United Arab Emirates said a drone strike on its Barakah nuclear plant had sparked a fire. Officials said they did not know who was responsible, but confirmed that the attack had come from Iraq, where some Iranian-backed militias are based.

It stinks

America's Justice Department created a \$1.8bn fund to compensate people who claim they have been targeted by federal prosecutors for political purposes, in effect allies and supporters of Donald Trump. This will probably include the rioters who stormed Congress on January 6th 2021. It forms part of the government's settlement with Mr Trump in a case related to his leaked tax docu-

ments. The president was seeking \$10bn in damages and has now dropped his claims.

A novice candidate backed by Mr Trump won a primary election for a congressional district in north Kentucky. Ed Gallrein defeated Thomas Massie, one of the few remaining Republican critics of the president, in the most expensive House primary election ever. Another thorn in Mr Trump's side, Bill Cassidy, was defeated in his Senate primary in Louisiana.

Andy Burnham, the mayor of Greater Manchester, emerged as the favourite to succeed Sir Keir Starmer as Labour's leader, and thus Britain's prime minister, in a putative leadership contest. But Mr Burnham is not a member of Parliament. The MP in Makerfield, a traditional Labour seat, stood down to enable him to contest it in a by-election. However, at the recent local elections Reform UK made big gains in the area, suggesting it is not a foregone conclusion that Mr Burnham will clinch the seat.

Peter Magyar, Hungary's new prime minister, visited Poland on his first trip abroad. Mr Magyar met Donald Tusk, his Polish counterpart, who also came to office by defeating an anti-EU government. Hungary's relations with Poland were strained under Mr Magyar's predecessor, Viktor Orban. Mr Magyar wants Hungarian-Polish relations to return to "normality" as the countries have much in common.

A NATO jet shot down a Ukrainian drone that had entered Estonia's air space. Ukraine blamed Russia's "electronic warfare" for redirecting the drone. Another Ukrainian drone crashed in Lithuania. Latvia also reported drone incursions. Latvia's prime minister, Evika Silina, resigned recently, after misdirected Ukrainian drones crossed over from Russia and damaged an oil facility. Latvia's defences had not detected the breach.

The world this week Business



SpaceX officially launched its blockbuster IPO. The space-rocket firm, led by **Elon Musk**, hopes to raise around \$75bn when it floats on the NASDAQ stockmarket, smashing the record for an IPO held by Saudi Aramco, which went public in 2019. Its target valuation is about \$1.75trn. In its prospectus SpaceX laid out ambitions to use its rockets to transport components to off-world data centres in space. It also mentions a vision of carrying people to Mars and of mining on asteroids. The IPO announcement came as the firm prepared the first test flight of its V3 Starship megarocket.

It was a mixed week for **Elon Musk**, who lost his courtroom battle with OpenAI and Sam Altman. It took a jury just two hours to reject the case, in which Mr Musk sought damages from OpenAI for allegedly reneging on a contract with him as a co-founder to run the firm as a non-profit instead of for-profit entity. Meanwhile, OpenAI was reportedly ready to launch its own IPO with a view to a listing in September.

Nvidia presented another stellar set of quarterly earnings. It recorded overall revenues of \$81.6bn, up by 85% year on year, with sales from chips for data centres taking in \$75.2bn, a rise of 92%. The company announced an additional \$80bn in share buy-backs.

A last-minute deal between management and unions averted a strike at **Samsung**. Samsung has reaped huge profits from selling its memory chips to data centres and the unions want their cut in bonuses.

Britain's annual inflation rate fell to 2.8% in April from 3.3% in March, mostly a result of the government lowering its cap on energy prices. Electricity prices fell by 8.4%. However the cost of motor fuel surged by 23%, prompting the government to postpone a rise in fuel tax this week. Economists expect the overall inflation rate to rise again in the coming months.

When in need

America extended a waiver on sanctions for buying **Russian oil** for another 30 days. The waiver applies to Russian oil held at sea and is designed to help poorer countries that have taken a hit from the stoppage of supplies through the Strait of Hormuz. Meanwhile, Britain eased sanctions on Russian oil refined into diesel and jet fuel in other countries. This will enable the country to import jet fuel from India.

Ryanair, Europe's largest airline, said it was "far too early" to give any meaningful profit guidance because of volatile jet-fuel prices. Europe is relatively well stocked with jet fuel, and Ryanair hedges 80% of its fuel costs, but it warned that spot prices had hit \$150 a barrel for unhedged fuel.

Adani Enterprises, part of **Adani Group**, agreed to pay \$275m to settle claims that it violated American sanctions by buying Iranian fuel between November 2023 and June 2025. The case was brought by America's Treasury Department. Meanwhile, the Justice Department decided to drop its criminal fraud case against Gautam Adani, Adani Group's chairman, and the Securities and Exchange Commission settled with Mr Adani on allegations that focused on investors being misled about his group's anti-bribery practices. Under the deal Mr Adani neither denies nor admits the claims.

Anglo American agreed to sell its coalmining assets in Australia for up to \$3.9bn to **Dhilmar**, a privately held mining firm. Anglo has now divested its coal, nickel and platinum divisions and is in talks with a consortium to sell off its De Beers diamond business.

A formerly unsolved **mathematical problem** has been cracked by a model from OpenAI, in what has been called a "milestone" in AI maths. The problem, which asks how many pairs of points can be exactly one unit apart

on a two-dimensional plane, was first posed in 1946 by Paul Erdos. The AI proof overturned previous best efforts, and was produced by a general model without any mathematics-specific training, said OpenAI.

How to make a quick buck

Swatch's Royal Pop pocket watch went on sale. Made in collaboration with Audemars Piguet, the watches combine the retail and luxury ends of the market. Hundreds of people queued outside Swatch shops around the world; some disturbances were reported as customers fought over a time-piece. Selling for \$400-420, some watches are being resold online for as much as \$3,000.

James Murdoch, son of Rupert and a former boss of News Corporation in Europe, is acquiring a roughly 50% stake in **Vox Media**. Mr Murdoch wants to be involved in "longer-form, thoughtful journalism", a contrast to the cut-and-thrust of the daily newspapers he used to run. One of the titles owned by Vox is *New York* magazine, which publishes articles on Gotham as well as stories of national interest, such as why "The Amish Are Falling in Love With AI".



A Starship enterprise

The listing of SpaceX is capitalism on rocket fuel

TO CHANNEL ONE of Elon Musk's favourite science-fiction authors, SpaceX is a product of infinite improbability. When Mr Musk began, few would have predicted that a start-up could design a liquid-fuelled rocket and put it in orbit. Nor that an engineer would be able to get a rocket's booster to return to Earth, land upright on its own tail and be re-used. And no one has yet recovered a rocket's second stage, which must withstand 1,500°C or so during atmospheric re-entry.

Having pulled off the first unlikely feat in 2008, six short years after its founding, and the second in 2015, SpaceX will try to use a launch due on May 21st to show that it can carry off the third, too. If the test fails by the time you read this, SpaceX will try again—and again, until it succeeds or runs out of money.

Success will be vital if Mr Musk is to realise his vision of dominating artificial intelligence, apparently by using space-based computers launched by SpaceX to satisfy AI's ravenous hunger for data-processing. To pay for his dream, on May 20th the company filed the prospectus that has started the countdown clock for an initial public offering next month worth about \$75bn. This will be the biggest listing in history.

In two ways, the IPO is infinitely improbable, too. It is inspiring in that Mr Musk aims to carry off yet more seemingly impossible feats of engineering. And it is worrying in that he is asking investors to trust their savings to a loss-making outfit with hardly credible financial plans over which he will have total control.

SpaceX's colossal listing will shake stockmarkets as much as its giant Starship rocket's Raptor engines now rattle the coastal plain beneath the launch site in Texas. SpaceX may instantly land among the world's ten most valuable companies, with a market value of as much as \$2trn. A fortnight later it is to be ushered into the NASDAQ index, ending up in countless tracker funds, pension pots and other stock portfolios. By then Mr Musk may have become Earth's first trillionaire, worth as much as all households in his native South Africa combined.

In a populist age, many people will see the concentration of power in trillion-dollar companies—not to mention trillion-dollar men—as a failure of capitalism. Yet as a story of risk-taking, competition, the ability to mobilise resources and thereby turn the improbable into reality, SpaceX is in fact capitalism at its most remarkable.

Sometimes Mr Musk has harnessed the state's resources, sometimes he has gone it alone. Government policy played a crucial role in getting SpaceX off the ground. But his drive has repeatedly solved problems that the state could not. He has already opened up space, and if he succeeds in his dream of cutting launch costs to \$185 a kilogram, 1% of their historical average, he will have utterly transformed it.

Getting there will be hard. SpaceX's business plan, as outlined in its regulatory filing on May 20th, has "risky" written all over it. The company is losing more than \$1bn a month, much of it on its AI business, xAI, which spent \$8bn in the first quarter and is still far behind the industry's stars.

The plan says that over 90% of SpaceX's potential sales, which it values at \$28.5trn—almost the entire GDP of America today—will come from AI. It is unclear whether that will be from making a success of his own AI business using data centres in space powered directly by the sun (and untroubled by terrestrial NIMBYS), or from selling data-centre services to others. Throw in some stardust about interplanetary missions, and asteroid mining and you can begin to grasp the scale of Mr Musk's ambition.

Yet even if he doesn't get all the way there, his fleet will open up new possibilities that he and other capitalists can exploit. The government encouraged William Boeing to do this after the first world war, using idle warplanes to ferry air mail. Jeff Bezos sold spare cloud-computing capacity when Amazon had built more of it than it needed for its e-emporium. Mr Musk himself has done it, too: Starlink was initially a way to fill up SpaceX's workhorse rocket, the Falcon 9, when external custom could not keep up with its billowing launch capacity.

He is also spurring competition. Blue Origin, founded by Mr Bezos two years before SpaceX, is working on a re-usable rocket of its own. Both it and Amazon have plans for their own satellite constellations to rival Starlink. In December a private Chinese challenger, LandSpace, test-launched a re-usable rocket; another Chinese company, called Space Pioneer, tried in April.

If anyone has earned the right to aspire to seemingly unattainable engineering goals it is Mr Musk. The financial risks—and the control he demands—are another matter. No other firm has tried to use public markets to raise as much as SpaceX will. NTT in Japan and Saudi Aramco were partial privatisations of state-owned assets. The nearest private-sector competitor, Alibaba in 2014, raised a third of what SpaceX wants.

SpaceX could, like any capitalist enterprise, crash and burn. Its valuation—as much as 100 times annual revenue—compares with 16 for Tesla. Mr Musk has a record of creating lots of shareholder value but little cash. SpaceX is on the hook for the loss-making xAI, and his social network, X, with which it merged earlier this year.

Don't panic

And by giving him and other insiders special shares to elect directors, SpaceX's structure makes Mr Musk unsackable. Investors are therefore saddled with his future business decisions as well as his obnoxious politics. He may face a backlash under the next Democratic administration for his racism and the ravages of DOGE under Donald Trump.

The thing is that Mr Musk comes as a package. He has needed a galactic ego and a cosmic appetite for risk to succeed—as did Henry Ford and J.P. Morgan before him. SpaceX is an imperfect company and Mr Musk an imperfect man. The marvel of capitalism is that it can harness their talents to create something extraordinary. While his investors take the risk, the rest of humanity can strap in for the ride. ■



The MAGA tax

American growth could be even better

MAGAnomics shows the world what not to do—but also what America keeps getting right

YET AGAIN, America's economy has shown up the rest of the world. For all the chaos of the past year, its GDP has chugged along merrily at 2% annual growth. True, that is hardly a blockbuster pace and—like everywhere else—consumers in America are grumbling about high prices. But Uncle Sam's peers are doing much worse. In Britain, France, Germany and Japan economic growth ranges from roughly zero to about 1%. America's outperformance began decades ago, but in the 2020s it has become vast. And it is likely to last. The latest IMF forecasts show American growth besting the rest all the way to 2030 and beyond.

Perhaps that makes this an odd time to look at what America gets wrong. But remarkably, its world-beating performance has lately come despite a handicap: Donald Trump's erratic and harmful policies (see Finance & economics section). Our analysis suggests that what we call the "MAGA tax"—a toxic mix of high tariffs, zero net migration and all-encompassing policy uncertainty—shaved around three-quarters of a percentage point off the rise in American GDP in 2025. That holds lessons for America and other countries desperate for growth.

America was teed up for a strong year when Mr Trump took charge in early 2025. Its robust economy should have been supercharged, especially by the boom in artificial intelligence and its consequences for the stockmarket. Instead, growth since then has been solid but, by American standards, unremarkable.

Indeed, outside AI-related sectors, policy uncertainty has pushed American business investment into a slump worse than Britain's after the vote on Brexit. See-sawing tariffs and hostility to immigration, even for skilled workers, have also weighed on the economy. Without this MAGA tax, growth could have been closer to 3% in 2025, not the 2% America got. The damage will continue this year, and probably worsen now that higher oil prices from Mr Trump's war with Iran are also a burden. In other words, the president's folly has slowed down a historic boom.

One lesson from this is to marvel at America's dynamism and resilience, even in the face of Mr Trump's vexing behaviour. Many of America's advantages are hard to emulate. The country's continental scale, single language, natural-resource wealth and the fiscal space that comes from issuing the world's safe asset give it a unique economic advantage over Europe. America's federalism is helpful, too. Misguided policies at the state level—like data-centre moratoriums or proposed wealth taxes—do not encumber the whole country. People and businesses can move to a different state.

But America also shows just how much other rich countries are failing to live up to their economic potential. Much of the recent heightening of America's advantage stems from its tolerance of disruption. America's shale-fracking revolution and liquefied natural gas exports have reshaped global energy markets and made its economy more able to withstand the effects of Mr Trump's war in the Middle East. Its embrace of AI data centres presages a shake-up of labour markets which, though

fraught, could propel its living standards still higher.

Unlike its peers, America has built, says Mario Draghi, Europe's growth czar, "an economy capable of preserving its growth from the disruptions it contributes to". Other rich countries, by contrast, are held back by conservative regulations, distrust of the financial industry and risk-aversion. They lose because they simply cannot build new industries so fast.

Yet another conclusion is that America should raise its ambitions still further. Imagine how astounding its economy would be without the drag of the MAGA tax. For Democrats, that question may soon stop being hypothetical. The party is still wrestling with what of Mr Trump's agenda to keep and what to discard, should it win back the White House in 2028.

Politically, rolling back tariffs may be tricky. Beneficiaries of protection will squeal and besides, many on the left have long been wary of free trade. Similarly, some Democrats now wonder aloud whether Mr Trump's penchant for bullying companies that displease him could be turned towards progressive ends, such as restricting lay-offs.

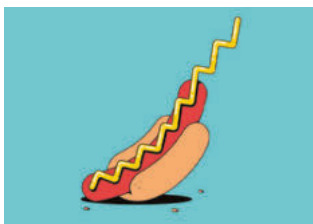
Be patient, say MAGA types; their policies will eventually yield benefits. In fact the misguided policies, if they endure, are likely to exact a bigger price over time. America is becoming less attractive to high-skilled migrants who make large

contributions to its dynamism; more vulnerable to a bond-market crisis as its debts mount; and more tolerant of corruption (see United States section). Mr Trump's style of government, if it lasts, will eventually sap the foundations that have made America so rich, by threatening the rule of law and encouraging firms to put lobbying for favours before innovation and the sound allocation of capital.

Since the MAGA tax inflicts great harm, unwinding it promises great benefits. Whoever succeeds Mr Trump should seize that opportunity. Economic growth is not a political cure-all—if it were then Kamala Harris, campaigning in a year with growth of 2.8%, would be president. But Democrats and Mr Trump's Republican successors will struggle to win voters' affections without raising living standards, as only fast growth paired with low inflation can. Listening to economists' advice is unfashionable these days, but there is a free lunch here and America's next leaders would be foolish not to tuck in.

The American experiments

The dangers of misreading MAGAnomics are even greater abroad. Populists everywhere have looked at America and been emboldened by the enticing conclusion that protectionism and industrial policy may be less harmful than free-market economists supposed. Claudia Sheinbaum in Mexico, the National Rally in France and Nigel Farage in Britain are all proposing to copy parts of Trumpist economics. Interventionism is in vogue; the barriers to trade and migration are rising. Even enemies of Trumpism increasingly answer it with a leftist statism of their own. Yet, if America has been dented by the MAGA tax, those policies would knock other countries sideways. ■



The transatlantic alliance

Work on the marriage, plan for divorce

Europe needs a Plan B for NATO. Mark Rutte, its boss, is wrong to discourage talk of it

THE FATAL flaw in a commander, suggested Carl von Clausewitz, a 19th-century Prussian general, is obstinacy, or “resistance against our better judgment”. For a modern example of this behaviour, look no further than Mark Rutte, the secretary-general of NATO.

For months Mr Rutte has refused to admit that the transatlantic alliance, the cornerstone of European security for more than seven decades, is close to falling apart. That has required wilful blindness.

Mr Rutte ignores Donald Trump’s repeated statements casting doubt on America’s readiness to honour NATO’s Article 5, which says that an attack on one is an attack on all. Mr Trump also talks about withdrawing some forces from Europe. Nonetheless, Mr Rutte, a former Dutch prime minister, has gamely insisted that he has no doubt America is “completely committed to NATO, completely committed to Article 5”. He maintains NATO has no need for a “Plan B” in case America were suddenly to pull out of the alliance. He has even banned any discussion of one within the NATO headquarters. On all of these counts, he is wrong.

In Mr Rutte’s defence, he is being diplomatic. Striving to keep America in the alliance, he has flattered Mr Trump, endured his tirades and called him “daddy”. Mr Rutte fears that Europe’s attempts to wean itself off America’s security guarantees may further offend its capricious president, and hasten the divorce that the rest of NATO wishes to avoid.

Other European leaders also resort to flattery and try to appeal to Mr Trump’s dealmaking instincts by spending billions on American weapons, many of which are to be sent to Ukraine. In some cases, they have little choice. For instance, Europe has few alternatives to America’s Patriot air-defence

systems (see United States section).

Even thinking about a Plan B is enough to give generals the collywobbles. Much of NATO’s deterrence comes from its ability to knit the armed forces of many countries into a single cohesive force under a unified command led by an American general. Without the superpower to keep good order, the rump of the alliance may be weakened by rows about who should be in charge—and doubts over whether they would actually have the authority to call the shots if Europe found itself at war.

Yet for all the risks of Europe openly planning for a divorce, it has no choice. Mr Rutte’s optimism has been overtaken by events. Mr Trump’s threats in January to seize Greenland from

Denmark, a NATO member, have spurred several European countries to begin secretly planning how to fight without America (see Europe section). Since then, Mr Trump has undermined his officials’ reassurances that the drawdown will be slow and orderly, as Europe takes responsibility for its own conventional defence. He has also announced unexpected cuts to the number of troops station-

ed in Europe and cancelled the deployment of others. On May 22nd America was expected to reduce the forces it pledges to send to Europe in a war. Its allies cannot be sure they will get the weapons they have bought from America, which is postponing deliveries in order to restock those used up in Iran.

Renovating the American-led alliance will be hard and costly: all the more reason to start now. Europe could Europeanise NATO’s structures; it could form a new alliance; or it could build on embryonic ones, such as the ten-member Joint Expeditionary Force, led by Britain, or the “coalition of the willing” that plans to offer security guarantees to Ukraine. Each would be risky—but less risky than doing nothing. ■



Global health

How to stop the Ebola epidemic

The outbreak in central Africa should serve as a warning about future pandemics

EBOLA IS A terrifying virus that kills 30-50% of the people it infects. Fortunately, the world has been getting better at fighting it. Since 2015 vaccines for Zaire, the most common Ebola strain, have limited its spread whenever hotspots emerged in Africa. Genetic sequencing of that strain has allowed rapid testing, which makes contact-tracing easier. African governments, helped by NGOs, have learned how to triage and isolate patients. Crucially, they have also won the trust of locals, who are horrified by haemorrhagic fevers.

But the latest Ebola outbreak is caused by a rarer strain, Bundibugyo, for which there is no licensed vaccine or rapid test. For months it swept undetected through eastern Congo. Aid cuts meant fewer health workers were watching out.

On May 17th the UN’s World Health Organisation (WHO) called the outbreak a public-health emergency of international concern (see Middle East & Africa section). By May 20th the tally of almost 600 suspected cases and 139 deaths was suggesting that this epidemic will be the worst Ebola emergency since at least 2018, when more than 2,000 people died in the same region. To avoid such a grim death toll—let alone that of 2014-16, when 11,000 perished in west Africa—urgent, co-ordinated action is needed. So far it has been lacking.

To win this fight, scientists need incentives to swiftly make and deploy a vaccine. Donors, especially governments in the rich world, should promise now to buy ample supplies of a vaccine as soon as it is authorised, to encourage drug firms to ►►

▶ work faster. On May 20th the WHO said two vaccines are being developed that show promise against the Bundibugyo strain. Although it will be months before either jab is ready, such efforts should be the priority.

Even with a vaccine, stopping Ebola requires co-ordination in different countries. In the past America, through its Centres for Disease Control and Prevention (CDC) and aid spending, would have led this effort. Despite severe cuts under the Trump administration, the CDC still has field offices in Congo and neighbouring Uganda, so it will play a role. But the most important institutions will be the WHO and Africa CDC, which is independent of its American namesake. Africa CDC did well during the covid-19 pandemic and can do so again if given enough outside expertise and funding. Humanitarian NGOs reduced their Ebola-prevention measures last year in Ituri, the province where the outbreak emerged, because of aid cuts. They urgently need funds to pay for protective gear and to recruit more health workers.

All this is made much harder by local politics. Eastern Congo is an ideal place for an epidemic to thrive. The state barely

exists, and is often predatory where it does. More than 100 militias terrorise the population. Much of North and South Kivu, two provinces where Ebola has been detected, is under the control of an armed group, M23, that says it wants to topple Congo's national government. Both the government and the militias have a sorry record of prioritising their own interests above the lives of their compatriots. M23 has resisted calls by NGOs to reopen airports in its territory to let in humanitarian supplies; it accuses the Congolese government of making things worse by keeping banks closed. Mediators, including America and Qatar, should tell both sides to let in supplies and workers; they should also press Rwanda, M23's patron.

This is Congo's 17th Ebola outbreak in 50 years. It will destroy both lives and livelihoods before it is contained. Fortunately, the virus is not airborne, so it cannot spread as fast as covid-19. Even so, it is a reminder of the need to prepare for the next pandemic. Vaccine research, the genomic sequencing of viruses and disease surveillance are the world's immune system, protecting it against wider, deadlier outbreaks. When that immunity is weakened, disaster beckons. ■

Political football

How a deflated Britain can bounce back

Lessons from the Premier League for the next prime minister

IF BRITAIN WERE a football team, it would be flirting with relegation. Its economy is flailing, its people are demoralised and another embarrassing crisis is unfolding at the top. Happily, one British industry provides a game plan for recovery. Forty years ago, English football was a laughing stock. Today, it is a global powerhouse. As we report this week, the Premier League is the most-watched sports competition on the planet and generates £10bn (\$13bn) in gross value added a year (see Britain section). Politicians who want to make Britain great again could do worse than learn from the Premier League.

One lesson is to play to your strengths. Britain invented football. It is also attractive to footloose capital and talent. The Premier League takes advantage of this, welcoming foreign investors to buy clubs and foreign coaches and players to make them excellent. It is far more open than its rivals. Openness has fallen out of fashion, but Sir Keir Starmer should know as an Arsenal fan—celebrating his team's league title this week—that unpopular tactics can work best. His government has adopted the opposite strategy, cutting work visas and keeping taxes hostile to rich foreigners. In footballing terms, he's got the formation all wrong.

Another lesson is to embrace competition. The Premier League is one of the world's greatest experiments in self-regulation, a case study of the value of ruthless rivalry and the free flow of cash. Yet a foolish cross-party decision to impose an independent football regulator passed into law last summer. English football is no stranger to cheating, from financial doping to a spying scandal at Southampton. But such cases are best dealt with in-house. If the regulator, as feared, puts up hurdles to investment, it would typify Sir Keir's failure to grasp what made the Premier League (and Britain) strong.

Whoever next leads Labour will hear plenty of voices favouring fiddlier rules and greater suspicion of foreign capital. If it is to be Andy Burnham, Manchester's mayor and the current favourite to unseat Sir Keir, he ought to visit his own constituents in East Manchester—a place transformed by the Emirati investors who took a liking to Manchester City. Britain desperately needs inward investment. The Premier League's formula of globalised, caution-to-the-wind capitalism is the best way to attract it. Not the nosy managerialism of the Labour left which, like a video assistant referee (VAR), only slows down the action and spoils the fun.

Last, Labour should strive for stability at the top, but not at all costs. The worst clubs get through managers almost as fast as Britain does prime ministers. The best have the longest-serving bosses. If Tottenham Hotspur's awful season holds any lessons, it is that sticking too long with a bad choice is a losing strategy.

Britain seems likely to avoid that mistake. Having lost the dressing room and the fans, Sir Keir is bound to be given the boot. Unfortunately, Britain cannot scout the world to find a brilliant new boss. Instead, it must pick a new prime minister from among the 400 or so Labour MPs. None of them could be mistaken for Pep Guardiola and some of them think the ideal game plan is to put everyone on the left wing and hope for the best.

Britain needs a better communicator in the national dug-out. Sir Keir hates giving pep talks and loathes the word "vision". He has a tendency to grumble about how hard his job is. His successor should instil a new sense of belief in a country prone to pessimism. English football's renaissance from mockable mediocrity to global dominance shows that a comeback is always on. It may be too late for Sir Keir, but not for Britain. ■



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Letters *Norway's economy, rich couples in urban areas, Marcus Crassus, Scotland's languages, the European Union*

Norway's labour market

A recent Free Exchange column (April 4th) highlighted two of the main challenges that the Norwegian government sees for the years ahead: increasing the supply of labour and raising productivity across the economy. As the population ages, pressure on labour supply will increase unless we succeed in bringing more people into work or producing more with fewer hands. We have succeeded with a pension reform in Norway, and the public sector is productive in many areas, but much more must be done.

That being said, parts of the column suggested that balance and precision have at times given way to the demands of a good story. As you noted, Norway spends a lot of money helping other countries, but the single most important contribution here is to Ukraine's defensive struggle. Norway, with fewer than 6m inhabitants, is among those providing the most military support to Ukraine. My reading is that *The Economist* generally regards such support as a good thing.

Norwegian young people perform well compared with other countries. Eurostat data show that unemployment among people in their 20s is substantially lower in Norway than in Denmark and well below the European average. The share of young people (aged 15 to 29 years) neither in employment nor in education and training is far below the European average.

Finally, although Oslo's coffee bars have enviable quality and expertise, it is not the case that 70% of baristas and others in unskilled jobs have a master's degree. Only 15% of Norwegian workers have a master's or equivalent level, and they have not piled up at the espresso machines. I recommend a visit to a coffee bar

when you are in Oslo, to experience the quality of coffee produced by skilled baristas without master's degrees.
JENS STOLTENBERG
Minister of finance
Oslo

The young and the affluent

In academic and activist circles "gentrification" is a pejorative word ("The kids in America", May 9th). However, we should celebrate the visible return of young affluent families to city centres; my research has found that wealthy families can be a net positive for urban neighbourhoods. For decades the greatest threats to our cities have been disinvestment and segregation. The arrival of "high-flex" families brings a stabilised tax base that funds public services, parks and infrastructure upon which all residents, regardless of income, depend.

The visibility of white toddlers in fancy strollers should not be viewed merely as a sign of neighbourhood vitality, but

as an opportunity for institutional renewal. The real test is whether these families invest in more than just their real estate. The goal is to see them commit to local public institutions, particularly schools. That is an essential first step towards the kind of durable urban integration we have long neglected.
KFIR MORDECHAY
Associate professor of education and policy studies
Pepperdine University
Malibu, California

Language lore

Describing the battle between Scots and Gaelic in Scotland you leaned on a fringe claim that Gaelic attracts greater government support because it is "uncontaminated by Englishness" ("Scots, wha hae wi' Wallace bled", May 2nd). In reality funding for Gaelic reflects decades of co-ordinated language activism since the 1970s and the existence of a developed institutional base. Gaelic speakers enjoy public broadcasting, a quango established by Labour in 2005, and around 90 schools offering Gaelic-medium education. Scots has no equivalents.

Rather than a nationalistic motive, Gaelic's prominence at an event to mark the Battle of Culloden is similarly prosaic. The commemoration is organ-

ised by the Gaelic Society of Inverness and has been for over a century. Nor was Gaelic "in effect banned" after the battle. The post-Culloden acts of proscription did not mention the language.

Despite this support, Gaelic is hardly winning what the article erroneously termed a "battle". "The Gaelic Crisis", a determinative, if depressing, sociolinguistic survey in 2020, found its usage to be in freefall in its few remaining heartlands. By contrast, Scots still counts around 1.5m speakers.

LIAM ALASTAIR CROUSE
*Isle of South Uist,
Outer Hebrides*

The deep-seated impulse to emphasise what distinguishes Scotland from England is a textbook example of what Sigmund Freud called the narcissism of minor differences. It may be an awfy shame, but I am afraid there is no cure.

DR JOHN DOHERTY
Vienna

Retail politics

Being myself on the verge of retiring from the European civil service, I felt a deep connection to Charlemagne's column on federalists' nostalgia for a Europe that never was (May 9th). I especially enjoyed the analogy between the pioneers of European integration and the builders of medieval cathedrals. Indeed I count myself as one of the "still many in Brussels who get teary-eyed as they describe their humble role in...this continental peace project, as worthy of admiration in their eyes as any cathedral". But alas, I can't help thinking that my late father, himself an MEP, was right when he used to fume: "They promised us a cathedral and built a supermarket instead."

ROLAND DEHOUSSE
Brussels

The folly of war

The *Telegram* (May 9th) compared Donald Trump to Caligula. As Caligula never started a war it would be more appropriate to compare Mr Trump to another Roman leader, Marcus Licinius Crassus. Crassus was a real-estate tycoon who made his fortune by buying the properties of those purged after a civil war. Frustrated that he had not received the credit he thought was his for having quelled Spartacus's slave revolt, he was eager to gain glory through a victory against a "real" enemy. With the help of his buddies in the triumvirate, Caesar and Pompey, he got his chance in 53BC. Crassus was assigned to subdue Parthia, an empire that covered today's Iraq and

Iran. Ignoring advice from his generals, Crassus rushed headlong into the desert and the Parthian armoured cavalry, resulting in one of the largest defeats suffered by a Roman army. The Parthians cut off his head and poured molten gold into his mouth.
DR PAUL VANDERBROECK
Geneva



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BY INVITATION

Gill Whitehead

The coming IPO wave will enshrine the AI gods' lack of accountability



AFTER WEEKS of waiting with bated breath, investors finally have sight of the prospectus for the initial public offering of SpaceX. OpenAI and Anthropic could join Elon Musk's firm in floating shares later this year. These IPOs will reveal that the race to control frontier AI is defined not only by their epochal vision and scale, but by the concentration of corporate control. All three firms have sought to govern cutting-edge technologies with custom-made rules, departing from contemporary corporate governance and its well-worn systems designed to thwart unfettered control by an individual through effective board oversight, risk-based incentive structures and robust succession plans.

Their logic is not without reason. The global financial crisis revealed that most failing banks actually met the criteria for good governance on paper. The recent backlash against heavy compliance costs and short-term market pressures—seen as a drag on innovation—has prompted a move towards founder-centric models.

However, there are causes for concern. The speed and scale of experimentation in corporate governance is unprecedented. Moreover, given the high degree of uncertainty as to the nature of AI risks, the tools of regulation, legislation, ethics and governance will be needed to manage the impact of these companies. There are three notable departures from traditional frameworks.

The first is the entrenchment of super-voting equity. Adopting the founder-control model pioneered by Alphabet and Meta, dual-class share structures bypass the “one share, one vote” principle. Founders retain voting control despite owning a minority of the equity, in effect silencing public investors. Provisions in SpaceX's IPO filings cement a structure granting Mr Musk and other insiders shares with ten times the voting power of ordinary shares, securing him a clear voting majority. This will be combined with the use of SEC-permitted “controlled company” exemptions that forgo an independent board or compensation committee.

The second departure involves jurisdiction-shopping and regulatory arbitrage. Because the scale of these listings is historic, American states and stock exchanges are jostling to secure them. States are competing by offering increasingly robust statutory protections to directors. Stock indices, meanwhile, have increas-

ingly skewed their listing standards to accommodate tech giants.

SpaceX is using this competition to its advantage. Following a Delaware court decision to void Mr Musk's Tesla pay package on the grounds of weak board independence (later overturned), he moved SpaceX to Texas and its director-friendly Business Court, later merging xAI into the rocket firm. Furthermore, SpaceX plans to list on Nasdaq, which would enable it to benefit from the exchange's new “fast entry” rule that allows newly listed large firms to enter the NASDAQ-100 after just 15 trading days, rather than three months. As a result, index funds are legally compelled to buy stock almost immediately, further weakening the leverage of active fund managers to extract governance changes.

The third departure is the use of hybrid structures. The anticipated IPOs of OpenAI and Anthropic may yet enshrine their novel corporate architectures, designed to insulate their technical missions from standard venture-capital incentives. In October OpenAI completed its restructuring into a for-profit Public Benefit Corporation (PBC) in which the original non-profit foundation now has a 26% stake. Although the foundation retains the legal authority to appoint and remove the PBC directors (and thus the chief executive), the intentional significant overlap in foundation and PBC board members—many of whom were installed after the reinstatement of Sam Altman as chief executive—indicates that operational power lies increasingly with the CEO.

Anthropic has made perhaps the most creative undertaking to enshrine its safety-first mission into its corporate governance. Its Long-Term Benefit Trust enables independent experts to elect a majority of directors over time. However, even under this model trustees serve only one-year terms and must consult the CEO on appointments. Crucially, a supermajority of shareholders retains the power to dissolve the trust without trustee consent. The trust provides an advisory guardrail, but the founders and their main backers (Amazon and Google) retain ultimate authority.

There is much to admire in these experiments; corporate governance must always be evolving. But they are untested in situations with such high stakes for humanity. The irony is that as industries scramble to engineer digital AI guardrails, many of their corporate-governance equivalents have just been dismantled.

The Musk v Altman trial spent weeks interrogating the trustworthiness of AI's elite, only to be dismissed this week on procedural grounds. Yet it exposed a raw truth: these historic IPOs and the humanity-impacting potential of their technologies now rely on the precarious scaffolding of a few founders' good intentions.

Investors have noticed. Institutional groups, led by public pension funds, are lobbying for governance changes such as clauses to limit the length of time for which dual-class structures apply. In the absence of change, responsible investors must weigh the immense financial upside of heavily oversubscribed IPOs against the lack of independent board oversight and accountability.

Governments and regulators should take notice, too. The hubris is not in believing transformative technology can be built. It is in believing that individual judgment is the appropriate final line of defence. Anthropic's Mythos preview last month showed that regulators are already struggling to keep up. Future AI rules must be designed with these corporate structures in mind. Backstops must not be removed precisely when they are most needed. ■

Gill Whitehead is a visiting policy fellow at the Oxford Internet Institute and a member of the advisory council of Frontier Economics.

Briefing SpaceX



Out of this world

SpaceX has initiated the biggest public offering ever

IT IS A launch of a more bureaucratic sort than normal for SpaceX, but still characteristically spectacular. On May 20th, after America's financial markets had closed, Elon Musk, the rocketry firm's founder and the world's richest man, lit the blue touch paper on the biggest initial public offering ever. Documents filed with American regulators pave the way for SpaceX, which has conquered space but is spending a fortune on artificial intelligence (AI), to make its debut on the NASDAQ exchange in June.

Mr Musk had hoped to mark the occasion with a test flight of the latest version of the firm's gigantic, troublesome Starship rocket, upon whose stainless-steel frame the company's astronomical ambitions rest. But the test, originally scheduled for May 19th, has slipped repeatedly. As *The Economist* went to press, Starship was due to fly on May 21st. Its success or failure could have a big impact on the IPO.

For many years SpaceX had dismissed the idea of an IPO. "We can't go public un-

til we're flying regularly to Mars," said Gwynne Shotwell, the firm's chief operating officer, in 2018. Mr Musk has long argued that public markets are too short-termist and unimaginative for a firm whose goal is eventually to build a city on Mars.

But getting to Mars is expensive. The \$75bn or so SpaceX hopes to raise from investors would far exceed the record sum raised by Saudi Aramco, an oil giant, when it went public in 2019 (see chart 1 on next page). SpaceX's target valuation of around \$1.75trn is more than 90 times the \$18.7bn in revenue the firm brought in last year. (Tesla, Mr Musk's electric-vehicle company, trades at a mere 16 times revenue.)

Business, but not as we know it

Since its founding in 2002—with little more than "carpet and a Mariachi band", as Mr Musk once joked—SpaceX has elbowed aside incumbents such as Boeing and Lockheed Martin and left government space agencies in the dust. These days it

runs the world's biggest space programme. Its pioneering re-usable rockets carry almost 90% of everything that goes into space (see chart 2). Starlink, its satellite-broadband service, boasts nearly 10,000 satellites, over two-thirds of the total in orbit, and more than 10m customers. As Mike Grace, the boss of Longshot, a space start-up, puts it, SpaceX "owns the solar system right now, and the rest of us just rent it".

Mr Musk thinks he has found an even bigger market for SpaceX to dominate: AI. He believes that AI's appetite for computing power cannot be satisfied by earthly data centres, which are often impeded by power shortages, red tape and opposition from a distrustful public. Along with several other AI bigwigs, he reckons those problems can be solved by putting data centres in space. He argues that SpaceX, with its unmatched ability to carry things off the planet, is the firm to make it happen.

The scale of his ambition is outlined in the firm's regulatory filings. Mr Musk's pay ►►

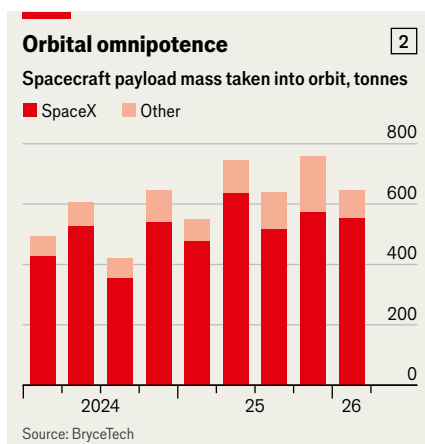
depends on SpaceX's valuation rising even further (to as much as \$7.5trn); on the firm putting 100 terawatts of computing power into orbit (about 1,000 times the total of every data centre on Earth today); and on building a Martian city with at least a million inhabitants.

Could anything remotely like that come to pass? The paperwork reveals a firm with a fast-growing money-spinner in the form of Starlink, which brought in \$4.4bn in operating profit in 2025—but also a money pit that is growing even faster. In January SpaceX merged with xAI, Mr Musk's AI firm, which lost \$6.4bn in 2025 as it raced to build computing infrastructure. Partly as a result, SpaceX's capital spending almost quintupled last year compared with 2023, to more than \$20bn. Mr Musk seems to have calculated that, even with Starlink, this bill could not be met with private money alone. But some investors may hesitate to pour cash into an AI lab that is a pipsqueak compared with such rivals as Anthropic and OpenAI. SpaceX's filings say AI makes up 93% of its claimed "total addressable market" of \$28.5trn.

If this gigantic bet on AI-from-space is to pay off, at least three things must happen. Starship, which is already behind schedule, must be made to work. Starlink must generate more cash. And orbital data centres must offer big advantages to an industry that is already investing astronomical sums in Earth-bound computers.

Start with Starship. The gargantuan rocket, the heaviest flying object ever built, is "vital to SpaceX's future", says Caleb Henry of Quilty Space, a firm of analysts. It is supposed to finish the revolution that the Falcon 9, SpaceX's current launch vehicle, began: transforming space flight from something that is rare, bespoke and expensive into something cheap, mass-produced and routine. Before SpaceX, most rockets were disposable, flown once and then dumped into the ocean or abandoned in space. Mr Musk likens that to building a plane, flying it once and then scrapping it.

Half of every Falcon 9, by contrast, is



designed to be re-used. After separating from the upper stage the rocket's lower portion can fly back to its launch site or a waiting ship and land on its own tail. It took SpaceX five years, and plenty of crashes, to pull off its first successful landing in 2015. But these days, it is routine. SpaceX conducted 165 Falcon 9 launches in 2025, a number that would have seemed impossible even five years ago. Some boosters have flown more than 20 times.

That has helped SpaceX slash the cost of going to space. It charges \$74m for a Falcon 9 flight, which can take 17.5 tonnes into orbit. That works out to about \$4,200 per kg—about 85% less than what incumbents such as United Launch Alliance, a joint venture between Boeing and Lockheed Martin, used to charge. (The actual cost to SpaceX is thought to be much lower, perhaps around \$1,000 a kilo.)

Starship aims to drive that down even further, partly by making the rocket's payload as big as 200 tonnes, but mostly by making the second stage re-usable as well. The filing documents suggest a goal of \$185 a kilo. But making it work has not been easy. "Starship was supposed to take over from Falcon 9 in 2023 or 2024," says Tim Farrar, who runs TMF Associates, a satellite-industry analysis firm. It is not just SpaceX that is frustrated by the delays.

NASA is waiting for a modified Starship to ferry humans to the moon as part of its Artemis programme.

SpaceX has on several occasions flown Starship's booster stage back to its launch site and caught it with robotic "chopstick" arms mounted on the launch tower. Re-using the second stage, though, is far harder. It travels much farther and faster, and must survive temperatures above 1,000°C as it re-enters the atmosphere at speeds of 8km a second or more. Speaking recently on a podcast, Mr Musk noted that several second stages have managed controlled splashdowns at sea, but their heat shields were damaged, he said, and so "not re-usable without a lot of work".

The Space Shuttle—the industry's previous big attempt at re-use—suffered from similar problems. Fixing the damaged heat-shield took weeks or months. Mr Musk, in contrast, hopes the same Starship might eventually fly several times a day.

Boldly going forward

Starship's main job will be to bolster Starlink, the fastest-growing and only profitable part of SpaceX's business. Since its debut in 2021 the service has signed up airlines, shipping firms and more than 10m individual customers, revolutionised warfare in Ukraine and spawned a secretive offshoot called Starshield for government customers. The firm's newest "direct-to-cell" (D2C) satellites can communicate directly with smartphones and other devices on the ground, with no need for the special pizza-box-sized antennas that its residential plans require. Starlink's revenue grew by 50% between 2024 and 2025, from \$7.6bn to \$11.4bn.

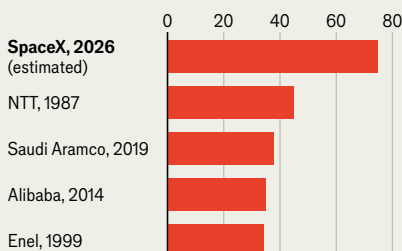
There is clearly more room for growth, says Mr Farrar. He notes that in one 80-day period in 2025 the firm added 1m new customers. He thinks Starlink may eventually sign up anywhere between 20m and 50m residential subscribers, although average revenues per customer are likely to decline as it adds users outside the rich world. At the moment, though, Starlink's growth is held back by the size of the satellites that SpaceX can fit onto a Falcon 9. It plans "version 3" satellites that will each add 20 times more capacity than the "v2 Minis" it currently uses. But they are so big they can be launched only with Starship.

SpaceX is investing heavily to keep Starlink growing. Last year it spent \$19.6bn to acquire spectrum from EchoStar, an American telecoms firm, to bolster the nascent D2C business, and has sought authorisation to launch 15,000 specialised satellites. PitchBook, which gathers data on venture-capital-backed firms, notes that these are expensive wagers. The EchoStar deal represents more than a year of SpaceX's revenues. But it expects Starlink to top 1bn subscribers over the next 15 years.

Stratospheric

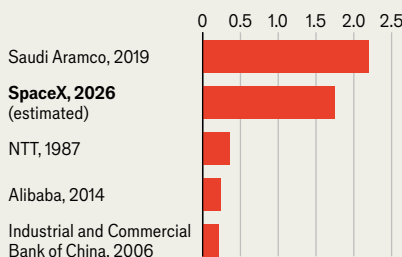
Largest IPOs, 2026 prices

By capital raised, \$bn



Sources: Bloomberg; press reports

By valuation, \$trn



► Analysts are divided over Starlink's potential. "I think D2C is just not going to live up to the hype," says Mr Farrar, arguing that it will be used mostly in places where there are no terrestrial phone masts and therefore few people. Simon Potter of BryceTech is more optimistic: "It could enable ubiquitous connectivity between devices using standard smartphone kit. That could be transformational in applications such as cars, agriculture, logistics—and maybe in military ones, too." Franco Garda, at PitchBook, thinks that Starlink and SpaceX's launch business could justify a \$1.75trn valuation by themselves.

Launching lots of satellites, however, will not use up all Starship's capacity. SpaceX is planning fleets of them. It is building a pair of huge factories, dubbed "Gigabays", which Mr Musk hopes will eventually produce hundreds of Starships a year. Even with the Falcon 9, points out Mr Henry of Quilty Space, there was not a "clear commercial case for a high-cadence launcher". In the end, SpaceX had to develop Starlink to give its rockets enough work to keep them busy. Its plans for data centres in space, he says, are an attempt to repeat the trick.

Tom Mueller, SpaceX's first-ever employee, the designer of its rocket engines and now the founder of Impulse, another space firm, summarised his old employer's plans last year. Building a settlement on Mars requires hundreds of Starships, he noted. But the movements of the planets mean they can travel to Mars only roughly every two years. "What do you do with the hundreds of Starships the other 25 months of the Mars cycle?" he wrote. "Fly data centres to space, paid for by investors."

SpaceX did not come up with the idea of putting AI in orbit. Last year Starcloud, a startup, flew a prototype (aboard a Falcon

9) to prove the idea could work. Google, a tech firm, hopes to test satellites of its own in 2027. (It is reportedly in discussions with SpaceX to launch them.)

Proponents of orbital data centres (ODCs) cite several advantages. One is an abundance of power. With no atmosphere to get in the way, sunlight in orbit is about 30% more intense than on even the sunniest day on the ground. Putting solar-powered ODCs in certain "sun-synchronous" orbits could provide almost permanent sunlight. Efficiency is another plus. A terrestrial data centre might use a quarter of the electricity it consumes to cool its servers. ODCs could instead rely on the freezing vacuum of space.

The final advantage is remoteness. The AI industry stirs fears of job losses and even human extinction. Many people take a dim view of the technology. Dozens of American state and local governments have banned new data centres or are considering it. Putting them in space might keep them out of mind as well as sight.

We can't find reverse

With typical bravado, Mr Musk has said that within two to three years, ODCs will be the cheapest way to provide computing for AI. (Sam Altman, the boss of OpenAI, thinks that timeline is "ridiculous".) The economics depend on how low launch costs go, how efficient satellites are at turning sunlight into data-crunching, and how much those satellites cost. SpaceX hopes to make use of its experience mass-producing Starlink's satellites by building ODCs, at least at first, on the same chassis. In January the firm applied for regulatory approval to launch up to a million of them.

If they can be made to work, the market could be absurdly lucrative. Capital spending on data centres is forecast to exceed

\$800bn this year. The revenues of the top AI labs are growing fast, from about \$17bn in 2025 to an anticipated \$90bn-100bn this year. Both Anthropic and OpenAI are planning IPOs at valuations of about \$1trn.

SpaceX could profit in two ways. The first is by boosting xAI, which has struggled to compete with its longer-established rivals. If being part of SpaceX gives xAI access to cheap, quickly deployable computing power, that could provide the boost it needs to overtake its competitors.

If that does not work, an industry-wide switch to space-based infrastructure could leave SpaceX occupying a similar position to Nvidia, which designs AI chips, or TSMC, which manufactures them—an indispensable provider of infrastructure. There have been hints of this latter approach already. On May 6th Anthropic signed a deal to pay as much as \$1.25bn a month for the next three years to use some of the computing capacity at xAI's existing, land-based Colossus data centre.

Giant rockets, AI in space—science fiction made real: will investors buy it? There is likely to be appetite, especially given what Gil Luria of D.A. Davidson, an investment firm, calls the "hopes and dreams" premium that Musk's futuristic endeavours tend to command. The IPO includes a five-to-one stock split, which reduces the cost per share, a boon for Mr Musk's legions of fans among retail investors.

But investors should brace for volatility. Many of the venture capitalists who made early bets on SpaceX will be keen to cash out as soon as possible. That could hit the stock price. The firm's two-tier share structure makes Mr Musk in effect unremovable, no matter how his plans go. Although some of his big bets have paid off—re-usable rockets, electric cars, Starlink—others have not, or at least not on schedule. Tesla's fleet of robotaxis has been just over the horizon for years. Its humanoid robots are also late for work.

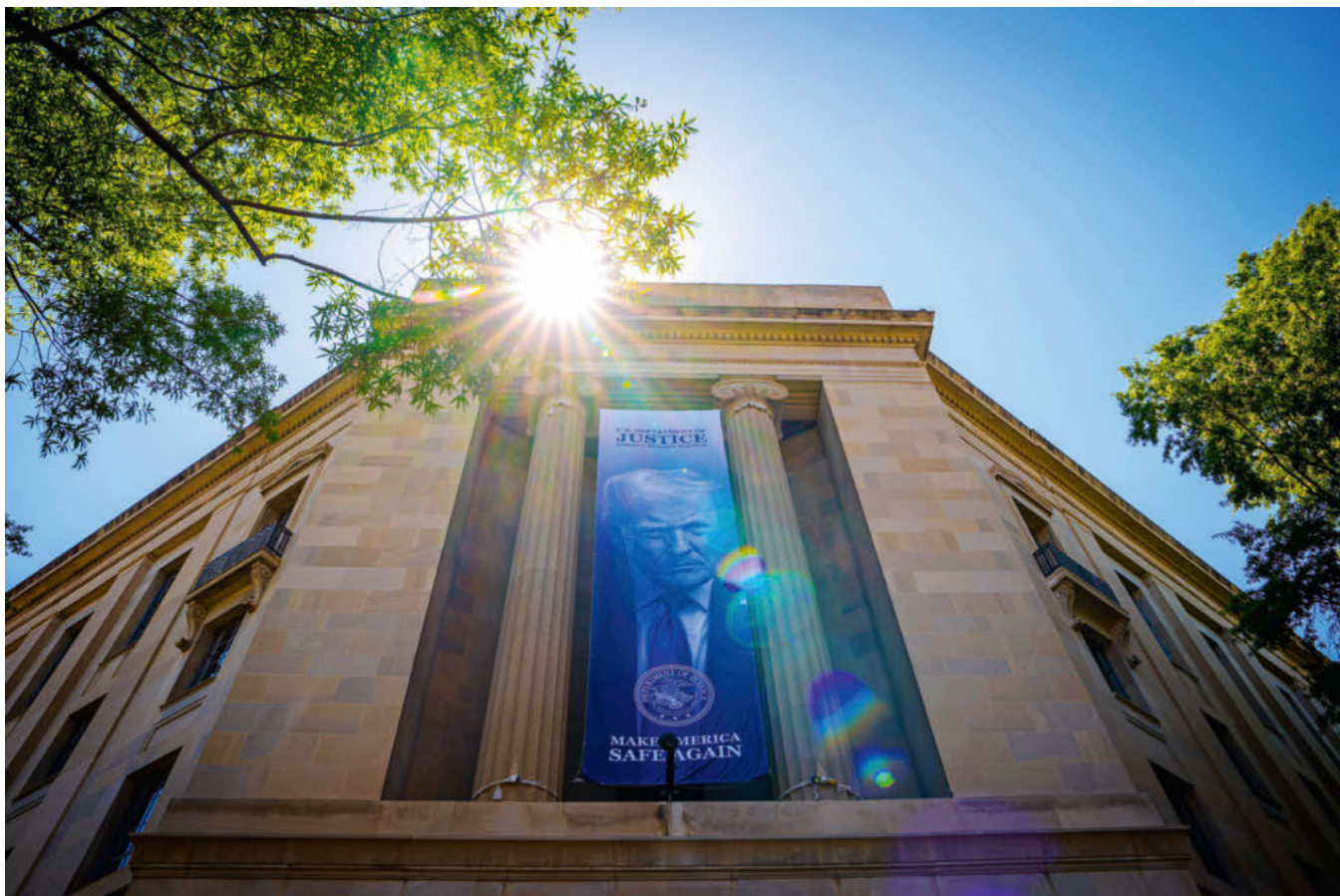
xAI, meanwhile, is losing billions. If progress on Starship slows, Starlink's growth will be hobbled, bringing in less cash to cover those losses. SpaceX's dominance in launch could be eroded by competitors such as Blue Origin, which is owned by Jeff Bezos, the founder of Amazon. Markets are high and nervy; even if AI proves as useful as its pioneers hope, many see current levels of investment as unsustainable. Where would SpaceX's plans stand if they slowed?

Mr Musk's lofty aspirations for SpaceX can be made to hang together, at least conceptually. As he is fond of saying when assessing a grand idea, nothing about them breaks the laws of physics. That means making it all happen is "merely" a question of engineering, both industrial and financial. But as Mr Musk knows, engineering is an unforgiving discipline. ■



The sky is not cloudless

United States



Trumpian justice

Payments for perps

WASHINGTON, DC

Taxpayers will soon pay compensation to the people who smashed up the Capitol on January 6th

AMERICAN COURTS are not supposed to be venues for performance art. If counter-parties in a lawsuit do not have adverse interests—say, because they answer to the same person—judges balk. So when Donald Trump sued his own administration seeking \$10bn in damages, the judge had questions. Rather than address them, on May 18th the president dropped his suit altogether. In exchange he secured a commitment by his own administration to compensate victims of government “law-fare”, to the tune of \$1.8bn.

In MAGA-speak, that means political allies prosecuted by Democrats: think January 6th rioters, pro-life activists and the like. The scheme is of a piece with the self-dealing and the shakedown that have defined this presidency. Rather than line his own pockets with taxpayers’ money—Mr

Trump is not taking a cut himself—this one will line those of his supporters.

The lawsuit originated with a genuine wrong to the president. Between 2018 and 2020 a contractor at the Internal Revenue Service (IRS) named Charles Littlejohn illegally leaked Mr Trump’s tax returns to the *New York Times* and *ProPublica*, an in-

vestigative outlet. Mr Littlejohn was prosecuted and sentenced to five years in prison. Other billionaires also had their information exposed, including Ken Griffin, a hedge-fund tycoon. Both Mr Griffin and Mr Trump sued the IRS. But whereas Mr Griffin only sought an apology, which he got, Mr Trump demanded \$10bn. Never before had a president sued his own administration. “Essentially the lawsuit’s been won,” he said back in February, a seeming acknowledgment of the stitch-up.

A standard payout for people with breach-of-confidentiality claims is \$1,000 per violation. Anything above that requires the plaintiff to show “extraordinary” damage, which should in theory match losses incurred. Mr Trump has not had to justify anything to anyone. Todd Blanche, the man defending the government, once worked as his private lawyer. “I love you, sir,” he told his boss last month. The melding of law enforcement with the president’s personal agenda is another hallmark of this administration. As part of the deal, the IRS also agreed to drop any pending audits of Mr Trump, his family or his firms.

Nothing about the fund’s structure is normal. The Department of Justice (DoJ) says anyone who “suffered weaponisation”

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▶ and lawfare” for their politics can file a claim and that payouts can be kept secret. Really, it is a taxpayer-funded “slush fund for people with no legally cognisable injuries that was never authorised by Congress or the courts”, says Maria Glover of Georgetown University. That it was set up through a private deal makes it a handy way to circumvent Congress, which controls spending under the constitution. Barack Obama’s DOJ did something similar when it sued banks and diverted the proceeds to left-wing groups without congressional approval. Those deals were criticised by Republicans in Congress.

There are no “partisan requirements” to seek a payout, according to the DOJ. In reality, the eligibility criteria are clear enough: only MAGA need apply. Already this year the Trump administration has paid over \$1m each to settle claims by Michael Flynn and Carter Page, two allies. Expect the Capitol rioters to be first in line. Mr Trump pardoned nearly all 1,600 of them on his first day back in office. Cynthia Hughes, a MAGA activist, has called those pardons a “start, not the finish”, and indeed some defendants were seeking civil damages before the announcement of the fund. They contend they were merely engaging in political protest that day, much like the crowd at the Women’s March. They skip over the fact that pussy-hat-wearing moms did not ransack the Capitol for all to see on TV, in the biggest act of American political violence this century.

Anti-abortion activists who obstruct or attack clinics have in essence been invited to seek payouts, too. Last year Mr Trump pardoned two dozen of them. In April the DOJ awarded \$1.1m to another who was acquitted at trial three years ago. Yet another cohort of claimants may well be violent cops. Through pardons, downgraded charges and reduced sentences, Mr Trump has undercut at least four cases against those prosecuted by the Biden administration for their excessive use of force.

Mr Trump’s crusade against “weaponisation” stems from his own experience. While campaigning in 2024 he was indicted twice by Joe Biden’s DOJ and twice by Democratic district attorneys. He was also sued in civil court. Some of those prosecutors ran on the promise to go after him. That was ill-judged. But Mr Trump has outdone them all with his baseless revenge-prosecutions targeting his critics.

Anyone who thought Democratic prosecutors were overzealous towards Mr Trump then should be boiling now. The president claims to want to end the weaponisation of the law. Of course the way to end it is to end it, not turbocharge it, notes Gregg Nunziata, a conservative lawyer. Conveniently, the anti-weaponisation fund expires before the next election, meaning Mr Trump’s targets will be ineligible. ■

Weapons sales

A disarming delay

WASHINGTON, DC

Europeans want more American-made kit but cannot get it

AMERICA HAS long been the world’s arms dealer of choice. From 2021 to 2025, its share of global arms exports jumped to 42%—more than Russia, China and the European Union combined. Over the same period, American arms exports to Europe were 217% higher than in the previous five-year period, according to the Stockholm International Peace Research Institute (SIPRI), a think-tank. As the old world rearms at its fastest pace since the end of the cold war, it has splurged on American warplanes, air-defence systems and long-range missiles.

These days, however, Europe could be forgiven for experiencing some buyer’s remorse. America has churned through munitions in the war against Iran and has told European officials to expect significant delays in the delivery of American-made weapons. Shipments are instead being diverted to replenish America’s own depleted stocks. That reportedly includes scarce munitions, like air-defence interceptors, as well as strike systems like the HIMARS rocket launchers. This has left European diplomats privately frustrated.

They have reason to feel aggrieved. White House officials have long pressed European governments to boost defence spending and also to buy American kit as a way of placating Donald Trump and keeping him engaged in NATO. It has not worked. He has repeatedly displayed con-

tempt for NATO allies, most recently over their refusal to send ships to reopen the Strait of Hormuz. The shortfall of munitions is starting to complicate NATO’s existing operational plans. Further delays in shipments might soon affect the flows of weapons to Ukraine, too.

Europe buys most of its weapons from America through the arcane Foreign Military Sales (FMS) process. Instead of being agreed on with contractors, FMS deals are struck directly with the American government, which has leeway to adjust the terms. It is best to think of FMS not as a standard procurement process but as a tool of American foreign policy, says Javier Ospital of Bruegel, a Brussels-based think-tank. The fine print on each agreement states that “materiel procured or stocked for FMS may be diverted to meet higher priority requirements”. America has redirected the delivery of some weapons in the past. Under the Biden administration, shipments of air-defences were diverted from other European countries to Ukraine.

The current problem is the sheer scale of the shortage and America’s inability to fulfil a giant backlog of orders. Take the Patriot air-defence system. America has burned through more than 1,300 interceptor missiles during the Iran war, which is equivalent to more than two years of production at current rates. At the same time, the Trump administration has fast-tracked the sale of Patriots to countries in the Gulf, which have also run down their stockpiles. That has pushed European orders even further down the priority list. Switzerland, for example, has said the delivery of five Patriots batteries, which it expected to receive this year, could be delayed up to seven years because of the Iran war.

Even before the Iran war, Trump administration officials had expressed mis-



For display purposes only

givings about selling certain weapons to Europe. Elbridge Colby, the Pentagon's undersecretary for policy, has long questioned the wisdom of some foreign military sales given existing production constraints. Each air-defence system sold to Europe, in his logic, is one fewer available to America to defend against China in the Pacific. But if the current delay in deliveries lasts much longer, it could come to hurt America in the longer run, too. "It really undermines trust in America's willingness to be a reliable supplier of arms," says Pieter Wezeman of SIPRI. Countries in Asia, for example, may think twice before buy-

ing American weapons, he adds.

European countries, for their part, are buying local to reduce their dependence on America. Last year Denmark opted for the Franco-Italian SAMP/T air-defence system instead of the Patriot. More recently, NATO's procurement agency chose a Swedish-Canadian aircraft as its future airborne warning and control aircraft, a programme which has been served by an American aircraft for 43 years. But Europe will struggle to quickly replace American capabilities like airborne surveillance or long-range missiles. Until then, it will have to follow the dealer's choice. ■

Church and state

Pulpit partisans

An analysis of nearly 90,000 evangelical sermons quantifies how many pastors are endorsing parties

THE STORY goes that Lyndon Johnson was so incensed that some non-profit groups had backed his opponent in a 1954 Senate primary election that he proposed a change to the tax code to prohibit such groups from endorsing or opposing political candidates. Today the Johnson Amendment means that churches can lose their non-profit status (and be forced to pay taxes) if clergy engage in political advocacy from the pulpit. In practice it is widely flouted. Back in 2016 Donald Trump vowed to "totally [knock] out" the amendment, which he has so far failed to do.

Yet surveys of churchgoers suggest that America's clergy regularly do engage in political advocacy. Despite this there has been little research into the content of such speech, or how common it is. Sermons have always been hard to study in a systematic way: they take place behind closed doors; congregants may self-censor when surveyed; and texts can be edited before being published.

Since 2020, though, evangelical pastors have been live-streaming their sermons and posting the recordings online, giving researchers lots to study. Electioneering happens across denominations: black churches, for example, regularly endorse and raise funds for Democrats. But no other church has documented itself in this way. Beyond the question of access, evangelical churches are worth studying in their own right. They are decentralised, meaning pastors are freer than Catholic priests or Mormon bishops to say what they please. And no other religious group has mobilised quite so enthusiastically around President Trump.

In a paper published this week in the

Proceedings of the National Academy of Sciences, a team of researchers analysed 63,683 hours of unfiltered audio recordings from 88,546 evangelical church sermons delivered across the country during the 2020, 2022 and 2024 election cycles, and a non-election control period.

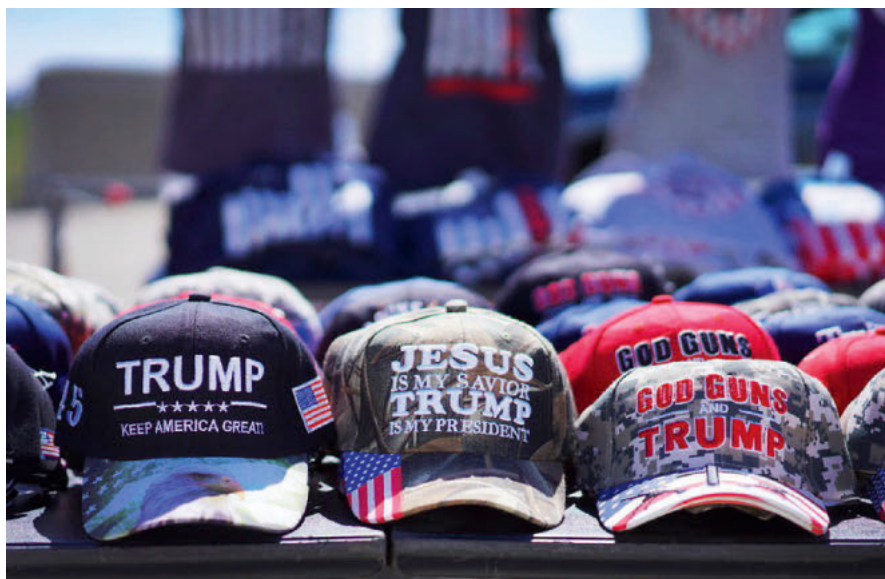
After transcribing the sermons, the researchers used a large language model (LLM) to screen them for electioneering. It first sorted political from non-political text by searching for words to do with politics, such as "elections" and "voting". It looked for references to political figures or parties, like "Trump" and "MAGA", and live questions such as "abortion". Generic references to "America", ancient governments

like "Judea" or spiritual mentions of "the nation" were excluded.

Text flagged as "political" was then passed through a second filter, which identified instances of "political advocacy" as defined by the Internal Revenue Service. The LLM was instructed to flag passages such as "Vote for Candidate X" or "Candidate X is God's choice" as well as any calls for campaign contributions. Mentions of general political issues or of "voting for God" were ignored.

A third step made sure that each instance of "political advocacy" referred to a specific party or candidate and classified each as either favourable or unfavourable. For example, the sermon that contained the passage "if you can vote for Kamala Harris, you better ask God to forgive you," was classified as "opposition: Democrats", while "please, oh Father, make us all vote Republican or libertarian," was classified as "favorability: Republicans". The study found that nearly 90% of references to Republicans were favourable, and more than 70% of references to Democrats were unfavourable. Mr Trump was the topic of over 50% of political discussion.

In presidential election years, around one in seven churches engaged in electioneering—six times the rate in non-election periods and twice that of the 2022 midterms. But when a human researcher reviewed a random sample of 500 passages that had been flagged as political, 380 were judged to contain political advocacy, compared with just 121 flagged as such by the LLM. This suggests that the true number of evangelical churches engaging in political advocacy during elections may be closer to one in two. In 2024 pastors in Republican counties were also more than twice as likely to engage in political advocacy as those in Democratic ones. ■



The church of latter-day saint

Georgia

Bottoms up

ATLANTA

The state's Democrats take a risk on the governor's race

THERE IS A certain kind of jubilation that comes from unexpectedly good results. When Keisha Lance Bottoms won the Democratic nomination for Georgia's governor outright on May 19th, avoiding the run-off many expected, her supporters in an Atlanta hotel ballroom broke into a line dance as if they were at a summer cookout. Women in impossibly tall stilettos and men in suits stepped and swayed in sync to "Ain't Nobody", a soul classic.

They aren't the only ones celebrating: Republicans across the state are licking their chops. In a midterm cycle where Democrats are thought to have their best chance of taking the governor's mansion in decades, Ms Bottoms, the former mayor of Atlanta, may be her party's most polarising general-election choice. She won this week because she has better name recognition than her opponents—Georgia's last Democratic labour commissioner and county executive, and a millennial state legislator with an Obama-like demeanour—and because a sisterhood of black women came out in droves for her. But her time in office was marred by problems that will dog her when she faces a Republican in November (a June run-off will decide which one). "The commercials write themselves," says Fred Hicks, a Democratic strategist. "Republicans don't even have to spend, they can just talk about her record."

Their simplest line of attack is that Ms Bottoms is soft on crime, an allegation that has bite with a conservative southern electorate. After George Floyd was killed in 2020, Atlantans looted stores and set cop cars on fire. Ms Bottoms spoke as if she had no patience for it. "When Dr King was assassinated we didn't do this to our city," she told rioters. "Go home." But more than a month later armed vigilantes occupied several blocks around a Wendy's in south Atlanta—and the city didn't stop them. On July 4th a gangster fired into a car driving past, killing an eight-year-old girl. Something seemed broken. Residents of Buckhead, a rich neighbourhood, began making plans to secede from Atlanta.

That autumn a city councillor chided the mayor for letting Atlanta spiral. "Stop minimising our concerns by telling us that 'crime is up everywhere,'" he wrote. "It will take a lot to turn this around. But here, in descending order, are the three things we need to begin: 1) Leadership 2) Some leadership 3) Any leadership."

Americana

Basqueing in it

BOISE

A corner of Idaho will forever be north-east Spain

INTRODUCING HOUSE BILL 561 to the Idaho Legislature, Ted Hill did not expect to stoke international controversy. The law, which originally banned local governments from flying the flags of non-states, was intended to stop Boise from flying the gay-pride flag. Earlier this year the president of the Basque Country, an autonomous region in Spain, sent a letter expressing concern about the effect HB 561 might have on the flying of the Ikurrina, the Basque flag, during Jaialdi, the 40,000-person Basque festival the city hosts every five years. Worried about flagging support for the bill, Representative Hill offered the Basques a carve-out for the Ikurrina.

Speakers of the language first came during California's gold rush, then moved from mining to sheep herding. By 1900 chain migration saw nephews follow uncles as Basque shepherds spread across federal land. They carved 25,000 Basque-language messages into trees across the West. Some with Basque ancestry tried to shed it. "My great-grandparents' generation said, 'Learn English, don't speak Basque.' But my mom's generation worked to get Basque back," says Olaia Urquidi Beals of Txantxangorriak, a musical group. On Tuesday nights they gather with *trikis* (accordions) and *panderos* (tambourines) and



Staying alive

sing in Basque. Afterwards, some musicians visit Ansots Basque Chorizos & Catering around the corner. Just down the road is Boiseko Ikastola, America's only Basque-language pre-school.

There was a time, in the late 1970s, when it looked as if the language and culture would fade away, says Dave Bieter, a former mayor of Boise. Now when he plays Mus, a Basque card game, he says a third of players speak Basque. There are about 40 Basque clubs in America, mostly in the West. *Jainkoak Amerika bedeinka dezala!*

City Hall insiders recall Ms Bottoms being absent during her term. Where she was perhaps most present was in her feuds with Brian Kemp, the popular Republican governor, over pandemic rules. In clashes with him she styled herself as a warrior for the underdog blue city, but ended up making enemies with the power-brokers who control state funds. In 2021 Ms Bottoms announced that she would not seek reelection and would take a job in the Biden administration later that year. "The last three years have not been at all what I would have scripted for our city," she said at the time. "It is abundantly clear to me today that it is time to pass the baton on." Some touchy locals felt she quit on them.

"She couldn't meet the moment and she didn't really try to meet the moment, but the question now is whether her campaign for governorship will be like that," says a former city official. Some Democrats argue the tailwinds are strong

enough to carry even a weaker candidate to victory this year. Crunch the numbers from Georgia's primary night and you can see that Democrats are more energised than Republicans: 150,000 more of them cast ballots. Ms Bottoms spent the past week on a bus tour far from her home city—a prudent sign, says Charles Bullock of the University of Georgia, that she knows she will need to run up the score in rural Georgia. As governor she says she plans to come out swinging to fight Donald Trump, expand Medicaid and preserve voting rights.

At her victory party Ms Bottoms admitted that when she started her campaign she had "no idea" where her support would come from. Joe Biden, her last boss, is her biggest endorser so far. While that may have helped in the primary, it could now be a liability. The former president describes Ms Bottoms as "battle-tested". No one questions that she has been tested. What is less clear is whether she won her battles. ■

Senate primaries

Three ways, not third way

DETROIT

Three Democrats in Michigan offer very different visions

YOU CAN learn a lot from the type of campaign stops politicians make. In early May *The Economist* went to Detroit to see the three Democratic candidates for Michigan's open-Senate-seat campaign on the same day. It provided an insight into three very different approaches.

First was Mallory McMorrow, a state senator, who visited a state-backed industrial project in the city's suburbs where she peppered staff with earnest questions about 3D printing. Next, in a left-wing coffee shop in the city, was Abdul El-Sayed, a former head of Wayne County's health department, who spoke for half an hour to your correspondent before giving a speech to 100 or so attendees at an event space nearby. Finally, there was Haley Stevens, a congresswoman from north Detroit. As far as can be told she did nothing at all.

Three months out from polling day, the race in Michigan is shaping up to be one of the most hotly contested Democratic primaries in the country. Gary Peters, the incumbent, is retiring. So unpopular is Donald Trump that our model gives Democrats a strong chance of taking the Senate, but to do so they will have to win Maine and then two Republican-held seats won comfortably by the president in 2024. What they cannot afford to do is to lose Michigan, which plumped for Mr Trump by a margin of just 1.4%.

Our model suggests they ought to win

by seven points. But the choice of candidate could change that substantially. Each of the three represents a different faction of the Democratic Party. Who wins will reveal a lot about how the party's voters want to approach the general election. The race is also a test of how much opposition to Israel, and to the war in Iran, will matter.

Start with Ms Stevens. Unlike the other two, she is already in Washington, representing a district covering Detroit's northern suburbs. Her most prominent backer is Chuck Schumer, the Senate minority leader, and his machine. After that is AIPAC, the pro-Israeli lobby group. Her list of endorsers is as long as Lake Michigan. But her campaign is deathly quiet: she has hosted at most a handful of open events this year, if that, and she was booed at the state party convention last month.

Going bluer

Ms McMorrow is louder (she arrived at the same convention followed by a marching band). She first gained national attention in 2022 with a powerful speech responding to a Republican colleague who had called her a "groomer" over her support for gay and trans rights. These days she styles herself as a policy technocrat who will disdain partisan warfare. She argues that if "MAGA Republicans" can be beaten, "more sane, rational, traditional Republicans" will re-emerge like crocuses. Her backers are the

sorts of Democrats who consider themselves to be sensible progressives, such as Senator Elizabeth Warren.

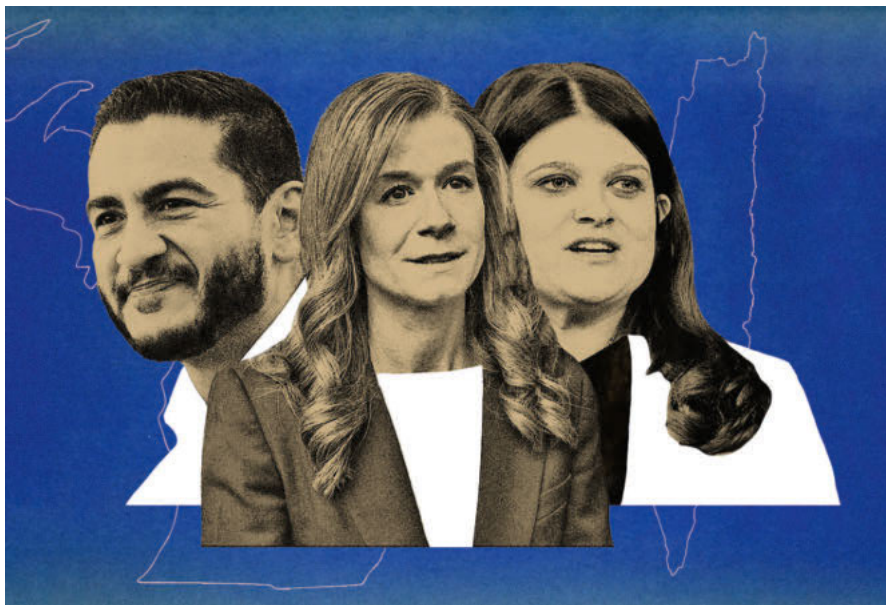
Most boisterous of all is Mr El-Sayed. The son of Egyptian immigrants, he studied medicine at the University of Michigan, but instead of pursuing a residency he went into public health (this has not stopped him misleadingly referring to himself as a "physician"). He wants to introduce universal health care, end all military aid to Israel and ban corporate political spending. The biggest problem with American politics, he says, is "the system that allows corporations and billionaires and special interests to buy and sell politicians". His most prominent backer is Senator Bernie Sanders.

In past elections Democratic primary voters in Michigan have been older, whiter and more female than the electorate in general, and have disdained radicals. In 2018, Mr El-Sayed ran for governor, and was defeated by 22 percentage points in the primary by Gretchen Whitmer, a more traditional Democrat. At the age of 41 he has served in political jobs for most of his career without ever winning an election.

But might the electorate have changed? Mr El-Sayed's events, including one with Mr Sanders on May 3rd, are drawing big crowds. "I feel more angry at the Democratic Party than [at] Trump", says Anne-Marie Carlson, a 28-year-old charity worker at his Detroit rally. Mr El-Sayed's fans reckon the party is in hock to rich donors, out of touch and barely deserving of power. Ms McMorrow and Ms Stevens are pinning their hopes on Democrats who are slightly less full of rage at their own side.

The wild card in the race is AIPAC. Ms McMorrow and Mr El-Sayed expect a wall of negative ads, paid for by the group, to hit them soon. After Ms Stevens, AIPAC's preference ought to be obvious. Ms McMorrow, who has a Jewish husband, says what Israel has done in Gaza and Lebanon is "an abomination". But she also worries that some voters are "not being anti Netanyahu, but anti-Jew". Mr El-Sayed by contrast has happily campaigned with Hassan Piker, a Twitch streamer who has praised Hamas. (Mr El-Sayed says he disagrees with many of Mr Piker's views.)

And yet AIPAC's influence is unpredictable. "They have a ton of money and use it stupidly," says one Democratic strategist in the state, speaking anonymously because of his (unrelated) campaign work. In a special election primary in New Jersey earlier this year, Analilia Mejia, another angry populist backed by Mr Sanders, beat a moderate critic of Israel attacked by AIPAC. Ms Mejia's House seat, however, is a safe one. What the party's stalwarts fear most in Michigan is that primary voters may be about to risk the Senate on a far-left, unelectable candidate. ■



LEXINGTON

The aquaculture warrior

Graham Platner is betting centrist and right-leaning voters want economic revolution more than culture war



GRAHAM PLATNER, the oyster farmer rocketing to Democratic celebrity nationally as he campaigns for Senate in Maine, is testing two hypotheses that could reset his party's direction: that the censorious left can sometimes ignore past, regretted transgressions such as fascist tattoos or bigoted social-media posts; and that voters in the centre and even on the populist right are far more drawn to economic causes such as universal health care than they are repelled by cultural ones he also believes in, such as welcoming transgender athletes into girls' sports.

Mr Platner's commitment to the second hypothesis has already helped him prove the first. Revelations of past misbehaviour have not slowed his momentum. A little hypocrisy in politics should surprise no one; it is simply more obvious on the high-church left, as it also is on the Evangelical right, because of the particular stress those movements place on purity. But just as the right has withheld its sanctimony from champions of its own, such as Donald Trump, the left has embraced Mr Platner because he looks to be their kind of winner.

Janet Mills, the sitting two-term governor, gave up a bid for the Senate late last month. Despite her own popularity with Democrats, she bowed to the inevitability that Mr Platner—a college dropout whose previous public roles were serving as a harbourmaster and chair of his local planning board—would crush her in the primary in June. He is the presumptive choice to run against Susan Collins, a Republican in her fifth term, in a race Democrats see as essential to winning a majority in the Senate.

Mr Platner is right to argue that, if anything, attacks over his tattoo (now inked over) and social-media posts (deleted) have helped him. They remind voters—or, at least, those Democrats paying attention—of what captivates them, that he has been hewn by hard knocks into a candidate struggling Americans may recognise as one of their own. As a Marine infantryman, Mr Platner, 41, served four tours in Iraq and Afghanistan. He got the tattoo, of a skull and crossbones, while on a break with comrades, not knowing it was associated with Nazism, he has said. He has said he wrote extreme statements on Reddit, a social-media site, in a time of alienation and depression brought on by his trauma from com-

bat. Returning to his hometown of Sullivan in eastern Maine to work on the water led him to “feel connected again”.

It certainly helps Mr Platner that, as he speaks in his gruff baritone, he radiates sincerity. It also helps that his explanations are reasonable (and that some of his posts were anti-fascist diatribes). “Rage-posting on Reddit was a thing, you know,” says Caitlin Murphy, a Marine, recalling her own return from service as she waited to hear Mr Platner speak at a veterans' hall in Portland on May 17th. “What do you expect? He's come a long way. He's grown.”

It probably most helps Mr Platner that he is not a heretic like Jared Golden, another Marine and Democrat who, four times, has swum against the political rip tides to win Maine's rural, Republican-leaning second congressional district. Partly because of vicious criticism over his conservative votes, Mr Golden is not running again. Mr Platner's brand of politics is usually more at home in Maine's first district, centred on prosperous, Democratic Portland. As fluent in the acronyms of the left as in those of the Pentagon, Mr Platner switches codes easily. He offhandedly recited the alphabet of persecuted identity—“LGBTQIA”—when, before a Democratic audience in Kennebunk, he lamented the party was failing to “fight tooth and nail for marginalised people”.

Mr Platner believes Democrats with progressive social values can win centrist and even right-leaning voters. To him, America's cultural disputes are used to divide the working class by “the enemy”—the oligarchy and the political establishment it pays for. In a brief conversation after speaking to union members in downtown Kittery on May 19th, he argues that economic concerns far outweigh social ones. “When the hospital closes in your part of rural Maine, and you now no longer have access to good health care for your family, the rest of it begins to not matter nearly as much,” he says. “We can argue about all the other stuff down the road *ad nauseam*, once we've got decent health care, once people's wages are keeping up with inflation.”

The shellfish and selfish

Mr Platner is a sure-footed, often fierce candidate with a potent account of Maine's, and America's, predicament. America had 90 billionaires in 1990 and now has 900, he says in his stump speech. “Do our paychecks go ten times as far?” he demands. When he was young, he knew people in Sullivan who earned enough digging clams to put their children through college. Now some of his neighbours work three jobs and spend more than 60% of their income on rent. That reality, he says, “is directly tied to the existence of Elon Musk”. Billionaires “stole” their wealth by distorting national policy. Mr Platner's story of disillusionment with the wars of this century, and of the price he and his comrades paid, lends weight to his account of establishment betrayal, and the war on Iran is now sharpening its edge.

What a strange twist of history that the Republican Party happened to beat the Democrats back to populism. In 2016 Mr Trump upended his party's establishment, while Bernie Sanders lost the nomination to Hillary Clinton. But the contest was just starting. Like other charismatic newcomers such as Zohran Mamdani, Mr Platner was inspired by Mr Sanders. Mr Platner would be just one vote in the Senate for such policies he favours as freezing electricity rates. But, he said in Kennebunk, his success has already caused “a bit of an earthquake” that may move Democratic senators his way. “The last thing they want to do is face a challenger like me next cycle,” he said. He is surely right about that. ■

The Americas



Cuba

No country for old men

MEXICO CITY AND MIAMI

Donald Trump is pushing towards the end-game in Cuba

FEW THOUGHT President Donald Trump would snatch Venezuela's strongman, Nicolás Maduro. He now languishes in a Brooklyn jail cell. Fewer still take seriously the notion that Mr Trump might seize Raúl Castro, the 94-year-old Communist Party grandee who is Cuba's most powerful man; his late brother, Fidel, brought communism to Cuba in the 1960s. But after the United States Department of Justice issued criminal charges against Mr Castro in Miami on May 20th, the possibility looks a little less remote.

The indictment accuses Mr Castro of ordering two aircraft flown by Brothers to the Rescue, a Miami-based exile group, to be shot down in 1996, killing four people. He was Cuba's defence minister at the time. He later became president and first secretary of the Communist Party, before formally retiring in 2021. Yet he is still

Cuba's de facto leader. All big decisions require his blessing. That includes any deal with Mr Trump and Marco Rubio, his Cuban-American secretary of state. Both are determined to force change in Cuba.

The charges against Mr Castro are not the same as those against Mr Maduro, but the political use of the law looks similar. The Trump administration called the raid that captured Mr Maduro a law-enforcement operation. Now the Americans are also arguing that Cuba, which lies 145km from Florida, is a national-security risk because of its status as a failed state, its support for China and Russia and its outflow

of migrants. American reconnaissance flights over Cuba have increased. Military planners are reportedly weighing options, from limited strikes to broader action.

The escalation follows months of coercion. Since removing Mr Maduro, the United States has blocked fuel shipments to Cuba. It has pressed countries across the region to cut off supplies of hard currency. The government in Havana has responded with limited concessions. These include letting private firms import fuel and promising to allow Cubans abroad to invest in the island.

Unsatisfied by that, Messrs Trump and Rubio are increasing the pressure. On May 7th the US Treasury's Office of Foreign Assets Control (OFAC) put sanctions on Gae-sa, the military-run conglomerate that controls much of Cuba's economy. It is widely regarded as the Castro-family business. Mr Rubio called it "the heart of Cuba's kleptocratic communist system".

A week later John Ratcliffe, the director of the CIA, flew to Havana to meet Raúl Guillermo Rodríguez Castro, Mr Castro's grandson. He warned that time is running out to make "fundamental changes" and urged the regime to accept \$100m in aid, to be distributed in co-ordination with the Catholic church. The meeting appears to

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► have gone badly. On May 18th OFAC placed more of Cuba's political and security apparatus under sanctions, including the regime's notorious spy service.

What next? The Americans are open about wanting economic reforms, prisoner releases and compensation for expropriated property. Though they do not say so publicly, they probably want more drastic changes, too, such as breaking up Gaesa and a transition to democracy. The Trump administration has alternated between saying it would work with the regime and hinting that it must be removed. On May 14th Mr Rubio took a hard line: "I don't think we're going to be able to change the trajectory of Cuba as long as these people are in charge in that regime."

A Venezuela-style operation may seem too extreme to the Trump administration. The sight of a 94-year-old man in handcuffs might not play as well with Americans as did Mr Maduro's seizure. Nor is there an obvious figure who could take over from Mr Castro and do the Americans' bidding, as there was for Mr Maduro. Mr Castro's grandson is a gatekeeper, not a natural successor.

Still, Cubans are not dismissing the possibility of the first American military action against the island for six decades. The regime seems to be playing it up. Civilians are getting military training. Cuba's defence forces are handing out leaflets telling people how to prepare for war. On May 18th President Miguel Díaz-Canel warned that an attack would produce "a bloodbath of incalculable proportions".

In part, Mr Castro's indictment is pitched at exiles in Miami who have been urging Mr Trump to be more aggressive. The date of its publication was symbolic: many exiles celebrate May 20th, the anniversary of Cuba's independence. But the Trump administration may also wish to use it as leverage in negotiations.

That will be tricky. "The government is wholly incapable of changing course or implementing reforms to dig themselves out of the hole they are in," says Ric Herrero of the Cuba Study Group in Washington, which favours engagement with Cuba's government. Try to see what comes next, he says, and "things get muddy real quick."

It can't last much longer

What is clear is that Cuba's situation is unsustainable. The sanctions on Gaesa will hurt: the conglomerate's revenues are thought to be more than three times the state budget. It also controls up to \$20bn in illicit assets. Foreign companies have until June 5th to wind down dealings with Gaesa or any entity it controls. Germany's Hapag-Lloyd and France's CMA CGM, the two big Western shipping companies that maintain operations with Cuba, have stopped accepting orders linked to the is-

land while they assess the risks.

In a country that imports roughly 70% of its food, that could be devastating. Cuba's energy minister has already said the island has run out of diesel and fuel oil for its power plants. Blackouts in Havana last up to 22 hours a day. For many, food is hard to come by. Services have broken down. "It's like living in a rubbish bin," says Yulietta Hernández Díaz, a Havana resident who owns a small construction firm.

Domestic pressure is growing. On May 13th police dispersed demonstrators in Havana protesting about the blackouts. Prisoners Defenders, a rights group based in Madrid, says Cuba has a record 1,260 political inmates. Pedro Monreal, a Cuban economist living in Madrid, reckons the economy may contract by 15% this year.

The regime says it is considering the offer of aid. To accept would be to admit its own failure. Rejecting it risks more protests as heat, hunger and blackouts worsen. Either way, the regime is running out of ways to say no to the United States. ■

Bolivian politics

On the block

After just six months in power, Bolivia's new president faces unrest

WHEN HE BECAME president of Bolivia in November, Rodrigo Paz was riding high. His election ended 20 years of almost uninterrupted left-wing rule by the increasingly autocratic Movement to Socialism (MAS). Now his centrist government is beset by violent protest. Dynamite-wielding protesters have blockaded La Paz, the seat of government. The city is running short of food and fuel. Banks are too frightened to open. Various groups are demanding that Mr Paz step down.

He promised to liberalise the economy while protecting welfare. The MAS had run annual fiscal deficits of around 10% of GDP, and the central bank was all but out of dollars. Voters ejected the party because of biting inflation and chronic fuel shortages. But these problems have continued. A batch of adulterated fuel damaged more than 10,000 vehicles, infuriating their owners. An agrarian reform passed in April angered indigenous people, who saw it as a sop to agribusiness. Mr Paz repealed it, but the general workers' union and peasant groups demanded his resignation anyway. The peasants set up the road blocks.

Mr Paz has responded with a mix of negotiation and force. He has reached agreements with some factions, including the teachers and some miners, but the block-

ades have not budged. On May 16th Mr Paz sent riot police to clear the roads. As *The Economist* went to press more than 150 people had been arrested, but the barriers were still up. On May 20th Mr Paz offered a cabinet shuffle in a bid to end the crisis.

Thousands of supporters of Evo Morales, a former president, arrived in La Paz on May 18th to join the protests. Mr Morales led the MAS to power through just this sort of street politics. He is hiding out in the coca-growing tropics, evading an arrest order for statutory rape. (Mr Morales says the allegations are politically motivated.) He and his new party, Evo Pueblo, want to exploit the moment.

Even if Mr Paz can defuse these protests, there will be more. The IMF reckons the economy will contract by 3.3% this year, with inflation rising to more than 20%. Mr Paz's relationship with the highland indigenous and peasant groups—whose votes were crucial to his election—has already broken down. Their blockades have toppled governments in the past.

The economic adjustment has barely begun. External factors have not helped. After Mr Paz cut fuel subsidies, the war in Iran sent prices soaring. The fiscal deficit in the latest budget was still 9% of GDP. At the official exchange rate the currency remains overvalued. Reforms to encourage investment in natural resources have yet to materialise. "The government is moving, but very slowly, very timidly," says Beatriz Muriel of INESAD, a think-tank in La Paz.

The Americans are watching. "The United States stands squarely in support of Bolivia's legitimate constitutional government," Marco Rubio, the secretary of state, said on May 20th. "We will not allow criminals and drug traffickers to overthrow democratically elected leaders in our hemisphere." ■



No peace for Mr Paz

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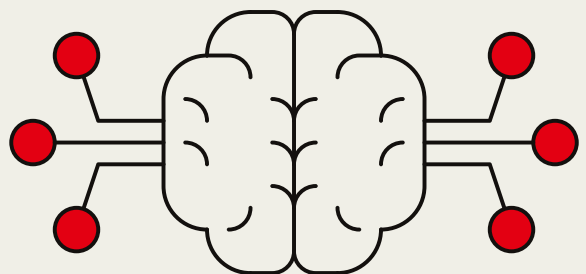
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**ECONOMIST
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Money for drugs

Brazil's bid for pharmapower

SÃO PAULO

Lula wants Brazil to spend its way to pharmaceutical might

IN RECENT MONTHS more than one million doctors and nurses in Brazil have been inoculated against dengue. The mosquito-borne disease is a perennial problem and vaccination normally relies upon expensive Japanese shots. But this year, for the first time, the jab is home-grown. Approved in November, Butantan-DV is the world's first single-dose immunisation for dengue (one dose is easier than two, especially when vaccinating poor, rural people).

It is also the first vaccine to be fully developed in Brazil. The Butantan Institute, a research outfit owned by the state of São Paulo, handled formulation, testing and manufacturing. On May 4th the institute's production of a vaccine for chikungunya, another mosquito-borne disease, was also approved. It is working on a jab for Zika, says Esper Kallás, Butantan's boss.

This is the kind of "medical sovereignty" that the left-wing president, Luiz Inácio Lula da Silva, wants. Lula, as he is commonly known, returned to power in 2023 after the covid-19 pandemic. That wave of disease exposed the weakness of the Brazilian pharmaceutical sector. The country's share of active pharmaceutical ingredients (APIs) that were produced domestically was just 5% at the start of the pandemic—lower than in Bangladesh—according to data from Abiquifi, a Brazilian trade association. Tight global supply chains left Brazil short of syringes and anaesthetics, let alone vaccines. A Brazilian jab, ButanVac, was abandoned because of poor performance. More than 700,000 people died of covid in Brazil.

This created political pressure for improvement. Lula took up the mantle. The pharmaceutical sector withered during his first two terms as president, from 2003 to 2010. After right-wing governments liberalised and reduced tariffs on the sector in the 1990s, it in effect gave up on research and development in favour of churning out generic drugs and doing final-stage "fill-and-finish" work for foreign brands. Lula was unable to reverse the decline. His efforts to create "state champions" struggled. Most firms were happy selling generics. A programme of loans for health-care firms created by BNDES, the country's state-owned development bank, was undersubscribed. A 2009 technology-transfer programme made little headway. China and South Korea raced ahead in R&D; India increased its output of generics. Brazil

languished in the middle, says Jorge Guimarães, a former boss of Embrapii, the agency for industrial innovation.

Butantan's recent success is the result of a new, more vigorous approach. More money is one element. Since 2023 the health ministry's budget has risen by 30%. In 2024 BNDES more than doubled the amount it lends to health-care firms (see chart). It is part of a broader industrial programme for the "health-economic-industrial complex" with a budget of around \$11bn. The government makes loans and grants to Brazilian firms, as well as taking equity stakes. It aims to increase the locally produced share of medicines and medical equipment from 45% in 2024 to 70% in 2033. A factory being built near Rio de Janeiro will quadruple Brazil's capacity to finish part-processed vaccines.

New laws have helped, too. The government changed the rules for clinical research in 2025 to speed up the approval of trials. This should help early investigations into novel medicines where the relative inexperience of Anvisa, the regulator, was choking innovation, says Reginaldo Arcuri of FarmaBrasil, a trade association. The Senate is also due to vote on a national health strategy that would give further preference to domestic manufacturers in public procurement.

Brain drain was another problem. Nearly 7,000 researchers left Brazil between 2015 and 2022, according to one think-tank. Lula's government responded with programmes to attract and retain scientists, as well as to foster partnerships with Brazilians working abroad. One of them, "Knowledge Brazil", had a budget of more

than \$100m to spend on research in 2025 and managed to lure 600 scientists back home. Another \$40m was earmarked for research abroad. This year the project will fund 1,000 doctorates and chuck an additional \$100m at, among other things, poaching foreigners. The United States is a juicy target, says Alexandre Padilha, the health minister, as Donald Trump's anti-science agenda is alienating researchers.

Lula is eyeing broader benefits, too. Brazil's economy runs on commodity exports. He hopes that cracking biotech will yield outsized returns and transferable skills for manufacturing as a whole. The government also hopes that boosting the domestic pharmaceutical industry might trim the steep cost of importing medicines for the national health service, the world's largest universal, free public-health system. The health ministry says the government saved \$2bn between 2011 and 2017 thanks to lower prices charged by companies using native APIs. It is not clear whether the government includes the costs of the hefty subsidies behind such numbers.

There are obstacles to a Brazilian pharma boom. Culture is the trickiest. Few pharmaceutical companies care about Lula's goals. Their generics business nets consistent profit at little risk. That the state is funding everything opens the door to accusations of unfair competition. The Trump administration has an investigation pending into Brazil's industrial policies, including in pharmaceuticals.

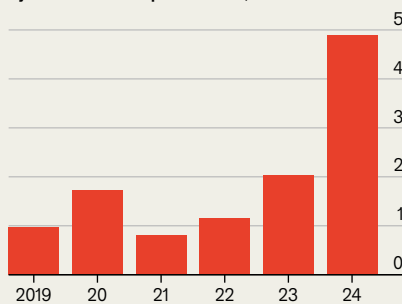
Your stuff, my prices

Lula's lax attitude to intellectual property could also be a problem. He reportedly considered escalating the spat with Mr Trump by terminating the Brazilian patents of American corporations. No drug company wants to invest in a country that might seize its property. Lula also has to deal with some trade-offs. High drug prices are unpopular, but using price caps to keep medication cheap can discourage private R&D. Autarky can bring more of an industry under state control, but it can also deter innovation.

The development of new drugs is happening at Butantan. But the institute benefits from an unusual alignment of research, regulation and manufacturing around the needs—and the money—of the state. Brazil cannot hope to replicate that across the industry. Merck, a multinational pharma giant, spends about \$16bn a year on R&D, roughly ten times Brazil's entire state pharma spending, even after Lula's expansion. At best, the state might develop a few products, such as vaccines, where the public-health economics stack up. A broader boom requires Lula to shake off his distrust of private enterprise. As he prepares to run for president for the seventh time in October, that seems unlikely. ■

A healthy loan book

Brazil, health-care credit approved by national development bank, bn reais



Source: Brazilian Development Bank

Asia



The Sino-Japanese spat

No end in sight

BEIJING, TAIPEI AND TOKYO

Why Japan and China will struggle to resolve their feud

RICHARD NIXON'S surprise trip to China to meet Mao Zedong in 1972 came to be known in Japan as a "Nixon shock". Donald Trump's choreographed meeting with Xi Jinping was not quite as shocking. But his recalibration of America's relationship with China has sent allies and foes alike scrambling. On May 19th Takaichi Sanae, the prime minister of Japan, jetted to South Korea to meet her counterpart, Lee Jae Myung. The two were doubtless keen to swap notes on America's shifting stance towards their imposing neighbour.

The implications are perhaps gravest for Japan's own troubled relationship with China. The two have been locked in a bitter stand-off for more than six months. After taking office last October, Ms Takaichi told parliament that Japan might have a role to play in a conflict over Taiwan, triggering an apoplectic response from China. Relations have deteriorated since; it has become clear that normalising ties will be

harder than during past feuds. Backchannels that once helped ease the relationship are bunged up. And growing warmth between America and China bodes ill for chances of a rapprochement.

The last major falling-out between Japan and China followed clashes over the uninhabited and disputed Senkaku islands in 2010-12. The focus of the current dispute is Taiwan, which Mr Xi defines as the "core of China's core interests". Ms Takaichi positions herself as a staunch friend of Taiwan, to China's chagrin. Japan also sees Taiwan's fate as deeply intertwined with its

own: hence Ms Takaichi acknowledged a conflict over it would probably constitute a "survival-threatening situation" for Japan. China has sought to make Ms Takaichi a "symbol" of what happens to those who cross its red lines on Taiwan, says Kawashima Shin of the University of Tokyo.

The maritime front line has become more tense in recent months. In December Chinese fighter jets locked their radars on to Japanese aircraft. A Japanese destroyer went through the Taiwan Strait in April, for the first time since Ms Takaichi became prime minister; China took extra umbrage at the fact it took place on the anniversary of the Treaty of Shimonoseki, which ended the Sino-Japanese war in 1895 and handed control of Taiwan from China to Japan. The following week two Chinese naval vessels passed closer to Japan's mainland than normal and China began building a new structure in the East China Sea, where the two have overlapping maritime claims.

Japan has been beefing up its defences in its south-western islands. That is part of a long-term build-up to counter China. This spring Japan's self-defence forces (SDF) deployed long-range missiles to a base in Kyushu, a southern island, which has put the Chinese mainland in range. Ms Takaichi recently loosened restrictions on exports of Japanese weaponry. China has denounced the moves, lambasting a "new

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▶ Japanese militarism". While such rhetoric may be inflamed, it betrays a genuine concern about the long-term trend of Japan becoming a more proactive player in the region's security.

People-to-people ties have also become strained. The number of Chinese visitors to Japan fell by more than half in the first quarter of this year, compared with a year earlier. Japanese pop stars have had concerts cancelled in China. Bombastic anti-Japanese videos have become a popular genre on Chinese social media: one viral clip features a Chinese archer shooting bows into the head of a mannequin that has been plastered with the Japanese flag. Offline, several disturbing, albeit isolated, incidents have occurred. In March a knife-wielding SDF officer broke into the Chinese embassy in Tokyo; on May 19th a Chinese man stabbed three people, two of them Japanese, at a restaurant in Shanghai.

Most worrying for Japan are Chinese curbs on rare-earth exports. Measures announced in January and February targeted dual-use items and Japanese firms involved in the defence industry. Though Japan diversified its rare earth supplies somewhat after previous run-ins with China, a prolonged cut-off would still pose serious problems. Publicly available data suggest that China has been choking off supplies of specific heavy-earth elements. But in contrast to the Senkaku crisis, there have not so far been large-scale boycotts of Japanese consumer product, perhaps because "this time the Chinese government is worried about the state of the Chinese economy," reckons Seguchi Kiyoyuki of the Canon Institute for Global Studies, a think-tank in Tokyo.

Nonetheless, a sense of confrontation has become entrenched. In the past Japan's ruling Liberal Democratic Party (LDP) could turn to high-ranking lawmakers that the Chinese Communist Party considered trusted intermediaries. When Abe Shinzo, a long-serving prime minister, wanted to begin a rapprochement with China in 2015, he reached out to one such figure, Nikai Toshihiro, who led a delegation of 3,000 executives to Beijing and hand-delivered a letter to Mr Xi. But Mr Nikai, like many of his peers, has since retired. This year a delegation of Japanese executives got the cold shoulder from Chinese officials and cancelled their annual trip for the first time in more than 13 years.

Younger politicians see little upside in cultivating ties with Chinese counterparts, given that more than 80% of Japanese voters have negative views of China. The China-friendly politicians who remain active are far from Ms Takaichi's inner circle. Komeito, a political party associated with Soka Gakkai, a Buddhist movement, was once an important conduit between Japan and China. Ikeda Daisaku, the movement's

founder, publicly advocated normalising Sino-Japanese relations following the second world war. When Komeito was part of a ruling coalition with the LDP, its leaders helped pass messages to Beijing. But Komeito left the coalition following Ms Takaichi's rise.

Mr Trump's overtures to Mr Xi (see next story) will only make a rapprochement harder. Relations between Japan and China tend to move in concert with those between America and China, and between America and Japan. When China's relationship with America is tense, it seeks warmer ties with Japan; when America and Japan drift apart, China seeks to drive a wedge between them. "China now has less reason to fix the relationship with Japan," says Sahashi Ryo of the University of Tokyo: "The triangle really matters." ■

Trump and Xi

Is Donald Trump selling out Taiwan?

TAIPEI

He hints he is bargaining with China

FOR A MOMENT, it seemed that Taiwan had dodged a bullet. Throughout the formalities of Donald Trump's two-day visit to Beijing, which ended on May 15th, the American president avoided talking publicly about Taiwan. That temporarily soothed Taiwanese fears that he might bow to Chinese pressure to curb arms sales to the island, or even change America's public position towards it by saying that he "opposed" its independence.

Then things started to unravel. In an in-

terview with Fox News recorded just before he left Beijing, and speaking to reporters on Air Force One after take-off, Mr Trump made a series of comments about Taiwan that deeply unnerved many on the island. He suggested that America's 44-year commitment not to negotiate with China over arms sales to Taiwan was out of date. He also said he had discussed a \$13bn arms package for Taiwan "in great detail" with Xi Jinping, China's leader, and saw it as a useful bargaining chip. And he portrayed Taiwan's president, Lai Ching-te, as trying to achieve independence and drag America into a war.

At the same time, Mr Trump surprised many listeners by suggesting that he would talk to "the person who's running Taiwan" about the arms package. There has been no direct contact between sitting American and Taiwanese leaders since 1979, although Taiwan's previous president, Tsai Ing-wen, called Mr Trump (when he was president-elect) to congratulate him on his election victory in 2016. Muddling the picture further, Mr Trump repeated his allegation that Taiwan "stole" America's semiconductor industry and said it should shift all of its chip manufacturing to American soil. He has said such things before, but rarely since Taiwan promised to invest \$250bn in America as part of a trade deal sealed in February.

Mr Trump insisted American policy had not changed. But his remarks suggested that Mr Xi had made some progress in swaying his thinking about Taiwan. Early on the first day of the meeting in Beijing, Mr Xi warned that mishandling the Taiwan issue could spark a conflict between China and America. The Chinese leader then spent most of that evening talking about the island, according to Mr Trump. "I heard him out," he added. Mr Trump's comments also suggested that he is taking a more transactional view of American relations with Taiwan than his predecessors, and is less committed to its defence.

Taiwan's immediate concern is the new arms package. Mr Trump approved a record \$11bn one in December. Ahead of the summit, it emerged that he had delayed a second one, worth about \$13bn, which has been approved by Congress. America has delayed arms packages ahead of meetings with Chinese leaders before. But Taiwanese officials will be deeply concerned if the new one is postponed for more than a few weeks after the summit. "I may do it. I may not do it," Mr Trump told Fox. "I'm holding that in abeyance, and it depends on China. It depends. It's a very good negotiating chip for us, frankly."

America has previously maintained that arms sales to Taiwan should not be included in broader negotiations with China, partly because America has been obliged under its own law since 1979 to help the is- ▶▶



Thinking twice

▶ land defend itself. President Ronald Reagan also promised in 1982 not to negotiate with China over arms sales to Taiwan. Asked about that, Mr Trump told reporters: “Well, I think the 1980s is a long way.”

That was especially disappointing for Taiwan’s government as, after months of wrangling with opposition parties, it has just managed to push through parliament a supplementary defence budget of about \$25bn to fund both American arms packages. Taiwan’s biggest opposition party,

the Kuomintang, or KMT, will no doubt point to Mr Trump’s comments as further evidence that the island can no longer rely on American support and needs to pursue closer ties with the mainland. The KMT hopes that that message will help it in local elections in November and in a presidential poll in 2028.

Equally troubling for Taiwan’s government was Mr Trump’s suggestion that providing arms to Taiwan might encourage Mr Lai to declare independence. “They have

somebody there now that wants to go independent,” he said. “If you kept it the way it is, I think China is going to be okay with that. But we’re not looking to have somebody say ‘Let’s go independent because the United States is backing us.’” In response, Mr Lai reiterated his position that Taiwan is already, de facto, a sovereign independent nation. The island would “never be sacrificed or traded away”, he said. The question now is whether Mr Trump agrees with him. ■

ASHOKA

Bread and circuses

India’s loudest political fight obscures a more urgent one

THE DISAGREEMENTS between India’s north and south are sometimes ill-tempered and often tedious. The north is poor, procreates prodigiously and reliably votes for Narendra Modi. The south is prosperous and less fecund and does not. The two have different historical experiences (Muslim empires made fewer inroads down the peninsula) and different linguistic lineages (their languages come from distinct families) and prefer different carbohydrates. The rice-eating south caricatures roti-munching northerners as barbarians imposing an alien culture upon them.

This schism was at the heart of recent shenanigans in Parliament. Mr Modi’s Bharatiya Janata Party (BJP) introduced a package of reforms that aimed to boost the number of seats in India’s lower house from 543 to 850, to redistribute them between states to reflect demographic changes, and to reserve a third of them for women. Opposition leaders—particularly from the south—called it a power grab dressed up as an effort to empower women. The bills failed.

Southern states are not wrong to worry about losing out. Under those recent proposals, their heft in Parliament would diminish while BJP strongholds in the north would grow more influential. Nor is it wrong to suspect the motives of the BJP, which put forward the measures without consultation. But the north has the facts and democratic fairness on its side. MPs in northern Uttar Pradesh on average represent a fifth more voters than those in southern Tamil Nadu. Eventually this imbalance will have to be fixed, probably after a fresh census is held next year. The sensible way to do it would be with cool heads and compromise. The BJP way is to steamroller opponents. The drama has a long way to run.

Yet the battles that the north and south wage over the electoral system obscure a problem that is just as important, but much less noticed: across India, voters in cities get far too little say. On average, metropolitan MPs represent a tenth more voters than those in the countryside, and in many places the disparity is much worse. Bangalore’s four seats hold as many voters as six average rural ones. Only one of the 20 constituencies with the biggest electorates is fully rural. Some of India’s most unpleasant towns—hello, Ghaziabad—are the most underrepresented.

India’s industrialised southern states complain that they are often treated shoddily despite the chunky contributions they make to national coffers. Cities can make the same claim, but better. Some 40% of Indians live in urban areas but they produce nearly 60% of GDP. In return they get broken footpaths, potholed roads, poor public transport, unbreathable air, sewage-clogged waterways and piles of festering rubbish. Imagine what they might be capable of if their towns were liveable.

A fair redistricting would boost the

number of seats by a sixth in India’s six biggest cities alone, according to *The Economist’s* calculations. And expanding the house would bring additional benefits. Suburban and semi-urban bits of the country are fast-growing, but often belong to large rural constituencies. A bigger house with smaller districts would see these broken into more sensible divisions. Together, redistricting and expansion would raise urban India’s share of MPs to something like its share of the population. Governments might finally start taking seriously the omnishambles of city life.

One of India’s absurdities is that its states run cities like colonies. A measure to grant urban areas more autonomy, enacted over three decades ago, was a miserable failure. They do elect councillors but actual power lies with short-tenure, state-appointed bureaucrats. Mayors are figureheads. City councils are decorative. Rural regions have the dominant voice in state houses.

Redistricting would take in state government, too, giving urban voters more influence in those houses. There is precedent for this: when states shuffled seats within their borders two decades ago, Bangalore’s provincial representation jumped to 28 from 16. The city is still dysfunctional, but it is one of the only places in India where the state is experimenting with a new metropolitan authority that might improve it.

Electoral reform will not magically fix all of urban India’s problems. But it will remedy at least one of India’s overlapping crises of representation. The noisy southern states command attention with their anti-roti grievances. Yet it is the cosmopolitan cities, where all carbohydrates are welcome, that have the bigger complaint.



The BRICS and the Quad

Busy doing nothing

DELHI

Indian diplomats are hosting the world—but what is getting done?

INDIA'S DETERMINATION to stay on good terms with all major powers often proves awkward. A traffic jam of meetings in Delhi this month is making its acrobatics look especially complex. On May 14th India welcomed foreign ministers from the BRICS—a bloc of 11 countries including China and Russia who like to complain about Western hegemony. On May 26th it will host foreign ministers from America, Australia and Japan—who with India make up the “Quad” of countries seeking to check China's rise. “May is a live minefield for India,” says Constantino Xavier at the Centre for Social and Economic Progress, a think-tank in Delhi.

The BRICS meeting was overshadowed by the war in the Gulf. These days the grouping includes both Iran and the United Arab Emirates, whose armed forces have been exchanging blows. So it is no great surprise that the talks concluded without a joint statement. For India this was probably a relief; it would not have approved language that condemned America for the violence. It much prefers BRICS to focus on things no one can dislike. One of its priorities as its chair in 2026 is to promote a “mission for healthy lifestyles”.

If core members of BRICS find India disappointingly cautious, the country has sometimes also been seen as a nervy partner for the Quad. Now that bloc is on life-support, in large part because Donald Trump, who is chasing Chinese business deals, appears to have lost interest in it. The other member countries, for their part, have lost trust in Mr Trump.

Foreign diplomats in Delhi think the Quad could yet achieve some modest advances in security. These would include building interoperable naval and communications systems, and working to break China's stranglehold on critical minerals and emerging technologies. But no one expects breakthroughs at the meeting in Delhi this month. The best result of the foreign ministers' talks might be to approve plans for another meeting. India hopes a gaffe-free gathering will persuade Mr Trump to come to its capital for a summit of Quad leaders later this year. Such a visit was supposed to have taken place in 2025, but was called off when Mr Trump began hiking tariffs on Indian exports.

Some in the foreign ministry and prime minister's office have begun asking if India has signed up to too many initiatives and

coalitions, inflating expectations of its capacity and courage. These days “there is less domestic benefit in playing global host,” thinks Dan Markey, an expert on China and South Asia at the Stimson Centre, a think-tank in Washington. Leaders of big powers are giving up the pretence of consulting large groups of the sort India likes to chair. While India was hosting the BRICS ministers, Mr Trump and Xi Jinping were meeting one-on-one in Beijing.

Others think India should become much more forthright about the big issues of the day. “We should be able to say to both America and Iran: ‘You cannot hold the world hostage like this,’” says Shivshankar Menon, a former Indian national security adviser and foreign secretary. For years the country's prime minister, Narendra Modi, sold India as a *vishwaguru*, or teacher to the world. Yet India very often seems reluctant to divulge what, precisely, it wants the world to learn. ■

Overseas Chinese

Crumbling pillars

GEORGETOWN

South-East Asia's clan associations must change to survive

TOURISTS IN GEORGETOWN, a city on the Malaysian island of Penang, love to pose by a well-known piece of street art depicting two children on a bicycle. Most of them miss the ornate building on the other side of the wall: home to Cheah Kongsi, one of Penang's richest Chinese clan associations, founded in 1810 (pictured). For two centuries such groups served migrants fleeing violence and poverty in China. Now they face a new challenge: the apathy of the young.

South-East Asia is home to about 80% of the world's overseas Chinese. Clan associations, along with Chinese-language schools and Chinese newspapers, are one of the three pillars of overseas Chinese society, according to Hong Liu of Nanyang Technological University in Singapore. Malaysia has more than 10,000 clan associations, with Penang a particular hotspot. They are also found in Myanmar, the Philippines, Singapore, Thailand and Vietnam, among other countries.

When they started in the early 19th century the associations were a kind of welfare state, employment agency and temple rolled into one. Migrants originating from the same clan, village or trade came together to socialise, pray to common deities, find housing and jobs, open schools, lend money and operate cemeteries. Some grew rich. The most influential ones in

Georgetown own homes and palm-oil plantations, among other properties. Cheah Kongsi still provides welfare, giving between 250 and 2,000 ringgit (\$55-450) per year to children of members and to retired people.

These groups have long had tight membership rules. Cheah Kongsi is open only to those who can trace their heritage to a particular part of Fujian province, in southern China. Gallingly, many of the most powerful clan associations in Penang still refuse membership to women. But now they are getting anxious about the future. “The main thing we worry about is that no young people are joining,” says Clinton Cheah, a trustee of Cheah Kongsi. What will happen down the line, he wonders, especially now that members often only have one or two children?

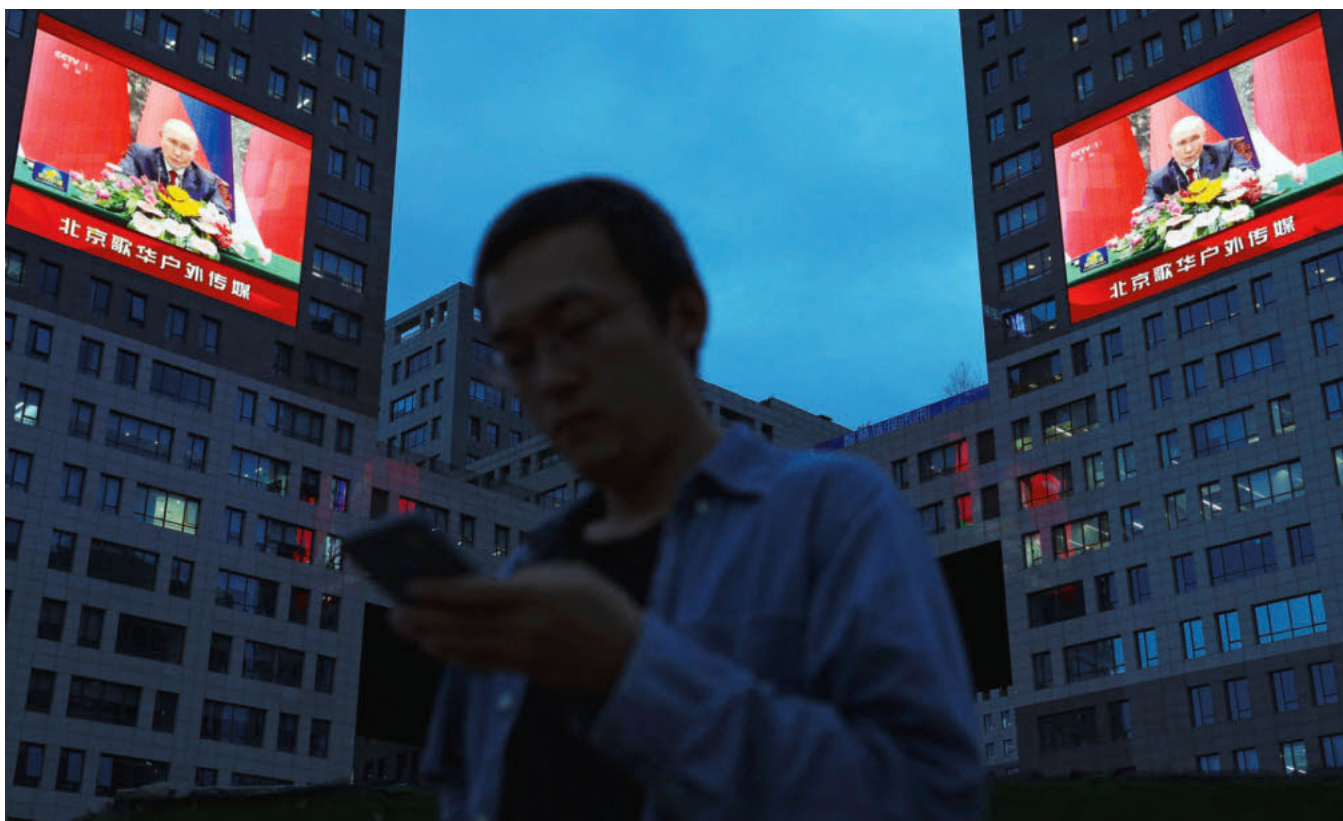
Migrants from China still move to Penang, often to work in its lively semiconductor industry. But they tend to join newer diaspora groups with more current ties to the Chinese mainland and less discriminatory membership requirements. Some regret the clan groups' diminished role in integrating newcomers. What it means to be Chinese is contested: with their long histories, the clan associations offer an alternative to the version the Chinese government exports.

At least one of the big clan associations in Penang is finally modernising. The Yeoh Kongsi decided to admit women only in February 2025. “It was a very sensitive decision, so the board gave its blessing only after seeking divine approval,” says Yeoh Seng Hooi, a trustee. “Should we admit women?” a member asked the gods, before throwing two crescent-shaped wooden blocks. The association has since welcomed 14 women, 35% of its new members. Thank heavens for that. ■



Dragons' perch

China



Vladimir Putin in China

Little brother's big visit

How Xi Jinping plays the world's great powers

FOR XI JINPING, a leader who aspires to place China at the centre of world affairs, it was a triumphant week of diplomacy. Just four days after Donald Trump concluded a visit to Beijing, looking for business deals and help with Iran, Vladimir Putin turned up, seeking assistance for his war in Ukraine. Russia's president, like America's, lavished praise on his host and left with a clutch of agreements. But like Mr Trump's, the deals were short on detail. And the enduring image at home and abroad was one of China as the fulcrum of global geopolitics, dealing with America as an equal and Russia as a junior partner.

China had not originally planned for the visits to be so close: Mr Trump's was scheduled for early April but delayed by the war in the Gulf. Still, the timing worked well for Mr Xi. It sent a clear message that better relations between China and America will not come at the expense of his "no limits" partnership with Mr Putin. At the

same time, Mr Xi showed the leverage that he has gained over Russia since its full-scale invasion of Ukraine in 2022. Despite talk of shared interests, Mr Putin ended his two-day visit on May 20th without a long-discussed deal on the building of another gas pipeline between Russia and China.

Mr Putin's visit was not as elaborate as Mr Trump's. China greeted the Kremlin's leader in the same way, with a military honour guard, a 21-gun salute and cheering children. Mr Xi and Mr Putin then held formal talks, followed by another meeting over tea, in the Great Hall of the People. But there was no equivalent of Mr Trump's

private tours of the Temple of Heaven and Zhongnanhai, the leadership compound. And Mr Putin and Mr Xi did not attempt any of the more personal displays of camaraderie that they have staged before, such as swigging vodka, making pancakes or watching a hockey game.

Instead they emphasised the breadth of their relationship and their shared opposition to many aspects of American policy. In a lengthy joint statement, they condemned American and Israeli attacks on Iran, as well as the abduction of heads of state (a clear reference to Venezuela's). They expressed opposition to sanctions on North Korea and to external interference in Latin America and the Caribbean. They criticised Mr Trump's "Golden Dome" missile-defence project as well as America's "irresponsible" nuclear policy. And they renewed calls for a multipolar world order, warning of a drift towards the "law of the jungle". As Mr Xi put it, in a thinly veiled reference to America: "The tide of unilateral hegemony is running rampant."

In his public remarks, China's leader put particular emphasis on the Middle East. He called for a "complete cessation of hostilities" there, warning that it would be "unacceptable" to resume fighting. That was sharper language than he used during Mr Trump's visit. The Trump administration has been considering the possibility

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► of resuming strikes on Iran, which have been suspended since a ceasefire was announced in mid-April.

Russian officials emphasised the 40 or so agreements that were signed during the summit, covering areas ranging from artificial intelligence and border infrastructure to co-operation in space and the protection of tigers and leopards. Mr Putin highlighted energy ties, noting that Russia was one of China's biggest suppliers of oil and gas and that nearly all their trade was done in roubles or yuan, protecting it from "external influence" (that is, American sanctions). He invited Mr Xi, whom he called a "dear friend", to visit Russia in 2027.

Yet there were few details of how the agreements would be implemented. And there was no mention in them, or in the two leaders' public remarks, of Power of Siberia 2—a 1,600-mile (2,600-km) pipeline that would carry up to 50bn cubic metres of gas annually from eastern Russia to northern China via Mongolia. Russia is keen to make progress on the project to secure a new market for gas it can no longer sell to Europe. But China has been driving a hard bargain on price, volumes and other terms. It has also been reluctant to rely on any single supplier for more than about 20% of its hydrocarbon imports, a level it has already reached with Russia.

Russian officials had hoped that China would show more flexibility since the closure of the Strait of Hormuz highlighted the vulnerability of its seaborne imports of energy. About 90% of China's oil imports still come by sea, largely from the Middle East, despite its efforts to diversify supplies and build pipelines across its borders. China and Russia seemed to make progress on the project when Mr Putin visited Beijing in September. Alexey Miller, the boss of Russia's state-run gas company, Gazprom, announced afterwards that they had signed a legally binding memorandum. China also committed to "advancing preliminary work" on the project in its latest five-year plan, unveiled in March.

In the end, though, Mr Xi appears to have stood firm on his demands for better terms. That underlined how Russia, once China's patron, has become dependent on Chinese economic support as well as dual-use technology for the war in Ukraine. "It's a pretty solid display of how much leverage China has," says Alexander Gabuev, director of the Carnegie Russia Eurasia Centre, a think-tank in Berlin. "Putin comes out not really empty-handed—everything that's working is working—but there is no additional mileage that China is willing to go to support him, unless it's really very beneficial for the Chinese." Mr Gabuev also said that the back-to-back summits played well for Mr Xi by suggesting that he has better relations with both Mr Putin and Mr Trump than they have with each other.

Other deals between China and Russia may not have been made public. One of Mr Putin's priorities was likely to have been securing access to more dual-use items, including drone components (see next story). Mr Xi, meanwhile, had been expected to seek more Russian help in developing China's military capabilities, especially in undersea warfare, and in accessing data and tactical insights from the front lines in Ukraine. The two leaders' statement did not mention such matters, saying only that the two countries would deepen defence co-operation, expand joint military exercises and patrols, and "work together to address various risks and challenges".

Further details may emerge in the days following the summit, just as they have trickled out since Mr Trump left Beijing. For the moment, however, Mr Xi must be celebrating a diplomatic double-bill that brought him closer than ever to the Sino-centric world he craves. ■

Nitrocellulose and more

On the wish list

How China quietly helps Russia in Ukraine

RUSSIA'S DEPENDENCE on China to sustain its war against Ukraine has steadily increased as Western sanctions have intensified. China maintains that it is neutral in the conflict, denying it gives Russia lethal aid such as finished weapons systems. But by supplying huge volumes of "dual-use" components and materials, it provides vital support for Russia's military-industrial base. During his state visit to Beijing on May 19th and 20th, Vladimir Putin was a supplicant.

Chinese microelectronics and semiconductors are critical to Russia's ability to churn out the precision-guided missiles and drones that pound Ukrainian cities. China also supplies most of the commercial first-person-view (FPV) drones and supporting technologies Russia needs. China also provides Ukraine with many of the same components—cameras, motors, transmissions—needed by FPV drones. But nearly all of their assembly is now Ukrainian. Russia, Ukraine knows, will always be first in the queue.

Russia's weapons and ammunition factories are almost entirely reliant on Chinese computer-numerical-control machine tools. The Jamestown Foundation, a Washington-based think-tank, estimates that around 90% of Russian machine-tool imports are now from China.

According to Western officials, Russia

also depends heavily on China for nitrocellulose, the cotton-derived feedstock that is the propellant for artillery shells, tank rounds and missiles. Russia produces its own at a handful of explosives factories (which the Ukrainians target) but not nearly enough for its wartime needs.

Turkey, through various intermediaries, provided about half of Russia's imports of nitrocellulose until recently, but the firms involved are now subject to heavy sanctions. Consequently China, allege the Western officials, is ramping up its sales of nitrocellulose to Russia through the Norinco Group, a massive state-owned armaments firm, which uses a pair of subsidiaries. The company has not commented publicly on this.

China insists that nitrocellulose is a product needed for paints and lacquer. To maintain the obfuscation, the four main Russian gunpowder factories reportedly run sister civilian operations. Norinco, Western officials say, tries to conceal sales behind shell companies and foreign intermediaries. China is also stepping up its exports of cotton pulp and cotton cellulose from which nitrocellulose is made. Nearly all of Russia's cotton pulp previously came from Uzbekistan and Kazakhstan. But packages of sanctions introduced last year and this year by the European Union on designated Uzbek and Kazakh exporters have begun to have some impact.

There is little likelihood of China reducing its support for Russia's war machine. Chinese firms are earning handsome profits while Russia provides cheap oil and gas in return. The polite fiction of "dual use" has been artfully maintained. Moreover, to the satisfaction of China's leaders, it seems certain that after the end of the war in Ukraine, Russia will remain a needy junior partner, dependant on China. ■



Never mind where it's from

Prices

The right kind of Japanification

HONG KONG

China can learn lessons from its neighbour about escaping deflation

JAPAN'S POST-WAR manufacturing miracle provided a template for China to copy. Japan's "lost decades" after 1990 served as a cautionary tale to avoid. Since the covid-19 pandemic and the war in Ukraine, Japan has left deflation behind, even as China has fallen victim to it. As China now contends with similar disruption from the war in the Gulf, could it learn lessons from Japan's example?

After its property bubble burst in the early 1990s, Japan's economy was stuck with heavy debts, lacklustre spending and falling prices. The country became synonymous with deflation. But in 2022 Japan and China traded places. China's inflation rate fell below that of its smaller, greyer neighbour and stayed there (see chart). The role reversal was later confirmed by the bond markets, which are acutely sensitive to the inflation outlook. Late last year yields on ten-year Chinese government bonds dropped below Japan's for the first time in data stretching back over 20 years.

This was not good news. Although nobody likes high inflation, its opposite can also be a sign of trouble. China's spell of deflationary pressure, like Japan's, follows a property bust. Developers defaulted on excessive debts and construction faltered. A deep decline in property values depleted household wealth and confidence. The result was a shortfall in domestic spending, putting downward pressure on prices. In October 2022 producer-price inflation turned negative. It remained below zero for the next 40 months.

Deflation can also be a cause of trouble. When prices fall, debts weigh more heavily, pay packets shrink, and households and firms hoard cash and defer purchases because they expect goods and services to be cheaper in the future. In Japan central bankers complained about the country's "deflationary mindset". Prices stagnated partly because everyone expected them to. Inflation did not loom large in pay talks.

It took two huge jolts to the global economy to shake Japan out of this mindset. The covid-19 pandemic and the invasion of Ukraine strained worldwide supply chains. The jump in global prices, amplified by a further drop in the yen, became impossible to ignore. Ahead of pay negotiations in 2023, Rengo, Japan's leading trade-union federation, asked for a 3% rise in base pay (excluding seniority adjustments), its highest demand in 28 years.



Cost increases, driven by global events, translated into higher wage claims, creating momentum for further price rises.

Will China follow such a path? Another big shock—the war in the Gulf—has driven up the price of energy and related goods. In March producer-price inflation turned positive for the first time since 2022. In April it increased to 2.8%, an even bigger jump than expected.

This "cost-push inflation", associated with the war, came on top of some "demand-pull inflation" created by the global AI spending boom. China's manufacturers have benefited from the rush to build data centres and power grids. They produce



Turning things around

much of the "unsexy but necessary" kit that supports high-end server farms, including pumps, fans and other electrical machinery, as Thomas Gatley of Gavekal Dragonomics, a research firm, has pointed out. Some forecasters think exports of goods could grow by 10% this year in dollar terms. In principle, this could create a virtuous cycle, by boosting employment, wages, spending and prices.

Unfortunately, the industries benefiting from the AI and energy "super-cycle" tend to be capital-intensive, notes Robin Xing of Morgan Stanley, a bank. Their success does not, then, generate big increases in employment. China's resilient export machine is no longer a reliable jobs machine. One consequence is that China's success abroad has not boosted morale at home. The property market (despite some favourable signs in Shanghai and a few other prosperous cities) is still in the doldrums. The job market is weak and consumers are wary. Retail sales rose by just 0.2% in April, compared with a year earlier, before adjusting for inflation.

China's reflation therefore remains narrow. It is largely confined to upstream inputs such as energy, petrochemicals and metals like copper. Industries that use these inputs are feeling the strain. The volume of crude oil processed in China fell by 5.8% in April compared with a year earlier. Industrial production overall slowed to 4.1% from 5.7% in the previous month.

Moreover, a monthly survey of managers in services and construction suggests that sales prices remain under downward pressure. The factory-gate prices charged for consumer goods are still falling at much the same pace as before. Workers may grumble about higher prices for petrol. But unlike Japanese workers in 2023, they lack the bargaining power to demand higher wages in compensation.

Another distinction from Japan lies in the temperament of policymakers. As inflation picked up in Japan, the government did its best to maintain the momentum. At the end of 2023 it introduced tax breaks, subsidies and handouts to ensure what it called a "complete exit" from deflation. Chinese officials do not share this determination. They are reluctant to ease monetary policy, because it could squeeze the profitability of banks. They are also wary of increased fiscal stimulus, given past excesses and rising public debt. Mr Xing no longer expects China's central bank to cut interest rates this year. Instead of trying to turn wartime price jumps into a broader, durable reflation, China's policymakers are treating it as an excuse to do less.

Japan's lost decades are a cautionary tale. Its recent efforts to ensure a permanent exit from deflation also hold useful lessons for Chinese officials. But they are in no mood to heed them. ■

Middle East & Africa



Ebola in Africa

A terrifying new outbreak

GOMA

Aid cuts, war and no vaccine mean the epidemic could be the worst in a decade

SINCE APRIL, we've seen people dying," says Sylvie Kabuo-Kinyoma, who sells vegetables in Mongbwalu, a gold-mining town in Ituri, a province in eastern Democratic Republic of Congo. At first she blamed the deaths on witchcraft. But then a nurse spotted a patient with a severe fever who was bleeding from the nose—a sign of Ebola, a virus with a fatality rate of up to 50% that spreads through direct contact with bodily fluids. "We're afraid," says Ms Kabuo-Kinyoma. "I don't want to lose my children."

On May 17th the UN's World Health Organisation (WHO) said the outbreak was a "public health emergency of international concern", only the ninth such declaration since 2005. It has not been called a pandemic, but it is shaping up to be the worst Ebola epidemic since at least 2018, when more than 2,000 people died in eastern Congo. Some experts worry it could be as bad as in 2014-16, when an epidemic in

west Africa killed 11,000. That is because of a fatal blend of bad luck, a country at war and a world cutting health aid.

By May 20th there had been nearly 600 suspected cases and 139 deaths in Congo. There have almost certainly been more, according to modelling by the MRC Centre for Global Infectious Disease Analysis, an institute at Imperial College in London. The WHO reckons the virus has been circulating for a couple of months at least. Two Congolese nationals have been diagnosed

with it in Kampala, the capital of neighbouring Uganda. An American doctor working in the area was also found to have the disease and has been evacuated to Germany. There have also been cases in Goma, the capital of North Kivu, a Congolese province to the south of Ituri (see map on next page). Rwanda has reportedly closed some border crossings with Congo. Public-health experts worry the disease could spread to Burundi and South Sudan.

Since the west African disaster, governments and international bodies have vastly improved how they respond to the spread of the virus. Vaccines against the most common strain, Zaire, have allowed medics to isolate infected populations. Genetic sequencing of this strain has enabled rapid diagnostic testing. Local health workers, backed by international NGOs and African governments, have got better at educating people about Ebola, earning their trust and isolating those with symptoms.

Yet none of that is likely to be much help with this outbreak, the 17th recorded in Congo since 1976. Bundibugyo, the strain that is causing it, is a bit less fatal than Zaire. But it is less common, and there is no licensed vaccine or rapid diagnostic testing for it. Samples taken in Ituri must be flown to Kinshasa, the capital of Congo, some 2,000km away. It can take days to receive a result. "This feels like we're back to

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square one,” says Bob Kitchen of the International Rescue Committee (IRC), a humanitarian NGO, likening the situation to the start of the epidemic in 2014, before vaccines against any strain were available.

The kaleidoscope of conflicts in eastern Congo complicates the picture. More than a hundred militias are active across Ituri and the Kivus. In the outbreak in Congo in 2018 some of these groups burned down clinics run by Médecins Sans Frontières (MSF), another NGO. “We are taking safeguarding very seriously,” says Alan Gonzalez of MSF. Armed groups—and some of the worst roads in Africa—make it hard for medics to get to remote areas.

The state in Ituri is weak, as in much of Congo. “We are very worried because, so far, the provincial authorities have taken no action,” says Machozi Mwanamolo, a university professor in Bunia, a city in Ituri. He laments that people from Mongbwalu have been “allowed to move about freely when they should be restricted to prevent the spread of the virus”. He is wearing a face mask and making ginger tea with lemon and garlic “to boost our antibodies”.

Farther south in the Kivus, the situation is even trickier. In the past two years M23, an armed group backed by Rwanda that is at war with the Congolese government, has taken control in much of the two provinces, causing many aid workers to leave. It has replaced officials across the local bureaucracy, including in the health ministry. When your correspondent visited a government-run clinic near Goma at the end of April, it was struggling to contain a measles epidemic. M23 has also declined to reopen the airport in Goma, forcing NGO workers to make journeys to parts of North Kivu via Rwanda and Uganda. Aid workers complain of supplies being held up by M23 officials demanding payment.

Then there are the aid cuts. Before 2025 American funding helped pay for surveillance (typically in the form of community health workers) and preparedness (education schemes and protective gear). But over the past year that has been slashed; IRC, for example, went from working in five areas in Ituri to just two after March 2025. On May 18th the State Department promised \$13m to help with the response, a fraction of what America spent to combat the epidemic in 2014-16. Other Western governments, including Britain's and Germany's, have also drastically cut aid spending.

Much will depend on whether the WHO, international NGOs and African organisations can fill the gap. Jean Kaseya, the head of Africa CDC, the continent's main public-health body, implicitly criticised America's decision to restrict travel by non-Americans who had recently been to the region. “Global health security cannot be achieved through borders alone,” he said. “It is achieved through partnership,



trust, science and rapid investment in preparedness and response capacity.”

Responding to Ebola was difficult even with more of those things around. In the 2018 outbreak it took two years, around 300,000 doses of vaccine and a well-funded aid response to end the epidemic. Without vaccines, the support of local Congolese for public-health measures is more important, just as there is less money for schemes that build trust. That sort of education matters, says Foibe Mbusi, a market seller from Bunia: “There’s a segment of the population, even female vendors like us, who don’t believe Ebola exists.” ■

Conservation in Africa

How to save the safari

MAASAI MARA

Balancing the needs of locals, tourists and animals is a difficult business

THE COTTAR family has faced a few challenges in more than a century of running safaris. Charles Cottar, who swapped the plains of Oklahoma for those of east Africa in 1909, survived a mauling by a leopard in 1922 but was eventually killed by a charging rhino in 1939. In 1965 his grandson Glen set up one of the first camps for tourists to photograph animals rather than hunt them. Not that the wildlife seemed grateful: that year Glen was gored by a buffalo.

Today, running a safari firm is less dangerous, but more complex. The industry faces a trilemma in balancing the demands of tourists, growing local populations and ecosystems. Some operators are trying new models to resolve it. That is crucial, because in some of the most spectacular parts of Africa, the current model of tourism, if unchecked, will spoil the natural splendour on which business depends.

Safaris are increasingly popular, attract-

ing some of the 81m tourists who visited Africa in 2025, a record according to the UN. The global industry is valued at \$38bn, estimates Grand View Research, which assesses it, rising to \$67bn by 2033. Most safari-goers are from rich Anglophone countries, notes Go2Africa, a travel firm, but the number from emerging economies like India is growing. In the Maasai Mara in Kenya, the number of lodges increased from 95 in 2012 to 183 at the end of 2025. In the neighbouring Serengeti National Park, in Tanzania, the number of tourism facilities has increased almost eight-fold over the past 20 years, notes Grant Hopcraft of Glasgow University.

Conservationists say the surge of tourists has affected wildlife. The population of large animals in the Mara has declined by 70-90% over the past 40 years. The famous migrating wildebeest spend approximately one and half months less in the core parts of the Mara and Serengeti than they did two decades ago, in part because they avoid buildings. Fewer cheetah cubs survive because safari vehicles that crowd around their parents’ kills attract lions and hyenas, which eat the young.

Rapid human population growth means that there are more people living around natural beauty spots. The population of Narok County, which manages the Masai Mara National Reserve (MMNR), has soared by 3,000% since 1947, from 38,000 to 1.2m. The number in the Mara-Serengeti region has doubled in the past 20 years and is forecast to double again in the next 20.

About half of Maasai, the pastoralist group in the area, live below the official national poverty line. Their efforts to make ends meet are putting more pressure on land they—and tourism firms—depend on. One third of the MMNR is degraded from overgrazing. Human-wildlife conflict is intensifying, with people killing elephants that trample on their land and, on occasion, poisoning carcasses to kill predators.

One way to reduce the pressure, at least from tourists, would be if places like the Mara became more exclusive. Only 100,000 tourists visit Botswana’s Okavango Delta every year. Rwanda charges \$1,500 per person to see its gorillas, and is considering raising the price. Many conservationists in the Mara argue that visitation it should become more expensive.

But it will be hard to change. “We would love to raise prices but we are competing with 250 other operators,” says the boss of one safari firm. Politicians are unlikely to embrace restrictions. Narok County receives more revenue per year than any other Kenyan county (including Nairobi) through leases and a portion of the “bed fees” paid by tourists. Local bigwigs also benefit from the fees. On the other side of the border Tanzania is promoting tourism via what it calls the “volun- ▶▶



A valuable sight

Israel's election

Bibi's last stand?

JERUSALEM

The legacy of October 7th and Israel's many wars will loom over its election

OBSERVERS HAVE repeatedly predicted the collapse of Israel's government over the past two and a half years. Indeed, many have wished for it. In the end, two ultra-Orthodox parties in the ruling coalition have prompted the moment of its own demise. On May 20th the Knesset, Israel's parliament, voted to dissolve itself. It was a preliminary vote and Binyamin Netanyahu, the prime minister, is still playing for time. But at best he can buy his government a few more weeks. This Knesset's term is about to expire: parliamentary elections must be held by October 27th.

The immediate trigger is a demand by the ultra-Orthodox parties for a law exempting religious seminary students from compulsory military service. Mr Netanyahu is willing to pass it but failed to muster the necessary votes for a measure that is deeply unpopular with most Israelis. Many of them believe the government's subservience to religious interests has come at their own expense. The issue will be among the most prominent in the coming campaign.

It is remarkable that the government has lasted this long, nearly completing its four-year term—a rare feat in Israel. After Hamas's assault launched from Gaza in October 2023, and Israel's failure to anticipate it, many thought Mr Netanyahu's coalition would fall. But the prime minister brazened it out. He blamed the armed forces and intelligence services and prom-

ised Israelis "total victory" in the wars that followed. Not only did he keep his coalition of hard-right and religious parties together; he persuaded others to join his government as Israel mobilised for war.

But victory has proved elusive. Israel controls a wasteland in just over half of Gaza; Hamas still rules the rest. Israel's other campaigns over the past two years have also failed to bring decisive results. The Iranian regime, which Israel has attacked twice alongside America, is battered but defiant. In Lebanon Hizbullah, Iran's proxy militia, continues to fight Israeli troops on the ground, despite losing most of its leadership to air strikes. Mr Netanyahu is still hoping Donald Trump will renew attacks on Iran. But few in Israel's intelligence community believe the regime is about to collapse. "It could still happen," concedes one Israeli Iran-watcher. "But it is unlikely to be in time for the election."

The shadow of October 7th and Israel's many wars since then will loom over the election. Naftali Bennett, a former prime minister who is leading a list of right-wing and centrist candidates, has promised that, should he form the next government, its first cabinet meeting would be dedicated to appointing a commission of inquiry into the failures that led to the massacre.

That does not, however, mean that this election campaign will be a national reckoning for the brutal conduct of the Gaza war. Over 70,000 people, mainly civilians, have been killed; Gazans were forced to the brink of starvation. Yet most Israelis still believe the wars in Gaza, Lebanon and Iran were justified and necessary responses to existential threats to the Jewish state. "We're likely to lose the election because the wars took so long and failed to yield decisive results," says a parliamentarian from Mr Netanyahu's Likud party. ➤

► tary relocation" of Maasai. Human Rights Watch, a global monitor, says the tactics equate to "forced evictions".

Is there a better way? Many safari firms now promise to "give back" to "communities". But Sue Snyman of the African Leadership University says some pay little attention to what local people actually want, which is often cash.

Cottar's Safaris pays to lease the land it uses from the Maasai-owned conservancy. The land is held in a community-owned trust, which distributes annual lease payments to 6,500 members, whose title deeds have been paid for by the safari firm. The trust consults households about whether they would prefer social projects or money—and nearly all prefer the latter.

Elsewhere in Kenya locally owned community conservancies organised in the Northern Rangelands Trust have deals with tourism operators. In South Africa & BEYOND, a luxury travel outfit, has returned 9,000 hectares to their ancestral owners as part of post-apartheid land restitution. In Namibia Damaraland Wilderness, another operator, set up a joint venture with a conservancy that involved the transfer of ownership. Kenya and Zimbabwe are changing conservation laws to encourage more such arrangements.

The big question is whether there is enough money to go around to resolve the trilemma. Beyond sparsely populated conservancies with expensive facilities, the dividends may be too meagre.

For some critics safaris are colonial cosplay, where tourists are part of an extractive business model—like mining, but with more gin and tonics. That is unfair. Most visitors are simply after the wonder these trips can stir. But without the right policies and business models, there will be ever less for them to look at. ■



▶ For the government's opponents, this election is about much more than the wars. They have not forgotten that before October 7th, Mr Netanyahu's coalition was trying to pass a controversial series of legal reforms that would solely weaken the Supreme Court. Hundreds of thousands of Israelis repeatedly took to the streets to protest against what they saw as the erosion of Israel's democratic foundations. They have watched the government lavish public money on ultra-Orthodox voters, despite their refusal to serve in the army and their limited economic participation.

While the election is unlikely to focus on Israel's economy, which is in surprising health (see next article), many Israelis worry that government spending is poorly controlled. Others warn that Israel's democracy is dysfunctional. In a recent survey, most centrist and left-wing voters said that losing this election to the Netanyahu camp would be "intolerable". More Israelis may choose to leave. In 2024 nearly 83,000 emigrated, a record, and in 2025 almost 70,000 followed suit. That could be an early sign of a brain drain.

Israel's economy, buoyed by its tech sector, and especially by the strength of defence-tech companies, has proved remarkably resilient. Still, most polls conducted over the past three years suggest that the parties in Mr Netanyahu's current coalition will fail to win a majority. But even if the opposition does, they will struggle to forge an effective government.

Mr Netanyahu's opponents include right-wing and centrist parties who refuse to join a coalition with Arab-Israeli parties. The opposition lacks a clear leader. Mr Bennett has joined forces with Yair Lapid, a centrist former prime minister. But, despite much talk of emulating democratic parties that united in Hungary to topple Viktor Orban, the other opposition parties have yet to recognise Mr Bennett as their candidate for prime minister. The lack of unity probably helps explain why polls still rate Likud the largest party.

Some are wondering whether, at 76, Mr Netanyahu, Israel's longest-serving prime minister, might at last call it a day. As well as struggling to keep his bloc together, he has been dealing with heart problems and prostate cancer. Retirement would not only avoid the potential of a humiliating electoral defeat, but would let him accept a plea bargain and end his long-running trial for bribery and fraud (charges he strenuously denies). But many believe he is incapable of giving up the fight. This is set to be his 12th election as leader of Likud. Over four decades in politics he has confounded the polls time and again. Another race may prove irresistible. And for all the grave questions facing Israelis in this election, it will almost certainly become yet another referendum on Mr Netanyahu. ■

The Israeli economy

Guns and unicorns

JERUSALEM

Its endless wars help explain why Israel's economy is booming

ISRAEL'S ELECTIONS are rarely focused on its economy. Questions of security and identity almost always dominate. The election expected in September or October will be no different. In a survey in Israel in February, only 11% of respondents said reducing the cost of living was the main issue that would determine their vote.

That is partly because, despite two and a half years of inconclusive wars in Gaza, Lebanon and Iran, Israel's economy is booming. The fighting has taken a toll. GDP shrank by 3.3% in the first quarter at an annualised rate after the country largely shut down in March during the conflict with Iran. But the Tel Aviv Stock Exchange and the shekel are both buoyant. Inflation and unemployment are low, partly because so many young Israelis have been drafted. Exports of natural gas from offshore fields in the Mediterranean are also helping.

Strong exports, especially from the tech sector, are helping restore growth after the wartime dip. Defence-technology firms, whose order books are brimming with contracts from both the Israeli army and foreign forces, have done particularly well. Long-time observers of Israel's economy are unsurprised. It was fast to emerge from the global financial crisis and the pandemic. The key then, as now, was an economy based on nimble tech firms trading with the world and a highly skilled workforce.

So far, there is little reason to think the

erosion of Israel's global reputation because of the ferocity of its war in Gaza will change this. Most Israeli tech wares are not sold to individual consumers, but as components, services and proprietary knowledge to larger international tech companies and as weapons systems to states.

Nvidia, a chipmaker with 6,000 employees in Israel, is planning a new base with 10,000 more. Jensen Huang, its founder, recently called Israel "Nvidia's second home", and promised the wars will not make the firm leave. Western governments are buying more Israeli weapons. In December Germany signed a follow-on €3bn (\$3.5bn) contract for Arrow 3 batteries, nearly doubling its investment in Israeli missile-defence technology.

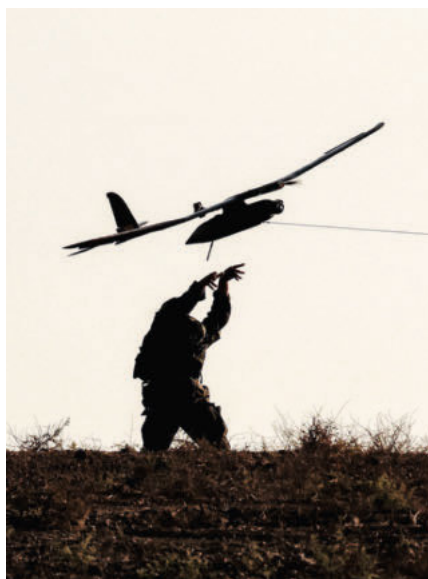
But economists worry that Israel's economic model—high-end technology, defence exports and a small skilled elite—is becoming demographically unsustainable.

Ultra-Orthodox Jews and Arab-Israelis make up over a third of the population and are having more children than other Israelis. They are barely represented in the tech sector. "Israelis' vote this time will decide the policies of the next government on integrating the ultra-Orthodox community into the workforce," says Yigal Newman, a professor of finance at Hebrew University. "If the workforce participation situation doesn't change, it could ultimately derail the resilience of Israel's economy."

The current government of Binyamin Netanyahu, Israel's longest-serving prime minister, has poured money into the religious communities represented by his key coalition allies. Only 54% of their men work. The rest spend their days in study. Their boys' secondary schools do not teach English, maths or science.

The share of tech workers in the labour force is the highest in the world. But it is still only 10%, and they account for over half of Israel's total exports. This leaves the economy exposed in the long term to a brain drain of its most mobile workers. "It's not just the economy which is based on a dangerously small number of people; it's national security as well," warns Dan Ben-David, head of the Shoshan Institute for Socioeconomic Research. "These are also the people who developed the defence systems protecting Israelis from missiles over the past two and a half years."

Israel's schools are deteriorating. On current trends, its future tech workers will be drawn from a shrinking reservoir. "This election can't be only about replacing Mr Netanyahu and his government," says Mr Ben-David. "It has to lead to a game-changing overhaul of the education system to draw in all Israeli children. Without that, Israel faces existential danger." Its economy has survived wars, boycotts and recessions. But, as in its politics, the gravest threat may come from within. ■



A high-flying sector

Egypt and the United Arab Emirates

Feeling surrounded

CAIRO

Why Egypt fears and needs the Emirates

WHEN ABDEL-FATTAH AL-SISI arrived in Abu Dhabi on May 7th, he might have expected a fulsome display of gratitude. Egypt had, belatedly, offered warplanes to help defend the United Arab Emirates (UAE) from Iran. But instead of a photo-op in one of his palaces, Muhammad bin Zayed, the UAE's ruler, took the Egyptian president for tea in a mall.

For Egyptians, the optics spoke volumes about their country's sinking prestige. They still think of it as *umm al-dunya*, the mother of the world, whereas the UAE is less a startup nation than an upstart one, newly arrived from the desert. They resent what they see as the UAE's encirclement of their larger, historically dominant country with separatist movements, militias and client rulers. They are bitter that Egypt's plight has forced it to sell prized assets and vast tracts of land to Gulf investors.

The Emiratis see Egypt as ungrateful. For decades, they have bailed it out. Yet rather than rush to the UAE's aid when Iran struck, many Egyptians cheered Iran. Instead of contributing to the fight against Iran, Egypt called for de-escalation and sent ministers to negotiate. Pakistan has paid for similar policies; the UAE demanded repayment of over \$3bn in loans and expelled 15,000 nationals. Some in Abu Dhabi mutter that Egypt, with 400,000 nationals there, should be taught a similar lesson.

Both sides have grievances. Egypt is alarmed by the UAE's support for the Rapid Support Forces, a militia accused of genocide in Sudan. (The UAE acknowledges providing some early support to the RSF, but denies it is still doing so.) Egypt also frets about Emirati-backed separatists in Libya and Somaliland and the UAE's support for a separate administration in Gaza.

Most troubling for Egypt is the UAE's growing alignment with two menacing regional powers: Abiy Ahmed's Ethiopia and Binyamin Netanyahu's Israel. Ethiopia's construction of the Grand Ethiopian Renaissance Dam—with the UAE's encouragement, say some Egyptians—threatens Egypt's life force, the Nile. Israel's continued squeeze on Gaza might yet tip its population, Islamists included, into Egypt.

The Emiratis believe they are owed more loyalty. In 2023 Egypt seemed close to economic collapse. Years of overspending on Mr Sisi's vanity projects, the army's gutting of the economy and mounting debt had shaken investor confidence. In-



Retail diplomacy

flation surged. The currency tumbled. The UAE provided billions of dollars, unlocking help from the IMF and the European Union, and has since poured in billions more.

Still, neither side will let the quarrel spiral too far. Mr Sisi has offered the UAE fighter pilots if war with Iran restarts and has told Egypt's media to stop criticising the Gulf state. But Cairo is also bolstering ties with Qatar, Saudi Arabia and Turkey. Egypt may not want to alienate the UAE. But nor can it afford to rely on it. ■

The Gulf war

Déjà vu all over again

WASHINGTON, DC

Donald Trump is still looking for a quick fix in Iran

BY NOW IT should be clear: Donald Trump is not eager to resume his hot war with Iran. For six weeks he has seized on any excuse, no matter how flimsy, to delay another round of military strikes.

Reluctance is not refusal, however. In the absence of a deal between America and Iran, the risk of renewed conflict is growing. Mr Trump is a man enamoured of quick fixes, and some of his allies are trying to persuade him that a few more air strikes will be just that.

His latest about-face came after a series of calls with regional leaders. He spoke first on May 17th with Binyamin Netanyahu, the Israeli prime minister. It seemed as if they had agreed to resume fighting. The next day, however, the president said he would hold off at the behest of leaders from Qatar, Saudi Arabia and the United

Arab Emirates. The strike, he claimed, had been “scheduled for tomorrow”. But he said the monarchs persuaded him that “a deal will be made, which will be very acceptable to the United States.”

Most Gulf leaders have indeed urged him to hold fire, fearing further Iranian strikes on their economies and infrastructure. But the president's announcement surprised them. Diplomats from two of the Gulf countries said they were unaware an attack was as imminent as Mr Trump claimed. They were equally bewildered by his talk of a forthcoming deal. “Everything is still stuck,” says one Gulf envoy.

Iran has hardly budged in weeks of indirect negotiations. It is willing to make a narrow deal in which both sides would lift their blockades on shipping through the Strait of Hormuz. Trickier issues, such as Iran's nuclear programme, would be left for protracted future talks. That is a hard sell for Mr Trump, since it would halt the fighting with no definite Iranian concessions beyond reopening a strait that was closed only because of his war.

Mr Trump is desperate for something to break the impasse—which, by now, has become a defining feature of the conflict. At first America thought assassinating Iran's leaders would compel the regime to surrender. Then it gambled on a blockade of Iranian ports, which took effect on April 13th and was meant to cause economic collapse within weeks. By May it had vested its hopes in Mr Trump's visit to Beijing: perhaps Xi Jinping would crack the whip and force the Iranians to accept a deal.

At every turn the administration thought it could find a silver bullet to end the war on favourable terms, only to be disappointed. A clutch of aides and outside allies, from Mr Netanyahu to hawkish journalists, now seeks to convince Mr Trump that a limited round of strikes on Iran's energy infrastructure will do the job. That almost certainly misreads how Iran would respond: it is more likely to hit back hard at its Arab neighbours than to capitulate.

Asked on May 19th how long he would wait, Mr Trump was muddled. “Two or three days. Maybe Friday, Saturday, Sunday, something. Maybe early next week,” he said. The calendar is complicated. The *hajj*, the annual Muslim pilgrimage to Mecca, will span the last week of May. The Saudis have begged Mr Trump not to restart a war that could leave thousands of pilgrims stranded. Less than two weeks later, America will be co-hosting the football World Cup along with Canada and Mexico. Some Washington hands think Mr Trump will not want to spoil the party (others doubt he will care).

The president may not wish to resume the war. But the longer Iran remains intransigent, the more it looks like a question of when, rather than if, he will do so. ■

Europe



European defence

The secret plan to replace NATO

HELSINKI, KEFLAVIK AND STOCKHOLM

Europeans are quietly working out how to fight if America quits the alliance

SOLDIERS OF the “Black Jack” brigade ritually furlled and packed their unit’s colours in Fort Hood, Texas in early May, as the tank unit’s 4,000 troops prepared to deploy to Poland to help defend NATO against the Russian threat. “When an armoured brigade combat team deploys forward, it sends a clear and unmistakable signal,” said General Thomas Feltey, the division’s commander, at the ceremony. Less than two weeks later America sent the opposite signal: the deployment was scrapped. It was the second time this month that Donald Trump had announced cuts to America’s military presence in Europe, reflecting his anger at the lack of European support for his war in Iran. On May 22nd America is expected to announce a reduction in the forces it pledges in the event of an attack, Reuters reports.

Mr Trump has been casting doubt on his commitment to NATO and its Article 5 mutual-defence clause since the start of

his second term. That has prompted a long-overdue increase in European defence spending. Yet in recent months he has gone further, announcing unexpected troop reductions and cancelling the deployment to Germany of a cruise-missile unit that was to plug an important gap in Europe’s defence. The rapid drawdown has upended Europeans’ assumption that they would have time to build up their own forces and replace vital American “enablers”, such as intelligence and surveillance assets. America’s huge expenditure of missiles in Iran is delaying shipments to Euro-

pean allies and Ukraine, as it restocks.

Some in NATO, shocked by Mr Trump’s threat in January to seize Greenland from Denmark, worry not only that America might sit out a war with Russia, but that it could actively thwart other members’ responses. The possibility is seen as remote. But interviews with senior officers and defence officials from several NATO countries reveal for the first time how seriously they take the risk. Some European armed forces are making secret plans to fight not just without America’s help, but without much of NATO’s command-and-control infrastructure. “The Greenland crisis was a wake-up call,” says a Swedish defence official. “We realised we need a Plan B.”

None of the officials interviewed would speak on the record, because of concerns that doing so could accelerate America’s departure. Mark Rutte, NATO’s secretary-general, “has literally banned talking about it because he believes it can add fuel to the fire”, says one insider. When Matti Pesu of the Finnish Institute of International Affairs (FIIA) co-authored a paper last year arguing for a Plan B, Finnish officials denied one would be considered. But the urgency of the threat has led several countries to start thinking about how, and under whose command, Europe would fight if NATO were to “malfunction”, as one official put it. “What chain of command can

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▶ you use if America is blocking NATO?” asks another defence official.

The question cuts to the core of the alliance's success. Most military coalitions look like a primary-school music practice: each country turns up, bangs its drum roughly in time with the others, and leaves. NATO, by contrast, was set up as a symphony orchestra controlled by a single conductor, the Supreme Allied Commander Europe (SACEUR), an American general who also commands America's forces in Europe. To conduct this orchestra, SACEUR has secure communications links to a network of permanent subordinate headquarters (see map), staffed with thousands of personnel ready to respond the moment a war starts. “US leadership is the glue that holds the alliance together,” says Luis Simón, the director of the Centre for Security, Diplomacy and Strategy at the Free University of Brussels. “Without them, we would see a fragmentation, probably, of the deterrence ecosystem.”

Thus a Plan B requires more than acquiring weapons; it means creating a structure under which Europeans would fight. The core, at least in northern Europe, would probably be a coalition of Baltic and Nordic countries, plus Poland. These countries mostly share common values, and all fear Russia. Several of NATO's bigger European members, such as Britain, France and Germany, have “tripwire” forces in the Baltics, and are thus very likely to be drawn into any conflict. Perhaps one-third of NATO members would “fight on day one” irrespective of whether Article 5 is triggered, says Edward Arnold of RUSI, a think-tank in London. “No one would be waiting for the Portuguese to turn up at the North Atlantic Council [NATO's highest decision-making body] to debate,” he says.

One often-mentioned alternative command structure is a British-led coalition of ten mostly Baltic and Nordic countries known as the Joint Expeditionary Force (JEF), with a headquarters near London. Established by Britain and six other NATO

members in 2014, the JEF was originally seen as a complement to the larger body that could provide high-readiness forces on short notice for circumstances that did not meet the Article 5 threshold. Its remit expanded when Sweden and Finland joined the coalition in 2017, several years before they applied for membership in NATO. It is now seen as a way to sidestep one of NATO's weaknesses: any member can block the triggering of Article 5, which requires a unanimous decision. The JEF, as its then commander, British Major General Jim Morris, said in 2023, “can react to situations on a non-consensus basis”. It has already been activated several times, for exercises and naval patrols.

“The JEF is the most established of the alternatives,” says Mr Arnold. Its headquarters already has capabilities in intelligence, planning and logistics, he notes. It has its own secure communications networks that, although limited, do not rely on

NATO. Britain's membership offers a degree of nuclear deterrence.

Yet the JEF focuses primarily on the Nordic and Baltic regions. It lacks major powers such as France, Germany and Poland. Some officials are anxious about Britain's defence preparedness: underfunding has left it with few ships, submarines and army units ready to deploy at short notice. “England is everyone's favourite uncle,” says one official. “But it is suffering from Downton Abbey syndrome. It keeps up the pretence, but it doesn't have the funds.”

Such problems might be mitigated if the group brought in Germany, which is enormously increasing its defence budget. For all its drawbacks, the JEF seems the best solution if European members are unable to take over the existing NATO framework. But Europe will find some form of defence framework to replace the Americans. A deterrent based on someone who may not show up is no deterrent at all. ■

Bulgaria and Romania

The wild south-east

BUCHAREST AND SOFIA

Corruption and inflation are boosting Eurosceptic populists

WHEN PETER MAGYAR beat Viktor Orbán in Hungary's election in April, most European leaders were relieved. “For the first time in years, there are no Russians in the room,” joked Donald Tusk, Poland's prime minister, at a European Union summit that month. Look to the south-east, however, and they have less to cheer. On May 8th Rumen Radev, a Russia-friendly populist, was sworn in as Bulgaria's prime minister after winning an election in late April. On May 5th Romania's liberal prime minister had been forced out in a no-confidence motion that was backed by both the centre-left and a rising nationalist-right party.

Might Bulgaria and Romania replace Hungary as the EU's awkward squad? Economic malaise plagues both countries. Last year prices rose by 3.5% in Bulgaria and 6.8% in Romania, above the EU average of 2.5%. Growth is strong in Bulgaria, but Romania's GDP was 1.7% lower in the first quarter than in the same period last year. This pushes voters towards populists.

Mr Radev, a former air-force commander, had served since 2017 as president, a mostly ceremonial role. He campaigned for prime minister on an anti-graft platform, vowing to dismantle Bulgaria's “oligarchic model of governance” and restore judicial independence. He linked the country's corruption, among the worst in

the EU according to Transparency International, a Berlin-based monitor, to its income inequality, the highest in the bloc.

After eight elections in five years, Bulgarians were looking to break the gridlock. Mr Radev's Progressive Bulgaria party won nearly 45% of the vote, crushing the corrupt parties of power and established anti-corruption outfits alike. He will be the first prime minister in decades with an outright majority in parliament. “Fatigue stemming from the lack of a single party offering a clear alternative” played a role, says Daniel Smilov of the Centre for Liberal Strategies, a Bulgarian think-tank.

The new government quickly demonstrated its populist bona fides. On May 11th it proposed “fair-price” benchmarks for consumer goods, forcing retailers to justify price increases or face fines. Economic populism runs through Mr Radev's approach to Russia as well. The prime minister claims that sanctions “harm the economies of Russia and the EU” and wants to increase imports of Russian fossil fuels (which the bloc wants to end by 2027). Europe, he argues, should stop selling arms to Ukraine and push for a quick peace deal. Dmitry Peskov, the Kremlin's press secretary, said he was “impressed”.

Romania's centrists are on the back foot, too. A year ago Nicusor Dan, a liberal anti-corruption campaigner, beat George ▶▶



► Simion of the nationalist Alliance for the Union of Romanians (AUR) in a presidential election. But the government he appointed lacked strong support in Parliament. For months it fought to cut Romania's budget deficit, the largest in the EU. After it passed a slimmed-down budget, the centre-left Social Democratic Party (PSD) left the coalition and teamed up with AUR to bring down the government.

The Social Democrats' patronage system has long channelled funds to local officials who drum up support for their party. They and AUR now present themselves as austerity's opponents. "The so-called pro-Europeans have delivered nothing but taxes, war and poverty," crowed Mr Simion after the no-confidence vote. His party is Romania's most popular, For Free Mags Check sastatus.com with support at 33% in polls (ten points ahead of the PSD).

Should the EU worry about a new Orban-style sub-bloc? Mr Simion has little love for Russia, but opposes aid to Ukraine, which he claims mistreats its ethnic Romanians. Should he win power, he could upend plans for Ukraine to build drones in Romania and disrupt the country's role in training Ukrainian fighter pilots. But a snap election that could allow Mr Simion to form a government is unlikely. Mr Dan says he wants a new coalition instead, perhaps led by a technocratic prime minister.

Mr Radev too may prove less troublesome than his words suggest. Bulgarian defence firms have boomed since the start of the war, selling Soviet-style munitions to Ukraine. Rheinmetall, a German armsmaker, is planning a €1bn (\$1.2bn) shell factory in Bulgaria, financed partly by an EU scheme. Mr Radev will hardly want to stifle the defence industry with an arms embargo on Ukraine. "Bulgaria sees EU countries as its natural family, but reserves the right to maintain its own positions," says Ivo Hristov, a deputy prime minister.

On corruption, Mr Radev talks a good game. He has pledged quick reforms of the country's judicial system and dismissed the deputy head of the national-security agency for alleged political meddling. He also plans a parliamentary anti-corruption commission. But fighting corruption is a long-term institutional commitment; one election victory may not change much.

Perhaps most worrying for Europeans in these days of transatlantic tension is his affinity with America. Mr Radev, a graduate of an American air-force fighter-pilot programme, often touts his American connections. His realist foreign-policy views have much in common with those popular in MAGA-world, and his desire to end the Ukraine war and get back to trading with Russia would go over well with J.D. Vance, Donald Trump's vice-president. Not quite "Russians in the room", then. But, for the likes of Mr Tusk, they are nonetheless too close for comfort. ■

Real Madrid

Pitch battle

MADRID

Europe's most powerful football boss faces a challenge

EVEN REAL MADRID's many detractors would concede that under Florentino Pérez, its president for most of the past 25 years, it has been the world's dominant football club. In that period it won each of the European Champions League, the World Club Championship and the Spanish league seven times. Its latest revenues of €1.2bn (\$1.4bn) are ten times higher than in 2000.

But the team has become dysfunctional, with no grand trophy for two years. This month two players quarrelled twice during training and were each fined €500,000. A recent spineless surrender to Barcelona gifted their arch-rivals the league title. And fans have taken to booing Kylian Mbappé, the team's highest-paid player.

Mr Pérez, a 79-year-old construction magnate, has faced other setbacks. His plan for a European league was squashed. An expansion of the Bernabéu stadium, originally budgeted at €575m, has cost €1.3bn and counting. On May 11th Mr Pérez suddenly called his first press conference in more than ten years. He gave a rambling and paranoid performance, attacking the press, referees and a campaign "in the shadows" against him. Only 16 months into a four-year term, he announced a fresh election for the presidency.

Real Madrid is a member-owned, non-profit organisation. Mr Pérez doesn't derive any direct financial benefit from it. But the prestige and other indirect benefits are huge. The club is one of Spain's most important institutions. "The presidential box at Real Madrid is one of the most coveted places to be," says a Spanish banker. "It's a good conduit for business."

Previously unopposed, Mr Pérez faces a rival in Enrique Riquelme, who owns an energy business. He is linked to Ignacio Galán, also an energy-firm chairman and an old foe of Mr Pérez. Under rules set by Mr Pérez, challengers must present by May 23rd a slate of directors and bank guarantees equal to 15% of club expenses. Mr Riquelme says he has the money but complains about the timing.

The club's future may thus be in the hands of its 100,000-odd members. Membership is cheap, at €179 a year, though a season ticket for matches costs up to €3,200 on top. There is a long waiting list. Members are not necessarily active fans: some sell their tickets on. Such is Mr Pérez's influence that few in or outside the club are prepared to call openly for his departure. Mr Riquelme's bid may fail. But he has put down a marker. Should he not win this time, he will no doubt get a rematch.



Hardly a banner year

Digital sovereignty

Bytebreak

BERLIN

Europe is trying escape the lock-in of America's tech giants

LOOK AT Europeans' software budgets and shows why they are worried about dependence on America. Germany's federal government pays almost half a billion euros a year in licence fees to Microsoft, an American software firm. A rough calculation by a French business association finds that large French companies buy more than \$50bn in software and cloud services annually from Uncle Sam's tech giants. Euro-zone imports of intellectual property services from America have ballooned to \$200bn a year and counting (see chart).

That is not surprising. Cloud and artificial-intelligence services could prove more transformative than online shopping, social media or internet search. And the four biggest American cloud and AI firms—Amazon, Google, Meta and Microsoft—are miles ahead of Europe. German politicians crow about a new €1bn (\$1.16bn) data centre near Munich; America's big four invested over 350 times that in 2025 alone. Of almost 100 notable AI models released in the past year, according to Epoch AI, a research firm, only one was from the European Union. The grip of American tech is, if anything, growing tighter.

Policymakers are keen to change that. On June 3rd the EU will unveil a tech-sovereignty package, including a cloud and AI development act. In April France announced it was switching all government computers from Windows to Linux, an open-source operating system originally from Europe. Germany is likely to task domestic firms with setting up a cloud for administrative data, and its domestic intelligence service has opted for ArgonOS, a French data analytics firm, over America's Palantir. Businesses are also diversifying. In 2022 most firms surveyed by Accenture, a consultancy, said they only considered American cloud providers. That share dropped to less than 20% in 2025.

Europeans have three big worries. The first is that sensitive data and services may not be safe. America's Cloud Act gives its government the power to request data from tech firms even when hosted by a subsidiary abroad. Some fear that America might wield tech as a geopolitical weapon, in the form of a kill-switch that can turn off services. That scenario is somewhat cartoonish, says an EU insider, but the dependence is real. Karim Khan, the chief prosecutor of the International Criminal Court, lost access to his email after being placed

under sanctions by America.

American tech giants and big cloud service providers, known as hyperscalers, respond to such fears with what they term "sovereign offerings". Microsoft promises European users never to cut them off, and to fight American data requests in court. Google has an "air-gapped" cloud with no connection to the public internet for clients with high security needs, including one of its newest customers, the German armed forces. But critics call such offers "sovereign-washing", as the firms are ultimately under American control. Europe needs a layered approach, argues Topi Manner, CEO of Elisa, a Finnish telecoms and cloud provider: "The most sensitive data [should be] kept in Europe, in highly secure data centres, and the big hyperscalers for the less sensitive parts."

A second worry is that American digital services spread harmful content, and dominate markets to the detriment of European businesses and media. The rise of AI has intensified that concern, as users turn to ChatGPT and its competitors for advice on everything from shopping to politics.

Europe has countermeasures, but they are controversial and slow. The bloc's Digital Markets Act (DMA) and Digital Services Act (DSA) let it punish anti-competitive behaviour and force tech giants to change their services. In April alone, the European Commission sent preliminary warnings to Meta, for breaching age restrictions, and to Google, demanding that it grant third parties access to search data. Other investigations are under way. One concerns whether Google's and Microsoft's cloud services are governed by the DMA—if so, it could

require them to unbundle their services.

Such measures may create a bit more room for competitors, but are unlikely to make Europe much less dependent on America. Fear of alienating America and putting NATO at further risk looms. Mario Draghi, a former chairman of the European Central Bank, argued this month that the EU must become more assertive, but "what is holding us back is security."

The third reason for Europe's digital sovereignty drive is growth. Europe is losing the technological race against America and China, including in AI, which reaches into many sectors where Europe remains strong. Building domestic providers is seen as necessary to stay competitive.

One question is whether users will be able to switch. Hyperscalers provide integrated services and the scale to make them efficient. "That bundling of services then leads to lock-in effects," says Svea Windwehr of D64, a digital-progress association in Germany: users cannot count on easily taking data with them. Quality and cost also matter. A survey of Danish firms, which use clouds more than any others in Europe, found that their top concerns (after security and legal compliance) were quality, ease of use and price.

That often argues for American providers. So does the fact that European firms' customers tend to use American tech, too. "For many industrial applications," says a European industrial executive, "there are no capable alternatives to software-as-a-service from American providers."

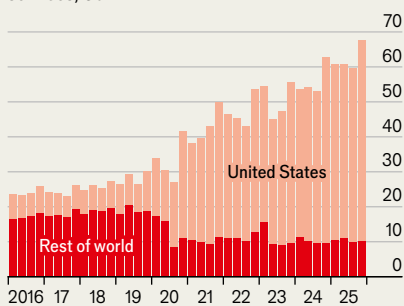
Chicken, meet egg

"Europe's challenge is one of industrial co-ordination and animal spirits," argues Georg Riekes of the European Policy Centre, a think-tank in Brussels. There is demand for sovereign solutions, industry representatives say. But building an ecosystem, from chips to data centres and services, is hard. America's strong economic momentum makes it harder still. Anne Hidma of ASML, the Dutch-based company that leads the world in the machines that make chips (but which sells only 1% of its machines in Europe), says Europeans start with their strong suit, innovative technical ideas. Then they see whether there is a market for them. In contrast, "America starts from the end market and builds its ecosystem from there," she says.

To get things moving, the EU wants to help build large data centres and strengthen the supply chain for chipmaking. The forthcoming package will probably include faster permitting procedures, some public funds to support investment and procurement rules that prioritise European suppliers. The hope is that this will create demand both for chip design and manufacturing, and for cloud and AI services made in Europe. It is a tall order. ■

Cloud cost

Euro area, imports of intellectual-property services, €bn



Source: ECB

CHARLEMAGNE

The long way back to Brussels

Bre-entry may be the next drama to grip the European Union



PICTURE THE scene, some time in the 2030s. Three dozen leaders of the European Union's member states crowd around a gilded desk in some history-laden hall—perhaps in Versailles, Sarajevo or Rome. The Spanish prime minister signs a thick vellum document, before passing the Mont Blanc pen to the German chancellor, then the Ukrainian president. The mood is jovial as, one by one, signatures are affixed on “The Treaty of Accession of the United Kingdom to the European Union (version 2.0)”. As the pen makes its way to the French president, she seems to waver for a second, perhaps recalling past tabloid headlines about bendy bananas and budget rebates. Finally it is the turn of the British prime minister of the day. In one stroke Brexit is reversed; the only member to have left the EU's embrace is now safely back inside. Amid the sound of champagne corks popping, few can make out the words of an elderly gentleman in the public gallery, one Nigel Farage, fulminating vaguely about the will of the people. *Welcome back, Britain! The prodigal nation returns!*

What would once have seemed like a Remainer fever dream no longer feels so absurd. For the first time since the Brexit referendum of June 2016, the idea of asking to return to the fold is on Britain's political horizon. Granted, for now the idea of Bre-entry is confined to Labour politicians, floating the notion to charm party members ahead of a possible challenge to Sir Keir Starmer; several hope to become prime minister in his stead. Granted, too, that the euro-bashers of Mr Farage's Reform UK party are far ahead in the polls. But rejoining the EU seems to resonate with many British voters, a majority of whom think leaving was a mistake. The notion is unlikely to go away soon. As more and more Westminster politicians declare they favour applying for re-entry, the EU may one day need to say what it thinks of the idea of Britain boomer-ang back into the union it left in a huff.

The current state of Britain-EU relations offers nary a hint of what response might emanate from Brussels. Since the divorce was officialised in 2020, the two sides have haggled over relatively small fry, such as aligning rules on food safety and whatnot. That is the purview of mid-level Eurocrats. In contrast, a bid for Bre-entry would be taken up by the EU's national leaders in Paris, Berlin

and beyond. Plenty would see the case for Britain to rejoin. Any union should welcome a G7 country with a seat on the UN Security Council, its own nuclear weapons and the continent's biggest capital-markets hub. The same factors that are prompting some in Britain to discuss a return—a chaotic global order, in which America and China vie for supremacy and leave middle powers scrambling for security—should also sway the EU towards Bre-entry. Across the union, voters say they would welcome Britain back.

Pas si vite, some diplomats might retort. Britain was always a half-hearted member of the EU. It insisted on opt-outs from some of the bloc's flagship measures, notably the euro and the Schengen passport-free travel area, not to mention demanding rebates on its contributions to the budget. Why bring the awkward squad back into a union they think has worked just fine without it? Since Britain's departure, the EU has pushed forward with federalising schemes that British prime ministers would no doubt have proudly kiboshed at Brussels summits. EU spending is now funded in part with money jointly borrowed by its 27 member states, surely a taboo in London. The union has adopted French ideas on industrial policy and “strategic autonomy”, in part thanks to not having to worry about British vetoes.

In theory, Britain's reaccession bid would be treated no differently from those of other countries, from Ukraine to Serbia, which have formally asked to join the EU. In practice it would be both easier and harder. As a past member whose regulation has not much diverged from the *acquis communautaire*, the hundreds of tomes of EU rules that all prospective joiners must accept, Britain would jump through hoops faster than most. The tough bit—apart from dozens of national governments each wielding a veto over the final decision—is that some might insist Britain could join the EU only if it also agreed to adopt the euro and Schengen. This could serve as a test of how seriously the aspiring newcomer *really* wants to be part of the club this time. In practice it would be easy enough for Britain to commit to join either one “when the right conditions were met”, only for said conditions never to be met. The budget rebate Britain once enjoyed would no doubt have to go, which might have happened anyway had it remained in the EU.

Breturn to sender

The main reason for reluctance about Bre-entry on the EU side is a niggling doubt as to whether Britain can ever be counted on to stay in. To have a member leave the union once may be regarded as a misfortune; for it to happen again would look like carelessness. In practice, says Charles Grant of the Centre for European Reform, a think-tank in London, the EU would want to see broad agreement across party lines in Britain before discussing re-entry terms, so that tomorrow's ministers did not unpick a deal reached by today's. That might entail the Conservatives backing a return (far-fetched for now), or Mr Farage's political fortunes waning.

Even then patience would be in order. Britain's initial bid to join the EU in 1961 succeeded only after 12 years (and despite two French vetoes). A similar timeline would imply more than a generation had passed between the 2016 vote and Bre-entry, a suitable timeframe for a nation to change its mind. The journey would be long, tortuous—and worth it. Brits would get to unpick an error that has left them poorer and more isolated. For the EU the prize may be even greater: to establish their union as the continent's manifest destiny, complete with chastened British officials once again haggling late into the night over fishing quotas. ■

Britain



The Premier League

World champions

Many things in Britain aren't working. Football is

BUS CONDUCTORS in Nigeria are typically Chelsea fans. In Goma, a city under the control of a rebel group that opposes the Congolese state, lots of bootleg Newcastle United shirts are on sale (the Geordies have a striker, Yoane Wissa, who plays for the national team). Tiny villages alongside Lake Victoria in Kenya may have no electricity and barely any roads, but they do tend to have a pub—a shack with solar panels on the roof, a fridge to cool beer and a small television with some way of receiving Premier League games. The FancyFree pub in Brooklyn overflows with Arsenal fans when there is a game.

On May 19th Arsenal clinched the league title. The presidents of both Kenya and Rwanda took to social media to congratulate the team. In Botswana the government had to debunk fake news about a bank holiday for Arsenal fans.

The English Premier League is a global cultural powerhouse. It is broadcast to 191

of the UN's 193 member states. More than 700m people may tune in to watch a single game between top teams. The American Super Bowl attracts just a third of that.

According to Google's worldwide-trends data, last year more people searched for Manchester United than for Taylor Swift and the Harry Potter book series combined. In 2024, the year of an American presidential election and an assassination attempt on Donald Trump, the Premier League was still a more popular

search topic than Mr Trump. In the past year it has beaten the Bible.

It seeps into language and politics, too. When South Koreans talk of someone's "Leeds days" (or golden age), they are using a football metaphor. In Sweden's parliament a former finance minister recently used the word "Spursy" (messing up despite being well placed to succeed) to warn that Sweden should not perform like Tottenham Hotspur. In Ghana an MP cruelly referred in 2022 to Harry Maguire of Manchester United, calling the vice-president an "economic Maguire"—always scoring "own goals".

The Premier League is not the world's biggest sports-league business: America's National Football League is far bigger, as are the major leagues for basketball and baseball. Unlike American franchises, where fans are richer and owners do not face the risk of relegation, Premier League clubs struggle to make money. Last year only four of its 20 clubs turned a pre-tax profit. Many, like Abu Dhabi-owned Manchester City or Saudi-owned Newcastle United, are kept afloat by foreign investors who splash out for prestige rather than profits. More than half of Premier League clubs are now owned or part-owned by Americans, whose motives tend to be more straightforwardly financial. Nor are English teams the most successful. Spanish ►►

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— Ungovernable Birmingham

► teams have won far more Champions League titles, the most prestigious European competition.

But this year English sides have reached the finals of all three major pan-European contests (Aston Villa won the Europa League on May 20th; Crystal Palace compete for the Conference League title on May 27th; Arsenal face Paris Saint-Germain on May 30th in the Champions League). Strikingly, in revenue terms the Premier League has in recent years comprehensively outperformed continental rivals, according to data compiled by Deloitte, a consultancy (see chart 1). EY, another consultancy, calculates that the league generates £10bn (\$13bn) a year in gross value added to the British economy. The Premier League is one bit of Britain that still rules the airwaves.

This was not inevitable. The modern game was invented in England in the 19th century and exported all over the world. But by the 1980s things were grim, violent and declining: English clubs were banned from European football for five years after Liverpool fans caused a crush that killed 39 people. Continental leagues—Italy's Serie A, Spain's La Liga—had more glamour and superstars. The English game cleaned up its act in the 1990s, banning drinking in the stands and clamping down on racism, and has been growing in popularity, at home and abroad, ever since. Stadiums, many new or revamped, are full every week.

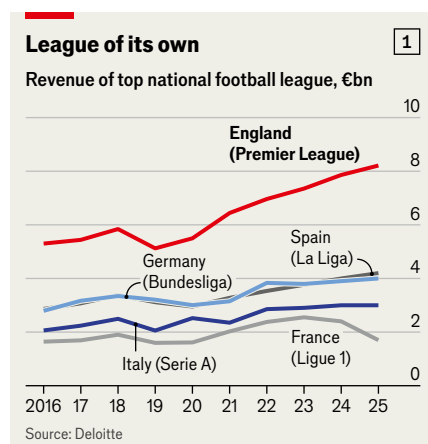
A winning hat trick

The passion of fans has proved infectious abroad. Manchester United alone has 362 official fan clubs across 94 countries. A member of the Chennai branch, Rathana Ganth, a 31-year-old working in digital media, watches every United game. When asked why, he refers to the team's managers from its glory days: "For Sir Alex [Ferguson]...for [Matt] Busby." Even in cricket-mad India, at least 70m people watched the Premier League in the 2023-24 season.

"Broadcast revenue is a proxy for following," says Kieran Maguire, a football-finance expert at the University of Liverpool. "And the Premier League generates twice as much from that source as La Liga, the Bundesliga and Serie A."

Three main forces explain this success: exports, imports and competition. On the export front, the Premier League benefited from first-mover advantage. For the league's first eight years from 1992 it made no profit on broadcast rights awarded to subscription channels globally, and built up fan bases around the world. "By the time other European leagues realised that there was a buoyant overseas market for live football," notes Mr Maguire, "the Premier League had established itself as a dominant player."

About half of the Premier League's rev-



enue comes from television rights. And the majority of TV revenue comes from foreign rights, according to data from Ampere Analytics, a research firm. The NFL, by comparison, gets 98% of its media-rights revenue at home (see chart 2).

Imports—not only of owners but of coaches and players too—are also key. Of the 20 clubs, 14 have foreign coaches, among them the league's most successful: Pep Guardiola at Manchester City and (now) Mikel Arteta at Arsenal. No English manager has ever won the Premier League.

Some 75% of the league's minutes this season were racked up by foreign-born players, compared with 62% in Germany and 44% in Spain. In total, 128 countries have been represented in England's top flight, from Iceland to Togo and Suriname to Venezuela. In the Netherlands' last competitive game, nine of the starting players played their club football in England. True, England's captain, Harry Kane, now scores his goals at Bayern Munich (a stunning 58 so far this season), and a potential successor, Jude Bellingham, plays

for Real Madrid, but the foreign stars playing in England bring with them a big following of fans from their home countries.

These players do not come cheap—which helps explain the clubs' precarious finances. The cash mostly goes into players' pockets in what a former Spurs chairman, Sir Alan Sugar, called the "prune-juice effect": in one end and out the other. Last season clubs spent, on average, 65% of their revenue on wages. Outside the "big six" elite clubs, that rises to 76%.

Competition, the third factor, also sets the Premier League apart. As with its rivals, a handful of the largest clubs tend to dominate, but less monotonously than elsewhere. The talent, and the money, are spread more evenly.

Crucially, the Premier League has redistributed its TV riches more equitably than most others. Spain has long diverted the lion's share of cash to the biggest two or three clubs. In England the pie is not just larger, but smaller clubs get a bigger chunk of it. Every team got more than £100m in TV cash in 2024-25.

Promoted clubs can now spend heavily. Sunderland, newly up, spent over £170m (\$230m, or €200m) and is safely in mid-table. By contrast, in Spain last year's promoted clubs—Elche, Levante and Real Oviedo—spent €17m between them last summer, according to Transfermarkt, a football-statistics website.

Dribble-down economics

Some of the fears about the knock-on effects of the Premier League's success have not materialised. England's extensive lower-league football ecosystem has not been killed by the big money going to the 20 clubs at the top (see next article).

What could go wrong? From time to time the threat of a breakaway by top teams to form a European Super League has caused concern, only to fade again. Scandals over rule-breaking spending could damage the league's reputation. So could controversy over the sources of the money behind some of the club owners or sports-betting sponsors—of the sort that forced Roman Abramovich, a Russian oligarch, to sell Chelsea in 2022. To mitigate those risks, the Labour government has established an Independent Football Regulator (IFR). It is currently taking soundings on a licensing regime for clubs, which it aims to have in force for the start of the 2027-28 season. The IFR itself calls this "the biggest change to the governance of English club football since the creation of the Premier League in 1992".

How it affects the sport remains to be seen. An overly zealous regulator could well make the league less vibrant. Yet for now, from the buses of Lagos to the bars of Brooklyn and beyond, the Premier League holds astonishing sway. ■



The lower divisions

Fever pitch

BRISTOL

Why football attendance is booming outside the Premier League

IT IS A cutting insult in certain parts of the world. “You’re just a shit Port Vale,” chanted the Bristol City fans as they celebrated a 2-0 win on the final day of the Championship season on May 2nd. Their words, and rude hand gestures, were directed towards an away end of the stadium full of Stoke City supporters—whose local rivals, Port Vale, had just been relegated to League Two despite a fairy-tale FA Cup run this season.

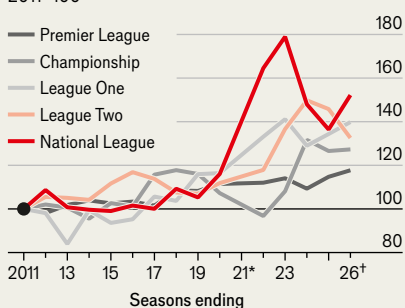
There are four professional men’s leagues in English football: the Premier League, the Championship, League One and League Two. The Premier League has the global glamour, but it accounts for only 41% of the more than 38m professional match-day tickets sold every year in England. The lower divisions are where attendance is growing fastest (see chart).

Over the past ten years attendance per Premier League game is up by 14%. In League One, as the third tier is confusingly known, it is up by 47%. The National League, the fifth tier of English football, home to a handful of semi-professional clubs, has seen a rise of 50%. In two of the last three seasons the Championship (the second tier) has been Europe’s second-best-attended domestic football league—behind only the Premier League.

Compared with the Premier League (where stadiums this season have been 97% full) the lower tiers have plenty of supply, and are far cheaper. Your correspondent’s team, Tottenham Hotspur, charge an average of £76 (\$102) for a ticket in their swanky new stadium. For the privilege of buying one, you must first fork out £50 for club membership. A ticket to Bristol City costs £20, and requires no sign-up.

Crowd sourcing

English football, average match attendance 2011=100



Source: Transfermarkt *Excluded due to covid-19 †To May 2025

It’s not all a matter of economics, however. “There’s a lot in this world that’s isolating,” says Ant Banks of the Forever Bristol City Podcast. Lower-league grounds, throwbacks to a grass-roots era of English football, are anything but. “People are looking for their tribe,” he says.

After the game Bristol City’s manager, 78-year-old Roy Hodgson, joined his players in a gentle lap of honour around the pitch, applauding the fans for their support. Their team had finished comfortably in mid-table. And the joyous involvement of Mr Hodgson, better-known for his four-year stint in charge of England’s national side, reflects the enduring attraction and

romance of the lower leagues.

A lot has changed since Mr Hodgson began his managerial career in 1976. One recent development, and a topic of endless debate among football commentators and cognoscenti, is the introduction of video assistant referees (VAR). Premier League games are increasingly decided by often controversial refereeing decisions made miles from the stadium, by officials watching a monitor. Critics argue that it slows momentum and blunts the emotion of football’s defining moment: a goal. Another reason people are attracted to non-Premier League football? One fan states it bluntly: “There’s no VAR.” ■

Manchester City

Playing by the rules?

The legal case hanging over the Premier League

IN THE OTHERWISE sunny story of the Premier League there is one big cloud. In 2023 Manchester City, one of England’s most successful clubs, was charged with well over 100 breaches of the league’s rules. Some of City’s alleged transgressions (all resolutely denied by the club) relate to non-co-operation with league officials. Most deal with failures to accurately report the club’s accounts, thus avoiding “financial fair play” regulations that could have curtailed the spending—over £1.9bn (\$2.5bn)—that took City from mediocrity to dominance. In the 14 seasons covered by the charges, Manchester City won seven Premier League titles.

To adjudicate the case the Premier League has employed an independent commission of three anonymous judges. The hearing took place behind closed doors in December 2024. A lengthy judgment is still being prepared. The long wait for a verdict has caused frustration and fuelled conspiracy theories among fans. Chief among them are unsubstantiated suggestions of diplomatic interference (Manchester City are owned by Sheikh Mansour, the vice-president and deputy prime minister of the United Arab Emirates).

In recent years Everton and Nottingham Forest have received points deductions (of six and four points respectively) for single breaches of the league’s financial rules. If City were to be found guilty, Premier League rules mean they could face a fine, a points deduction or even expulsion from the league. Given the sheer number of charges, the scale of



A time of trophies

any potential punishment would almost certainly invite an appeal and legal challenges from the club. If City are cleared, receive a light punishment or strike a favourable plea bargain, fans and clubs alike will question the sporting integrity of the competition. It is without doubt the most consequential case in English football’s 163-year history.

An initial decision, rumoured for the summer, is unlikely to put the matter to bed. Any sanctions would have to wait until after the inevitable appeal. That could easily take another year. This looks destined to stretch into extra time plus plenty of injury time.

BAGEHOT

Makerfield and the new politics of paradox

Hate Labour? Vote Labour!



IF YOU VOTE Reform in three weeks' time," said the Labour activist, "Keir Starmer will still be prime minister." It was a threat. Eileen, a Reform UK voter, from Winstanley in the constituency of Makerfield, was laying out her problems with the current Labour prime minister. The only way of removing him, said the Labour activist, was to vote for Andy Burnham, the Labour candidate who intends to win the seat on June 18th and replace Sir Keir as prime minister soon after.

"I don't believe you," replied Eileen.

Why would she? It is absurd. A Labour activist begging a Reform voter to back Labour to give Sir Keir the boot? Pull the other one. Yet the by-election in Makerfield, a collection of towns and suburbs in a former mining area just outside Wigan, is riddled with such oddities and contradictions. The battle for Makerfield has already had commentators reaching for "The Road to Wigan Pier", George Orwell's journey into working-class life in the area in the 1930s. In fact, "1984" is a better guide to the current situation, where paradox and doublethink are the order of the day and Labour offers a very strange pitch: Hate Labour? Vote Labour!

Politics in Makerfield is upside down and inside out. Words take on their opposite meaning. After the 2016 Brexit vote, one writer broke Britons into "Somewheres", who were rooted, and "Anywheres", who were mobile elites. Candidates boast of local connections. Yet Makerfield is "not really a place", in the words of Josh Simons, its former MP who stepped down explicitly so Mr Burnham could challenge Sir Keir. It is a sprawling, awkward map born out of electoral convenience rather than geography. This current emblem of Somewhere is nowhere in particular.

Terms such as "left behind" have become contorted and no more so than in Makerfield. Parts of it are bashed up; most of it is fine; some of it is flourishing. If Makerfield is left behind, so is the rest of Britain. In 2025 median weekly wages were £762 (\$1,028), or a fraction below the national figure. A child in Makerfield is no more likely to be in poverty than one picked at random from the rest of England. Their chances here are often better. Winstanley College, a local sixth-form college, is one of the best in the country and sends as many students to Oxbridge as Dulwich College

in south London, a public school that charges £32,000 a year.

When Mr Burnham says Makerfield and its ilk have been "long forgotten", he really means the opposite: they are impossible to ignore. In truth, British politics has catered to their supposed whims for a decade now. Their demands, whether departure from the European Union or drastic cuts to immigration—overseen by the tail end of the last government and by this one—have been met. No matter. In the apogee of post-Brexit politics, a collection of northern towns will alone decide the fate of the government. If Mr Burnham wins, the "king of the north" expects a coronation in Westminster; if he loses, Labour endures its own version of "The Anarchy". The "long forgotten" decide which path he takes.

And why not? It is, after all, a Labour "heartland". The area has returned Labour MPs since 1906, when the party first emerged. Except the term "heartland" has inverted: often it is a place where no one votes for you any more. The Tory heartland of Oxfordshire now votes Lib Dem; the Labour heartlands of the north-west and north-east voted in reams of Reform councillors in the local elections on May 7th. Labour is in danger in Makerfield, but it is in danger everywhere, from Anywheretown to Somewhereville. Yet Makerfield matters more, if only in the heads of Labour MPs.

Today geopolitical realities come second to Makerfield. Mr Burnham, who was once rather bombastic about rejoining the EU, now dismisses the idea of rejoining any time soon, in order to pander to these voters. Yet believe the polls and Makerfield, a "heartland" of the Leave vote where two-thirds voted out, would vote to stay in the EU if the vote was re-run. Ten years is a long time and demography—Leave voters were overwhelmingly older—is a powerful thing. Politicians have grown used to the old labels. A place that was once Leave is Leave for ever, opinion and actuarial tables be damned. Maybe the voters of Makerfield are actually being ignored this time.

Perhaps the most fateful figures in the politics of paradox are non-voters who vote. They are electoral dark matter, weighing heavily on British politics, shaping it invisibly. When they do emerge, as they did in 2016, they can mould it for a generation. At the local elections this month in the area, they turned up. Hundreds of people on estates who usually never bother turfed out long-standing Labour councillors and ushered in Reform ones. Of the 25 seats up for election on Wigan council, Reform won 24. Whether they swell booths again in June will determine the next occupant of Downing Street.

He's a good northern lad/His eyes are big and sad

That depends on what they make of Mr Burnham. Who better than he to rule the politics of paradox? He personifies it. A man who decries 40 years of misrule was in government—whether as a bag-carrier or high-flying minister—for a good chunk of it. The lifelong politician is now an outsider, having spent the past decade as the mayor of Greater Manchester. A man firmly embedded in the establishment will become the tribune of people who want to smash it. If voters are willing to embrace paradox, they are willing to embrace him. But it is a tough sell.

Just how tough becomes apparent with Eileen. If the prospect of sticking it to Sir Keir by voting Labour does not convince, what about an appeal to basic competence? Do you think Reform would do any better at running the country, asked one activist. "No," she replied. "Hate Labour? Vote Labour!" is a strange slogan in British politics. But it is no weirder than the rest of it. ■

International



Education

No place like home

In the wake of the pandemic, more parents are home-schooling their children

A COUPLE OF months after two children attacked Enna Pink's son with sticks at kindergarten, he began pleading with her to stay at home. "I didn't want to force him to go," says Ms Pink. She and her husband, who both worked at a startup, thought home-schooling would be a better fit for their son, who is "hyper-sensitive". But it is illegal in Germany, where they lived. So they moved to Costa Rica, where home-schooling is illegal for locals but there is little oversight for digital nomads.

Now her children, seven and four, do not follow lesson plans; instead they learn by playing outside, joining other children in local activities and travelling around the world. She thinks all this fosters curiosity and confidence. "We feel that what our society needs in the future is not what the school system can offer," she says.

Home-schooling has long been associated with oddball parents, awkward children and shaky pedagogy. But it is growing

swiftly. Numbers were rising before the pandemic; they have since surged in countries like Britain, Australia and Canada (see chart 1 on next page). In America 3.2m children, or 6% of the school-age population, were home-schooled in 2024—more than double the number in 2019.

As home-schooling has grown, the families who take it up have changed as well. Take America, where home-schooling, once a fad on the counter-cultural left in the 1970s, was by the 1980s driven by conservatives who decried schools as "Satanic hothouses". It is still associated with white evangelical Christians.

But nowadays home-schooling parents in America look broadly similar to the rest of the population, says Angela Watson,

head of home-schooling research at Johns Hopkins University. In fact, she says, home-schooling is rising quickest among families of colour, many of whom worry about discrimination and culturally insensitive curriculums. For perhaps similar reasons, in the Census Bureau's Household Pulse Survey in 2022-23, a higher percentage of home-school parents identified as LGBT than did public- or private-school parents. Ms Watson notes that most families "mix" types of education: around half of home-schooled children in America are taught that way only for one to three years.

The reasons for home-schooling are changing too. Parents now are more likely to say their primary concern is their child's physical and psychological safety. "Since [my eldest] was born, the world has gone a little bit crazy," says Rebecca Hardman, a British mother. She and her husband planned to send their young son to school when the pandemic lockdown ended. But they began to see home-schooling as a long-term alternative. Children at school seemed more exposed to pressure from peers and the corrosive influence of social media. "All this stuff has changed so rapidly that every moment that I thought, 'Oh, maybe it's time', I'd just be like, 'God, actually, what would he be learning?'"

Parents also worry about the rat race. The UN's World Health Organisation, in a ▶▶

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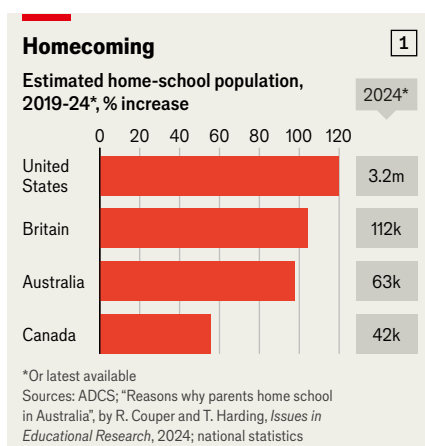
▶ survey of 280,000 young people across 44 countries, found that the share of 15-year-old girls who felt pressured at school had risen from 54% to 63% between 2018 and 2022. That is a particular worry for children who have learning difficulties, are autistic or suffer from poor mental health. (One in six of the 126,000 children educated at home during the autumn term in England last year cited mental health as the main cause.) Traditional education is “rigid, unwelcoming, stressful, bureaucratic and plainly unbearable”, says Hanna Lippi, a parent in Slovenia. She home-schooled her children before regulations were tightened in 2024 and is considering a move abroad, so she can continue to do so. Because of the focus on academic assessment, she says, “Families are burned out.”

Other parents simply think they can offer better teaching. Many see national curriculums as behind the times on everything from race to artificial intelligence—or as too hidebound for a fast-changing world. Issy Butson, who hosts a popular home-schooling podcast, began home-schooling in New Zealand after 20 years working in software companies and start-ups. He spurns traditional subjects if his children do not enjoy them. “Our eldest is a storyteller: he’s a writer and a reader and an illustrator, and he’s always been into that. So it’s absolutely crazy to think we would ever sit down and teach him maths. It would just seem completely pointless.” Instead he uses online tutorials to teach topics like coding, which he thinks are more useful. Schoolchildren have “been trained in a totally different model for a totally different world”, he says.

Rejecting schools worries many. Some studies conducted by home-schooling advocates find that home-schooled children outperform their peers. But the practice looks worse in some other studies which take family background into account.

In 2025 Cardus, a Canadian think-tank, published research which factored in childhood poverty, whether the respondent grew up with both biological parents and whether they were in a religious household. The paper, by Ms Watson and Albert Cheng of the University of Arkansas, found that American adults who had been home-schooled were less likely to work full-time or have a household income above the median wage. A 2014 study, using data from America’s National Survey on Drug Use and Health, found that home-schoolers aged 12 and up were two to three times more likely to report being behind their grade level. And in 2020 a meta-analysis by Robert Kunzman and Milton Gaither found that home-schoolers tended to perform well on verbal tests but fall behind on mathematics.

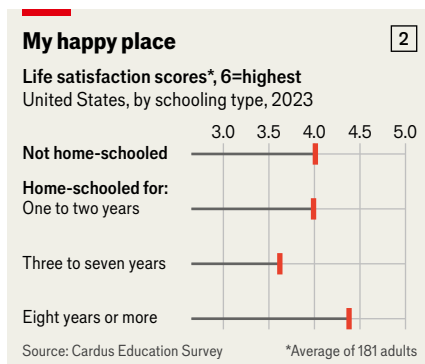
Studies of mental health and social integration also paint a mixed picture. Much



of the data is collected through self-reporting, and most home-schooling families are inclined to say they are well socialised. But the duration of home-schooling seems to matter a lot. The Cardus report found that pupils taught at home for eight years or more reported the highest levels of optimism and close social bonds. But those taught that way for one to two years reported the highest levels of anxiety, and those who were home-schooled for three-to-seven years had the fewest close social bonds and lowest life satisfaction (see chart 2).

One aspect that is difficult to measure is the degree to which children may be isolated and vulnerable to mistreatment at home. That is a focus of advocates for more home-schooling regulation, like the Coalition for Responsible Home Education, an American outfit where many staff were home-schooled themselves.

There is no peer-reviewed evidence linking home-schooling to higher rates of abuse or neglect. But globally teachers are the most frequent reporters of cases to child-protection services, so reduced contact between pupils and school staff may enable mistreatment to go unnoticed. Some experts worry further that children may be removed from school by abusive parents on the pretext of home-schooling. In 2024 a study by England’s Child Safeguarding Practice Review Panel found that home-schooled children were “less visible” to safeguarding agencies, though it also



stated that most children educated at home “have happy and safe lives”.

Many countries have extremely strict home-education regulations. School attendance is compulsory in places such as China, Germany, Greece, the Netherlands, Spain and Turkey; exemptions are rare. In some other countries, including South Korea and Singapore, it is subject to tight restrictions. This year Britain passed the Children’s Wellbeing and Schools Act, which sets up a national register of children who are not in school and provides more local oversight of home-schooling; previously parents were required simply to ensure their children receive a full-time education “suitable” to their age. And in 2021, after a spate of terror attacks, France passed a law restricting home-schooling to exceptional circumstances in a bid to combat extremism and protect secular values.

But in other countries the rules are remarkably relaxed. America, the home of home-schooling, has seen waves of deregulation, in part thanks to decades of lobbying by the Home School Legal Defence Association, a conservative group that now operates around the world.

Those who can’t, teach

In almost all American states guardians can educate at home even if they have a conviction for violent or sexual crimes against children. In 42 states there is no minimum qualification threshold for parental education. Only eight American states require all home-schooled children to take academic assessments, and 27 have no home-school testing requirements at all. Eleven states do not require families even to notify districts that they are going to home-school. And some states now subsidise the home-schooling of children, allowing parents to spend taxpayer money on services such as tutoring.

In extreme cases the lack of regulatory oversight can enable pernicious ideology to spread unchecked. In 2023 a home-schooling network in Ohio that connected as many as 3,000 white supremacists over social media was exposed; Hitler’s “Mein Kampf” was a cornerstone of its curriculum. Officials investigated the group but ultimately found that no laws had been broken, as the state’s home-schooling laws do not regulate curriculum content.

Such may be the stereotypical hazards of home-schooling, but more broadly the practice is finding support across the ideological spectrum. Ms Watson notes that in a 2024 survey American home-school parents were only slightly less likely to identify as liberal or moderate than their public-school counterparts. Parents of all stripes now are given to say—notwithstanding the mixed findings on performance and social adjustment—that they simply want what is best for their children. ■

THE TELEGRAM

Israel the lonely

Israel is not as isolated as its critics hope, but it is alienating some true friends



“DID YOU watch the Eurovision Song Contest last night?” asked the barber in a Jerusalem back street, razor in hand. Confessing that he had unaccountably failed to tune in, your columnist knew why the question was asked. Protest politics intruded on this year’s Eurovision—a contest cherished by fans for terrible songs, corny lyrics and bad-taste costumes—after Israel made it to the finals. Citing Israel’s iron-fisted military occupation of Gaza, five European countries boycotted the competition. Mingled boos and cheers greeted Israel’s second-place finish.

Israel’s isolation is not a frivolous matter. Increasingly its scientists report unexplained refusals when international research grants come up for renewal. Parents worry about youngsters being ostracised if they enrol at foreign universities, especially after completing national service in the Israel Defence Forces (IDF).

Boycotts do not intimidate Israel, though. Five days of interviews with serving and retired military officers, government officials and elected politicians reveal a country that still sees itself as fighting for survival in a dangerous neighbourhood. Such a nation can endure solitude. With a world-weary pride, an official quotes a Bible verse from the Book of Numbers, calling the Jews “a people that dwells alone; not reckoned among the nations”. Moreover, Israel is less alone than critics and enemies hope.

Israel’s image has taken a battering since the Gaza war killed over 70,000 Palestinians, mainly civilians, women and children. In Israel, global criticism is called unfair and dismissive of Israel’s right to self-defence after the surprise attack by Hamas terrorists on October 7th 2023, which saw 1,200 people murdered, mostly civilians, women and children. Led by the prime minister, Benjamin Netanyahu, many Israelis accuse critics of antisemitism.

Israel can ignore bad-faith attacks, defenders say, because relations with America have reached new heights under President Donald Trump. There is talk of unprecedented military co-operation with American and Israeli officers working “shoulder to shoulder” in Israeli operations rooms. Some predict that the country will become a “fortress” for American troops and hardware in the Middle East, long after the war with Iran. American officials call their collaboration with Israel a model for working with partners such as

Japan in a conflict with China over Taiwan, this columnist heard.

Israel’s defenders make an angrier claim, too. Many governments condemned Israel for using disproportionate violence in Gaza and for tolerating violent attacks by Jewish settlers in the West Bank. This year several have deplored what they call indiscriminate Israeli air strikes on Lebanon, aimed at Hizbullah, the Iran-backed Shia militia. Foreign leaders have rebuked America and Israel for launching an ill-judged war against Iran. But some finger-wagging governments, notably in Europe and the Gulf, take a different line in private, it is said. They admire Israel’s toughness, and ask to share its intelligence and buy its defence kit.

Israel’s closest Arab partner, the United Arab Emirates (UAE), has “crossed the Rubicon” and made an irreversible decision to align with Israel, whose air-defence systems are now shielding the UAE from attacks by Iran. Jordan is called a close partner, as is India, whose prime minister, Narendra Modi, made public his country’s deep ties to Israel when he visited the country in 2017.

For all its criticisms of Mr Netanyahu, Europe remains Israel’s largest trade partner. Led by Germany, there are vast two-way flows of air-defence systems, submarines and the like. Greece and Italy dream of being connected to energy pipelines running through the Holy Land. Israel, then, is not all alone. Yet shared interests are not the same as trust or admiration. As a result, some high-ranking Israelis say their country is dangerously lonely.

Many ordinary Israelis revere Mr Trump as their country’s stalwart friend. Members of the security establishment are less confident, expressing fears that Israel dazzled Mr Trump with plans to kill Iran’s supreme leader and top aides in one strike. That primed Mr Trump to expect a swift regime change. Instead, Iran remains defiant. On May 18th Mr Trump halted planned strikes following pleas from Arab rulers. Mr Trump’s vanity will not let him admit that he was led into an unwinnable war, says a source. But senior figures worry that many elected Democrats and Republicans will blame Israel for dragging America into disaster. They know America cares more about opening the Strait of Hormuz than Israel does. They dread a skittish Mr Trump making a deal that fails to end Iran’s nuclear programme, Israel’s priority. Israel used to enjoy bipartisan support in America. Now many Democrats are sharply critical and young Republicans are increasingly hostile.

Security is an illusion, if people have no hope

Some in Israel argue that foreign relations will reset if elections this autumn end Mr Netanyahu’s long reign. Others see a need for larger changes. Ami Ayalon is a former naval commando, head of Israel’s navy, director of Shin Bet, its security service, and government minister for the Labor Party. Leaders cannot safely ignore public opinion, he counsels. Egypt’s president and Jordan’s king may wish to preserve ties with Israel. But if Israel cannot offer Palestinians a state of their own, “we won’t be able to preserve peace with Egypt and Jordan.” The old warrior compares terrorism to water rising from a spring, rather than a target that can be bombed. Any child of 12 or 14 who sees his father killed may “take up a knife and be ready to die”. Most of the international community “believes in Israel, alongside a Palestinian state” he says. “They don’t hate us, but they will if we don’t change our policy.”

Israel, an embattled and divided country, is not about to grant statehood to Palestinians. On the Palestinian side, there is no sign of a negotiating partner of vision and authority. Until the world sees grounds for hope, Israel will be an ever lonelier place. ■

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Almost every day for the past 15 years, Ben Proud had to make his whereabouts known to the doping authorities. Travelling for work? His hotel would have to be logged on a clunky website. Staying over at a new girlfriend's place? That would have to be recorded, too. It didn't make for the most spontaneous of lives. But it's what you have to do if you want to be an Olympic athlete.

In 2024 Proud had won a silver medal in the 50-metre freestyle swim at the Paris Olympics. It had been his third Olympics and the high point of his career. But last November he was in a slump. He was 31, old for a competitive swimmer. His knees hurt and his back was shot. There was a persistent, dull pain in the tendons around his elbows.

One Wednesday at 6am he heard a knock at the door of his flat in Stratford, east London. He opened it

DOPE AND GLORY

Can performance-boosting drugs stretch the limits of humanity? Barclay Bram and Natasha Loder get the inside track on the Enhanced Games

to find a man and a woman, sent by UK Anti-Doping, Britain's drug-testing body. They were there to check that Proud was complying with what is known among Olympic athletes as "the Code", a set of regulations from the World Anti-Doping Agency that includes an agreement not to use certain substances.

Proud knew the testers might pay him a visit that day, but hadn't expected them to come so early. He had gone to the toilet just before they arrived, so they would all have to wait before another urine sample could be collected. The three of them didn't make small talk; rather, they sat on the sofa and looked silently out of the window at the view.

Nearly two hours later, Proud was ready. The male tester followed him into the flat's only bathroom, an ensuite; they had to creep quietly through the bedroom to avoid waking Emily Barclay, Proud's partner, who is also a competitive swimmer. Proud found the experience profoundly awkward. "You stand there with your trousers around your ankles and your bare arse out like a schoolboy," he said.

After the testers left, Barclay emerged tentatively from the bedroom. She passed Proud her phone to show him a piece of news. "Kristian has just broken the world record," she said. Proud stared at the screen, confused. He frequently competed against Kristian Gkolomeev, a Greek swimmer. They'd tied for fifth place at the Tokyo Olympics in 2021, but Proud had comfortably beaten him in Paris. He couldn't believe that Gkolomeev had somehow become the fastest 50-metre swimmer in history—in the off-season, no less.

Reading on, Proud learned that Gkolomeev had signed up with the Enhanced Games, a company that was seeking to disrupt conventional sport by allowing athletes to compete while on performance-enhancing drugs. Later that day, Enhanced (as the wider company is sometimes known) released a documentary about Gkolomeev's swim, which he had done alone in a pool in North Carolina. Not only had he taken the world record—Enhanced had paid him \$1m for the achievement. "The swimming world is going to hate every second of it," Brett Hawke, Enhanced's head swimming coach, says in the documentary. "This is where human performance is going. And I think a lot of people will start to embrace it over time."

As the couple watched the documentary, Barclay had to pause it several times because Proud had been so overwhelmed with emotion. His pursuit of Olympic glory had come with steep costs. He'd grown up in Malaysia and moved to Britain to train when he was 16, leaving most of his family behind. There had been years when he had scraped by on nothing more than the £28,000 (\$38,000) stipend provided by the national aquatic sports association to elite swimmers. He wasn't sure what was in store for him after his swimming career came to an end—which, at his age, could be as soon as his next injury.

If Proud joined Enhanced—and adopted its controversial drug protocols—he could keep swimming at the highest level, perhaps even faster than before, and potentially earn a life-changing amount of money. But he knew it was a Faustian bargain: taking part would tarnish his reputation, and he would probably be barred from mainstream competition for ever.

Proud asked Barclay if she would judge him if he threw in his lot with the Enhanced Games. When she said she wouldn't, he called his agent. He was ready for something new.

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On May 24th around 50 athletes—in swimming, weightlifting and athletics—will gather in Las Vegas for the first Enhanced Games. Up to \$25m in prizes will be awarded, with \$250,000 going to those who win first place and bonuses of up to \$1m to world-record breakers. Around 2,500 people will watch from the stands of a purpose-built arena, where they'll also be treated to a Super Bowl-style closing ceremony—The Killers, a rock band, are playing—and have the chance to mingle with scientists, crypto investors and celebrities. The games will be streamed on YouTube and Roku: Bryan Johnson, a venture capitalist who has become famous for trying to chase immortality by tweaking every aspect of his body and lifestyle, will be one of the commentators.

This spectacle is an advertisement for Enhanced's other business: selling performance-enhancing products to non-Olympians. The company has pumped its competing athletes with drugs over the past three months in the hope that their achievements will inspire viewers—from ordinary gym-goers to grandparents trying to keep up with their grandchildren—to also get enhanced. A dizzying array of injectable drugs is already for sale on Enhanced's website, promising to support everything from "sleep, energy and sharper mornings" to "healthy ageing".

Ultimately the real purpose of the games is to push the limits of what the public sees as the acceptable use of performance-enhancing drugs. Some of these have long been a part of the high-street gym scene: anabolic-androgenic steroids, such as testosterone, and human-growth hormones are widely used to improve muscle strength and performance. Increasingly, though, bodybuilders, "looksmaxxers" (people who go to extreme lengths to become more physically attractive) and Silicon Valley coders are supplementing their regimens with a new generation of drugs called peptides. Many peptides are unlicensed and produced by dubious-looking manufacturers in China. Some claim to speed recovery and healing; others to thicken hair, build muscle, deepen a tan or even improve libido. Gym enthusiasts and their coaches are now experimenting with combinations of chemicals aimed at improving strength, longevity or looks—nicknamed "stacks"—all without medical supervision.





The Enhanced Games emerged from this subculture. Just before Christmas in 2022, Aron D'Souza, an Australian lawyer in his 30s, was working up a sweat in an upmarket gym in Miami when he overheard the buff people around him chatting about being “enhanced”. After D'Souza asked them what they meant, they cheerily discussed the contents of their stacks with him. For D'Souza, these enhanced gym-goers were prototypes for a healthier, more ambitious humanity. He began to wonder: what if the Olympics allowed for open enhancement? What feats could humans achieve if they could do whatever they wanted with their bodies?

D'Souza was well-connected with rich people who liked big ideas, including Peter Thiel, a co-founder of PayPal and a venture capitalist. At Thiel's annual new year's party D'Souza pitched his idea for an Olympics on steroids; Thiel later wrote him a cheque. Balaji Srinivasan, a former executive at Coinbase, a cryptocurrency exchange, and Christian Angermayer, a German entrepreneur, followed suit. Their shared interest in the Enhanced Games stemmed from similar political orientations: all three men believe that people should be able to do what they like with their money and their lives with as little state interference as possible—preferably none. (Thiel has characterised regulators as “the Antichrist”.) They also all have close ties with Silicon Valley, where there is a tradition of viewing physical weakness, disease and even death as problems to be engineered away. D'Souza's pitch was perfectly calibrated for this mindset.

To draw attention to his nascent company, D'Souza began courting controversy. His main target was the International Olympic Committee (IOC). He argued

New highs
Ben Proud, a swimmer, is taking part in the Enhanced Games

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that the IOC exploited athletes commercially while paying them nothing. (An early version of Enhanced's website accused the IOC's president of jetting to fancy hotels to visit billionaires and dictators while Olympic athletes slept in their cars.) D'Souza also called for national anti-doping agencies to be defunded for their “discrimination” against enhanced athletes.

There are some grounds for criticising anti-doping agencies, especially on the issue of consistency. Tough bans have been imposed on individuals while groups of athletes from powerful countries such as Russia and China have been shown leniency. Few would go as far as D'Souza in suggesting that anti-doping rules be scrapped entirely, but many observers agree the system is broken. John Hoberman, a professor at the University of Texas at Austin who studies the history of doping in sports, told us that doping has been “out of control for a long time” and that frequent testing doesn't appear to be a strong enough deterrent. Some experts have even argued that medically supervised enhancement could be a safer option for competitive athletes, given that so many of them take black-market drugs anyway. But even critics of the current anti-doping regime viewed the Enhanced Games with concern—Hoberman, for instance, has called it a “dishonest and careless project”.

D'Souza knew his idea would need scientific credibility to get off the ground. In 2024 he started to approach potential scientific advisers and organised two gatherings grandly billed as “conferences on human enhancement”. The company also needed to find athletes willing to put their bodies on the line—so it offered \$1m to anyone who broke a world record in certain disciplines while on an enhancement protocol.

In February 2024 James Magnussen, a 32-year-old retired swimmer from Australia known as “the Missile”, was the first to take the bait. Having won silver and bronze medals at the London Olympics, he had been trying out several alternative careers, as many former Olympians do: gym-owner, sports commentator and “Dancing with the Stars” contestant. But after hearing about the Enhanced Games, he was now ready, as he put it on a podcast, to “juice to the gills” and break the world record in the 50-metre freestyle.

Magnussen asked Brett Hawke, who had coached two of the previous world-record holders in the discipline, to train him. He then started a stack composed of anabolic steroids, testosterone and human-growth hormone—as well as some peptides—using dosages agreed upon with his own doctor.

At the time, no one knew what it would take to transform a retired swimmer into a world-record breaker. But it became clear that the drugs did work—almost too well. Magnussen's recovery between sessions was so quick that he asked Hawke to push him harder. His lifts in the gym became heavier and heavier. By the time he swam for the world record in May 2025, he had put on 20kg of muscle; in photos, his back bulged out of his swimsuit like the Incredible Hulk.

The record attempt was a flop—Magnussen's gains were so large they had slowed him down. “Unfortunately I didn't actually grow gills,” he said, “because with all that muscle I was very low in the water.” Even so, he had proved that an enhancement protocol could help a retired elite athlete return to sport faster and more successfully than anyone had imagined. Perhaps the athletes just needed fewer drugs—and protocols informed by science.

In December 2023 Guido Pieleles was on his way to a medical conference when he received an unexpected email. It was from D'Souza. He wanted Pieleles, an expert in sports cardiology and congenital heart conditions, to advise Enhanced. "Hang on, really? Are they joking?" Pieleles recalls thinking.

But he was interested enough to take a call with D'Souza anyway. "I had a lot of questions," he remembers. Pieleles wondered whether the athletes' enhancement could be run as a clinical trial. Not only would this be safer—because the athletes would be under medical supervision—but the data gathered from the trial could then drive further research into the use of enhancement drugs in ordinary people, with the goal of improving their long-term health and longevity.

Pieleles agreed to join D'Souza at his first enhancement conference in February 2024. By the end of the event, he had convinced D'Souza and Maximilian Martin, one of Enhanced's co-founders and, at the time, its chief strategy officer, to the merits of a trial. Pieleles was then dispatched to assemble an independent team of experts, which would design the trial, pick the drugs, decide the doses and supervise the athletes' health. (This team ended up including both mainstream doctors and scientists as well as enhancement specialists, whose work tends to be on the fringes of sports science.) The experts stipulated that the athletes could use only drugs that had been approved for humans—ruling out most peptides.

They also knew that the possibility of receiving a placebo would deter athletes from taking part—no one chasing a cash prize would accept such uncertainty about the contents of their protocol. Instead, the trial would measure the outcomes over time of a group of athletes, most of whom would take enhancements. Enhanced wouldn't be able to prove whether any effects on health or performance were caused by the drugs, the intense competition, or both. But with blood tests every two weeks it would be possible to track how athletes responded to the drugs and changes in dosage. This close observation could also mitigate the known risks to the athletes' cardiovascular and endocrine systems of taking enhancement drugs.

Enhanced decided to run the trial in the United Arab Emirates (UAE), which offers both world-class medical facilities and a strong, but suitably flexible, medical regulatory system. When we visited Abu Dhabi in early February, the trial had yet to be approved by the Ministry of Health, so the athletes had not started taking the drugs. But Pieleles, an athletic-looking 52-year-old, was ready to answer our questions.

There would be a menu of enhancement drugs on offer rather than a set prescription, Pieleles said, allowing each protocol to be tailored to the specific demands of the athlete's sport. Testosterone and anabolic steroids would be made available to build muscle; human-growth hormone to repair tissue; erythropoietin to boost red-blood-cell production; meldonium to enhance endurance (whether it actually does this is disputed); and modafinil and Adderall to sharpen focus and reduce fatigue. Dosages would start low but would be gradually increased to levels that Pieleles said would exceed those used in standard medical care, while staying within what he considered safe limits. (These levels would also typically be lower than those seen in some users of high-street gyms.)

Pieleles said that people in the enhancement business laughed when he told them what he planned on



Fuel injection

"I don't know why there is so much fuss about it," said Guido Pieleles, the chair of Enhanced's independent medical commission, about the drug protocols

giving the athletes ("Your stack is so boring!"). He thought they had a point. "I don't know why there is so much fuss about it," he told us.

In February around 40 Enhanced athletes moved to Abu Dhabi to begin training under their specially designed drug protocols. Their new home and training centre was EARTH, a luxury hotel that is being converted into a state-of-the-art sporting complex as part of the UAE's plan to become a leading venue for global sport. Building works were shielded from view by placards declaring, appropriately, that the facilities were being "enhanced".

Even so, EARTH already had all the equipment the athletes needed: an enormous gym, a running track, a 50-metre indoor pool. "I've never experienced having all your wants and needs taken care of like this," said Hawke, the head swimming coach. The athletes also had access to physios, sports massage and doctors; a private beach; and the hotel buffet, where the chefs had reimaged the indulgent menus to make them leaner and protein-rich. "This meal might cost me \$75 back in Austin," said Emmanuel Matadi, a sprinter, sitting at an outdoor table shielded from the strong sun by an umbrella. He shook his head in disbelief at the mountain of grilled salmon and vegetables on his plate. "I often don't get all the calories I need simply because I don't have the time or the money to cook at this level."

Normally elite athletes pay for their own food, coaches, training facilities and travel. Ivan Rojas, an experienced weightlifting coach who now works for Enhanced, told us that when he coached the American team one year at the world weightlifting champion-

Magnussen had proved that an enhancement protocol could help a retired elite athlete return to sport faster and more successfully than anyone had imagined

ships he realised they hadn't been given travel uniforms. He ended up buying them embroidered polo shirts with his own money. There are modest prize pots and performance fees on offer at established competitions, but if athletes have to drop out they lose this income. "That's how track is," Marvin Bracy-Williams, a 100-metre sprinter, told us. "You have a bad day, somebody ain't eating."

To make ends meet, Bracy-Williams had spent years as part of the practice squads of National Football League teams, where there is steady pay on offer. But after he broke his arm in 2019 he once again took up competitive sprinting: "It's like gambling. You're spending money to be a part of something, hoping that you win big."

Bracy-Williams had been offered a sponsorship deal worth \$120,000 ahead of the Paris Olympics, conditional on his being selected for the American athletics team. But he pulled a leg muscle during training and in the qualification trials ran two seconds slower than his personal best. He didn't make the cut. "I watched the money walk away," he said.

Desperate to get back into the game, Bracy-Williams contacted someone who could hook him up with performance-enhancing drugs. He knew the risks—a probable ban from the sport, being dropped by his sponsors, social stigma—but felt as though he had no other choice. "I was willing to do whatever it took to just not hurt any more," he said.

Bracy-Williams started taking small doses of drugs, primarily testosterone. But they didn't make much difference: his times didn't budge and the pain in his leg grew worse. In agony one day, he confessed to his coach. The coach was sympathetic, but told Bracy-Williams he'd have to report him to America's anti-doping agency. The next day inspectors turned up and tested Bracy-Williams. His sample was positive, and he was suspended from the sport until November 2027. "The hardest is when it's gam and you know all your friends are at practice," he told us, "and you're just sat there with nowhere to be."

For Bracy-Williams, the appeal of the games is clear. He's one of six Enhanced athletes who have faced drug-related suspensions in their careers, leading some critics to dub the games the "Cheaters'

ERTH already had all the training equipment the athletes needed. "I've never experienced having all your wants and needs taken care of like this"

Olympics". Athletes who were known for being squeaky clean, like Proud, the 50-metre freestyle swimmer, have more to lose. Anyone—be it a competitor, doctor or coach—who takes part in the Enhanced Games could face a long-term ban from professional sports.

Proud sometimes seemed to be trying to justify his participation. He argued that Enhanced was actually levelling the playing field, which he believes is skewed in favour of athletes from countries that turn a blind eye to doping. "One of my close friends narrowly missed a gold to one of those Chinese swimmers," he said, referring to a doping controversy at the Paris Olympics, in which 11 Chinese swimmers who had previously tested positive for a banned heart medicine were nevertheless allowed to compete. "You have to suck it up and get on with it, because as swimmers we don't feel we have the power to stand up and say that something like this is clearly wrong."

He also seemed glad finally to be making money. "I always said, growing up, you never get into swimming for the money...that's fine to say when you're 20, 23 years old, but when you get to 28 or 30, it's a very different story." Since joining Enhanced he has earned both a signing bonus and a salary (he didn't disclose the exact figures, but the total is believed to be in the hundreds of thousands of dollars). And he still stands to earn \$1.25m if he both wins his race and breaks the world record.

In late February Proud stood in his hotel room, shaking as he inserted the needle into the soft flesh just below his waist for the first time. He felt a small prick and pushed the plunger. Now he knew he could never go back.

Whether the inaugural games marks the launch of a viable business or a one-off PR stunt will largely come down to Enhanced's leadership, particularly Christian Angermayer, who has become its main backer. D'Souza left the company at some point—it's not entirely clear when, but his departure appears to have been amicable.

Maximilian Martin, the company's former chief strategy officer, seemingly became chief executive last August. Under him, Enhanced has become less provocative and more polished. Instead of proclaiming, as D'Souza did, that athletes today have to hide their doping like gay people had to hide their sexuality 50 years ago, Martin has tried to reassure the public that performance-enhancing drugs are "not that dangerous...under the right clinical and medical supervision".

But thanks to Angermayer's expanded influence, Enhanced still possesses its founding techno-utopian spirit. In the past few decades he has funded projects researching longevity medicine, brain-computer interfaces and the use of psychedelics to treat mental-health conditions (Angermayer is a proponent of hallucinogenic drugs, which he also believes will help humanity adjust to the mass joblessness that AI will create). Taken together, Angermayer's preoccupations speak to a vision of the future in which medicines, rather than being simply used to treat disease, can extend human longevity and enhance well-being.

We recently met Angermayer in his penthouse flat near Old Street, a trendy area of London. He has been on testosterone-replacement therapy (TRT) since he was 30; now 48, he is built like a gym bunny. Sitting at the head of a table the size of an aeroplane-landing strip, he offered us cannabidiol-infused sodas from a



firm he invests in. On the wall was an artwork made of Ecstasy pills, arranged in a colourful halo.

"I was always very vain," Angermayer recalled. "Even in my 20s, I was already thinking: how can I slow down ageing?" Now he believes he has found a way to bring the drugs he has been experimenting with to a mass market. The business model he described sounded similar to that of Red Bull, an energy-drink brand: stage a series of high-profile sporting events (in Red Bull's case, extreme ones like cliff diving or motor-sports) and use them to publicise a product.

Earlier this year, Enhanced launched a range of personalised performance and longevity medicines, as well as services like blood panels and telehealth consultations. Angermayer envisions the games as a way to market these products. The venue in Las Vegas has been built using a modular system, which means it can be reassembled anywhere in the world (it cost about \$8m to create, with \$6m spent on the pool alone). He has big ideas for future iterations of the competition, musing that the athletes could one day race against humanoid robots or celebrities.

The games under Angermayer are well-suited to the political moment, in tune with the cultural and ideological energies of Donald Trump's second term ("A hundred percent the reason they're happening in the US is because Trump won," Angermayer has said). Trump's son Donald junior is an investor through his work at 1789 Capital, a venture-capital firm that aims to stimulate a MAGA-adjacent "patriot economy". The administration has also proved friendly to ideas that Angermayer and others interested in body-optimisation and longevity medicine have long promoted. Earlier this year an expert panel from the Food and Drug Administration (FDA) recommended making TRT more widely available. The FDA is also poised to review its restrictions on some unapproved peptides (Angermayer says Enhanced will only sell peptides that have credible data on their safety in humans).

Still, Enhanced has yet to turn a profit. According to its filings it lost \$4.7m in 2024 and \$26.7m in 2025. Lucrative media rights and big-name brand sponsorships have largely not materialised, perhaps because companies fear damaging their relationships with traditional sporting bodies.

In early May the company went public by merging with a special-purpose acquisition company (SPAC), a shell company that is already listed. (SPACs have become a popular backdoor route for companies looking to avoid the bureaucracy of a traditional listing.) The merger documentation acknowledged that Enhanced possesses an "unproven business model" but nonetheless valued it at \$1.2bn. Even so, most shareholders in the SPAC decided to sell their shares before it went public—taking most of the \$200m in cash the combined firm might have held. Perhaps these investors didn't believe in Enhanced's vision or balked at the structure of the merger, which gave Angermayer's investment firm around 97% of the voting power.

Enhanced is just one project in Angermayer's portfolio—his most publicised attempt yet to bring his drug-fuelled vision of humanity's future into being. If the company fails, it's only his finances that will take a hit. But almost everyone else involved in the games has burned bridges, risked their future livelihoods or their health. Understanding the long-term side-effects of the enhancements used in the protocols will take time. Yet with the launch of Enhanced's consumer business,

"I was always very vain," Angermayer recalled. "Even in my 20s, I was already thinking: how can I slow down ageing?"

more and more people may soon be wagering their bodies on a chance to roll back the clock.

Just weeks after we visited the Enhanced facilities in Abu Dhabi, America and Israel attacked Iran, causing turmoil across the Middle East. Suddenly, the athletes were training to streaks of missiles across the horizon and the flash of interceptors. "It's funny how quickly you grow accustomed to it," Shania Collins, a 100-metre sprinter, told us. She didn't leave the ERTH complex for over a month, and her enhancement protocol was delayed because the lab processing the results of some preparatory tests was hit by debris from an intercepted missile.

Proud remembered the first night of the war as "awful" (he said that a swimmer from Ukraine "who has been through all of this before" helped the other athletes find a place to hunker down). Soon though, the explosions became background noise. As a side-effect of the testosterone in his protocol, Proud has been sleeping more deeply than ever before, and he often didn't wake at the warning sirens.

Most of the athletes have managed through their protocols well enough. Collins was generally pleased with how she was feeling—her arms were more toned, her biceps more defined, and she was smashing her personal bests in the gym. (She also noticed a difference in her fellow athletes, especially the swimmers: "I looked at them and thought, wow, are these guys modelling now or swimming?")

But she was worried about the potential side-effects of taking testosterone; people online had speculated that doing so would turn her into a man. Her doctors assured her that her dosage was far below that needed for a gender transition. Even so, she had downloaded a pitch monitor on her phone to make sure her voice wasn't getting any deeper.

Other athletes were concerned about what would happen to their careers and sponsorship opportunities after the games, and whether their bans from mainstream sport would continue. But despite these worries and the sounds of war, the ERTH complex had become something of a Neverland: a place where athletes whose best days were behind them had another chance to do what they loved, and in the competition of their lives.

For months, Proud has been visualising his race—that 20-second chase for a life-changing amount of money. As he approaches the starting block, he will be intensely focused. "I'm not daydreaming or letting my attention go anywhere else. I focus on my block and on the process...You feel your heart pounding, you're dizzy and you don't know if you can stand up straight. But your body has this method of keeping its control."

At the sound of the buzzer, he will dive into the pool. If everything has gone well with his enhancement protocol he will spring forward a little farther than usual, and cut through the water more quickly. As Proud crosses the mark on the bottom of the pool signalling the last 15 metres, he will kick harder towards the finish. "The last thing you have to think about is to make sure your touch on the wall is at the perfect distance, at just the right time," he told us. "And then everything else is history." ■

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Business



Artificial intelligence (1)

Everything engines

SHANGHAI

How AI super-apps are remaking China's internet

TO HAVE A coffee delivered to an office in Shanghai, simply ask one of China's artificial-intelligence super-apps to choose a brew on your behalf, press "confirm" and the beverage will be on its way. Delegating such important decisions brings risks, of course. When your correspondent asked one popular AI app to deliver a "special coffee", he received a rose-petal-vinegar-flavoured one.

Nevertheless, the pace with which such services are being adopted in China is remarkable. Already more than 600m of its people are thought to have used some form of so-called agentic app. China is speeding towards a future in which AI chooses, purchases and delivers many of the goods and services people consume, upending its digital economy in the process.

Chinese netizens have so far lived through two distinct internet eras. From the early 2000s most turned to Baidu, a search engine, as their window to the web.

When Google was forced to exit the country at the end of the decade, Baidu, in effect, became a monopoly. But as it sought to monetise its service more aggressively, ad-driven recommendations took over, frustrating users. That brought a backlash against web-based search and, thanks to the spread of the smartphone, pushed

China into a second internet era dominated by mobile super-apps that combine functions such as shopping, entertainment, communication and payments.

One consequence is that China's biggest technology firms—including Alibaba, an e-commerce titan, ByteDance, an entertainment giant, and Tencent, a gaming-and-messaging colossus—possess sprawling portfolios of digital services and logistics networks that can be used to develop agentic offerings capable of performing a wide variety of tasks for users. Another consequence is that any of these companies could conceivably emerge as the leader of this new, third era in China's internet. Already the competition is ferocious. On May 13th Pony Ma, founder of Tencent, warned that a messy "land grab" for AI services was in the offing.

On May 11th Alibaba announced that it had fully integrated Qwen, its chatbot, with Taobao, its shopping app. With a few simple commands the AI can now procure all manner of products and services (including rose-petal-vinegar-flavoured coffee). ByteDance is preparing to release a similar integration between Doubao, its chatbot, and Douyin, its short-video app (which also incorporates shopping).

Tencent has been a dark horse in the race. Its investment in AI models started more slowly, but the company says that

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► over the past six months it has completely revamped how its AI team works. A new model, Hy3, is in a testing phase and has performed well. Tencent is now slowly integrating this model with WeChat, its ubiquitous messaging-and-payments app. Millions of businesses have created “mini-programs” within WeChat that could be woven together with AI.

For the tech giants, AI super-apps may offer an attractive new source of growth at a time when consumer spending in China is anaemic. Adjusted operating profit in Alibaba’s Chinese e-commerce division was down by 40% year on year in the first quarter of 2026. And although the firm’s cloud business is booming as enterprises spend more on its AI infrastructure, that growth requires enormous capital investment, weighing on cashflows.

In private China’s tech giants say that their AI super-apps do not recommend products based on ad spending. But to make these profitable, they may eventually have to. Although ByteDance recently launched a paid tier for Doubao users wishing to access certain features, most chatbots in China are free to use. In February, to coincide with the Lunar New Year, many tech giants offered generous promotions to encourage people to start using their agentic services. With competition so intense, it seems unlikely they will start charging for the services soon.

The rationale for the tech giants may also be defensive. Some in the industry worry that the emergence of a device powered by AI with agentic capabilities embedded in its operating system could displace existing super-apps. OpenAI, an American lab, is reportedly working on such a device. ByteDance tried this in December when it launched a smartphone with ZTE, a device-maker, which came pre-loaded with an AI assistant. But despite an initial burst of enthusiasm, with around 30,000 of the devices produced, the project was a failure, in part because Alibaba and Tencent blocked it from using their payment platforms.

That will not stop others from trying. In March Xiaomi, a gadget-maker that has expanded into electric vehicles (EVs), announced the release of new AI models that would be embedded in its smartphones and cars. Huawei, another tech giant that makes both smartphones and software systems for EVs, could also muscle in. Auto executives reckon that many young Chinese will access agentic services through voice discussions with their car.

Xiaomi and Huawei already have hundreds of millions of users apiece. If Alibaba and Tencent were to try to hobble a push into agentic offerings by either company, local regulators might well intervene. As China’s internet enters its next era, a bitter fight for dominance looms. ■

Artificial intelligence (2)

Agents for the masses

MOUNTAIN VIEW

Is Google the new king of consumer AI?

THE AMPHITHEATRE complex where Google holds its annual software-developers’ conference has a cheesy, fair-ground feel. RVs are parked on site. Employees whiz in on the tech company’s multi-coloured bicycles. There are stands and sideshows everywhere. On stage, Sundar Pichai, its boss, tells a corny joke about Google’s overworked chips, known as TPUs, doing “teraflops into bed”.

The event is by no means as sleek as Apple’s developer jamboree held in June, which tries to retain some of the minimalist chic of the late Steve Jobs. But when it comes to artificial intelligence, Google long ago upstaged the maker of the iPhone (its models will power many of Apple’s AI features going forward). It now looks as though it may steal the consumer AI crown from OpenAI, maker of ChatGPT, as well.

On May 19th Google unveiled a new line-up of AI agents powered by its latest Gemini 3.5 Flash model. It included AI coders to rival those offered by OpenAI and Anthropic, but also agents designed to perform various tasks for regular folk going about their daily lives. Some will appear in the Gemini app, used by 900m people every month. Others will be embedded directly in Google Search, used by more than 3bn. The company is, in sum, bringing agents to the masses.

As is common in Silicon Valley, the examples executives used on stage raised plenty of eyebrows. How often would someone need an AI agent to make a slide presentation for a bouncy-castle party? Yet the tools unveiled also show promise. An agent called Gemini Spark will be able to do things such as scan emails or organise

group trips even after a user has closed their laptop or put down their phone, while “information agents” in Google Search will be able to keep tabs on sports tournaments, shopping sales or the stockmarket.

All this looks particularly troubling for OpenAI, which has so far led in consumer AI. Shortly after Google launched its Gemini 3 family of models in November, Sam Altman, OpenAI’s boss, issued a “Code Red” emergency to galvanise employees into speeding up improvements to ChatGPT. Since then, the focus at the lab has shifted towards its coding agent. But the release of Gemini 3.5 Flash, which Google says is four times faster than other frontier models, and the new suite of agents is likely to raise fresh questions about what OpenAI is doing with its flagship chatbot.

Investors are certainly bullish on Google’s prospects. The market value of Alphabet, its parent company, is now within a hair’s breadth of \$5trn, having passed \$4trn only in January (see chart). Yet Google’s success in AI is also creating problems. According to Mr Pichai, the number of tokens—Silicon Valley’s favoured measure of AI usage—consumed by its services has risen to 3.2 quadrillion a month, up from 480trn a year ago. Each token requires computing power, and therefore money, to generate, which is why Google’s capital expenditure this year will be up to \$190bn, six times as much as four years ago. Moreover, that money does not go as far as it used to, because everything from chips to energy has become more expensive. Even for Google, there are limits to how much it can afford to spend.

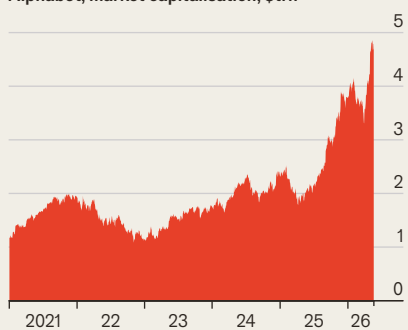
There are a few potential solutions. One is to lower the cost per token by making the technology more efficient, which Google will surely do. Another is to put limits on AI usage. Such restrictions were not announced at the event, but Gemini subscribers were alerted afterwards that they would apply, albeit with higher limits than for non-subscribers, according to Richard Windsor of Radio Free Mobile, a research firm. Usage caps may also encourage more people to pay for a subscription.

A third solution is to lean more into ads. Google reckons that the greater detail in AI queries will appeal to marketers. Although it has not yet incorporated ads into its Gemini app, it has been interspersing them into the AI responses its search service now spits out, and soon it will put AI product explainers alongside these ads.

Mr Pichai noted at the conference that some companies are “already blowing through their annual token budgets—and it’s only May.” Consumers are not “token-maxxing” to anything like the same extent. But the more they use agents, the more providers of AI will need to come up with novel ways to make money from them. ■

Hot spell

Alphabet, market capitalisation, \$trn



Source: LSEG Workspace

Fashion

Loco for Coco

Chanel's creative revival is paying off

FEW FIGURES have been as influential in shaping modern haute couture as Karl Lagerfeld, creative director of Chanel from 1983 until his death in 2019. His 36-year stint at the *maison* brought not only fresh designs after years of creative stagnation, but also extravagant—and occasionally bizarre—fashion shows that wowed magazine editors and wealthy shoppers alike.

His long-time deputy, Virginie Viard, proved an uninspired choice as his successor. Although sales rose by half during her five-year tenure, that growth came largely through price hikes during the post-pandemic luxury boom. Fashion critics soured on the brand. Amid a wider downturn in the industry, sales at the world's second-biggest luxury label fell by 4% in 2024, with operating profit down by 30%.

Lately, however, Chanel has been regaining its shine. On May 19th the private company, which is owned by brothers Alain and Gérard Wertheimer, reported that its sales and operating profit were up by 2% and 5%, respectively, last year. In the first quarter of this year Chanel for the first time topped a ranking of the hottest labels in fashion maintained by Lyst, an e-commerce site. Two of its wares, a pair of square-toe pumps and a chunky handbag, numbered among the ten most desirable products as measured by social-media chatter. During the most recent Lunar New Year holiday in China, analysts at Bernstein, a broker, counted a 130% increase in foot traffic at Chanel's stores in luxury malls. Its shops in New York and London have experienced a similar up-swell in visitors.

For the change in Chanel's fortunes, the Wertheimer brothers can thank Matthieu Blazy, who took over as the label's creative director in April last year. Fashion commentators have dubbed the rapturous response to his collections "Blazymania". His success has served as a reminder that, in the world of high-end fashion, creativity and clever marketing are as vital as ever.

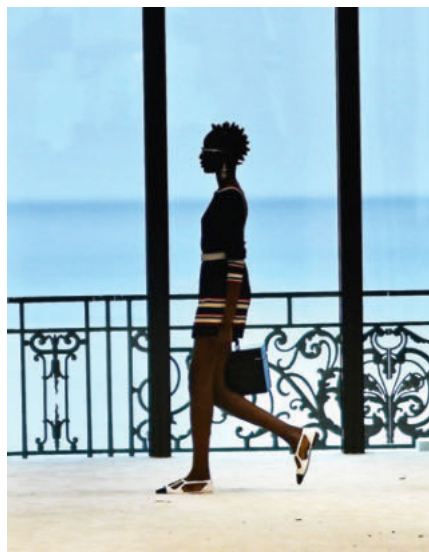
On April 28th Mr Blazy held his latest fashion show at the French seaside town of Biarritz, where he took house signatures such as tweed and lightened them up, sending semi-transparent suits and fringed skirts down the runway. Before the event, the company held lavish preview events for its "Very Important Clients", the high-rollers who account for an outside share of total sales. It also made sure that

new products, including some less expensive items, would enter stores quickly to make the most of the online buzz. Mr Blazy has proven adept at refreshing the brand's image in order to attract new customers without alienating existing clients by abandoning its hallmarks altogether.

Meanwhile, lots of Chanel's competitors continue to struggle. Many had hoped that 2026 would be the year in which spending on luxury items returned to growth. But the war in Iran has dampened consumer sentiment, not least among big-spending shoppers in the Gulf. Sales in the fashion division of LVMH, the colossus that counts Louis Vuitton, the world's biggest luxury label, among its *maisons*, were down by 2% year on year in the first quarter. For Kering, which owns Gucci, they fell by 3%. The buzz around Mr Blazy's new designs hasn't helped rivals. "Chanel is vacuuming most of the consumer attention," says Luca Solca of Bernstein.

Even Hermès, which had seemed immune to the luxury downturn, now looks to be under increasing pressure. Its sales in the first quarter of 2026 were up by 6% year on year, far slower than they had been growing. The brand has cultivated an air of exclusivity by limiting production of its coveted Birkin and Kelly bags, forcing shoppers to join lengthy waiting lists. But beyond the release of new colours each year, Hermès has not refreshed its core product range in decades. Now there are signs that demand may be waning. The resale price of Birkin and Kelly bags has been dropping.

By contrast, demand for Chanel's bags has been booming on second-hand markets. In the week after Mr Blazy's first show for the brand, searches on Vestiaire Collective, a resale platform, for its signature flap bag increased by 132%. Might Chanel have found its new Lagerfeld at last? ■



Making waves

Hospitality

Striking a new chord

The strange fate of Hard Rock Cafe

WHEN THE first Hard Rock Cafe opened in 1971 in London, the musical genre to which the restaurant chain pays tribute had just begun to hit the airwaves. Over the following decades, as the ranks of head-bangers swelled, hundreds of the eateries were opened in cities around the world.

Yet these days a visit to one of the restaurants feels as passé as an aged rocker performing yet another farewell tour. Patrons are treated to long-forgotten hits and underwhelming fries.

Several of the restaurants are now making losses, and some are closing. Last year Chicago's Hard Rock Cafe shut its doors after nearly 40 years. In January the branch in Manchester, opened in 2000, also closed, joining those in Paris, Sydney and elsewhere.

What is more surprising is that so many remain open. Still today there are Hard Rock Cafes selling pricey burgers and branded T-shirts in more than 50 countries. Some are run by independent franchisees, but many are operated by the company behind the brand, Hard Rock International. Against all odds, it looks to be thriving.

Back in 2007 Hard Rock International, which had fallen on tough times, was purchased for just under \$1bn by the Seminoles, a Native American tribe in Florida. They proceeded to build a hospitality empire across America and beyond that now encompasses 15 casinos and 36 hotels which use the brand. Visitors can play blackjack at the company's casino in the Dominican Republic or sip cocktails at its resort in the Maldives.

Such venues have become the real money-spinners for Hard Rock International, with the ubiquitous restaurants serving as a way to keep the brand alive in the minds of consumers. The launch in Florida in 2023 of an online-gambling app called Hard Rock Bet has presumably helped boost growth. It has since been rolled out in nine other American states.

In 2025 Hard Rock International generated \$7.9bn in revenue, estimates *Forbes*, up by a third from two years before. The golden age of rock and roll may be over. But don't expect Hard Rock Cafe to disappear soon.

Star Wars

Yodanomics

Disney turned its flagship film franchise into TV fodder. Can it go back?

SIX YEARS after the twin suns last set on Tatooine, one of Hollywood's most valuable movie franchises is returning to cinemas. Star Wars has not had a theatrical outing since "The Rise of Skywalker", which grossed \$1.1bn in 2019. With the release of "The Mandalorian and Grogu" on May 22nd, Disney will discover whether the brand has survived its long absence from the big screen.

After acquiring Lucasfilm and the Star Wars brand from George Lucas for \$4.1bn in 2012, Disney cranked out five new movies in the five years to 2019, grossing nearly \$6bn at the worldwide box office. But then it abruptly put the cinema franchise in the deep freeze, like Darth Vader suspending Han Solo in carbonite.

The long pause is partly the result of creative mistakes. "The Rise of Skywalker" had a plot that both irritated purists (the evil Emperor Palpatine, last seen plunging to his doom on board the Death Star in 1983, made a miraculous recovery for a final battle) and failed to tee up a sequel. Follow-ups were mooted, but directors and writers found Disney overprotective of its priceless intellectual property.

Yet the lack of movies also reflected a deliberate strategic choice by Disney. Star Wars may have been absent from theatres, but its production has gone into hyperspace on television. Starting with "The Mandalorian" in 2019, Disney has made seven live-action Star Wars series as well as

various animated ones. The shows have recruited millions of subscribers to Disney+, helping the legacy entertainment giant survive the streaming wars. Nielsen, a data company, estimates that Americans watched 33bn minutes of Star Wars content on TV last year (though for all Disney's recent efforts, the most viewed was the original film first released in 1977).

With "Grogu", Disney now faces the inverse challenge of turning a TV franchise into a spin-off movie. This presents several difficulties. Casual fans are struggling to keep up with the expanded Star Wars universe's many interwoven plotlines. A plethora of series has diluted the magic of the brand. And whereas America, where nearly half of households have Disney+, has fallen for the Mandalorian and his little green sidekick, other markets are less familiar with the material. "Grogu" is on track to make less in its opening weekend than "Solo: a Star Wars Story", another spin-off released in 2018 that was seen at the time as a bomb.

The new film's reported budget of \$165m, modest by Star Wars standards, at least lowers the bar for financial success. But Disney's new boss, Josh D'Amaro, who took over from Bob Iger in March, will be watching its performance anxiously. A bigger-budget Star Wars movie, "Starfighter", featuring A-lister Ryan Gosling, is due out next year. "Grogu" is an indicator of how it may fare.

What is more, for Disney, movies are not just about selling tickets at the box office. Success for "Grogu" would lead to more sales of \$40 Yoda dolls, more fans queuing for Star Wars rides at Disney's theme parks and more converts to a religion whose followers have kept the faith for 49 years. If Star Wars' long-awaited return to the movies fails, Disney will feel a great disturbance in the Force. ■

The electricity business

Power move

A new mega-deal shows how AI has turned utilities into hot property

AMERICAN UTILITIES are enjoying a "golden age of power demand" thanks to artificial intelligence, declared John Ketchum, boss of NextEra Energy, last year. That is why on May 18th his company, already the world's largest listed utility, announced that it would acquire Dominion Energy, a smaller power provider based in Virginia whose territory covers the largest data-centre cluster in the world.

The mega-deal, which values Dominion at \$124bn and the combined company at \$420bn, including debt, is the latest in the spate of supersize transactions that have been announced since Donald Trump returned to the White House in January last year. It demonstrates how AI has put the dull utilities sector at the heart of the world's hottest technology—and brought a political firestorm.

Perhaps no company has navigated America's power politics better than NextEra. Founded in 1925 as a local energy supplier in Florida, in the 21st century it reinvented itself as America's leading developer of wind and solar farms. More recently, with Mr Trump railing against "windmills", the company has rebalanced its efforts towards natural gas, nuclear power and battery storage.

NextEra's all-of-the-above approach has helped it win business from AI companies desperate for energy. In October it agreed to reopen a 615-megawatt (MW) nuclear plant in Iowa to power Google's data centres there. Soon after it announced 2,500MW-worth of generation and storage contracts with Meta.

Now it wants to go further. Since 2021 NextEra has increased its annual capital expenditure by more than half, spending \$25bn last year. The merger will allow it to make investments in Virginia that Dominion, which is weighed down by \$50bn in debt, has lacked the cash to fund. The combined firm could shell out more than \$220bn in capital expenditure between 2027 and 2030, reckons Fitch, a credit-rating agency, a significant increase on their current spending (see chart on next page).

A tie-up with Dominion would be helpful for other reasons. Rating agencies want at least 70% of a utility's business to be in regulated electricity markets, which provide stable cashflows. But NextEra's unregulated business—under which most of its AI projects fall—has been growing at roughly double the rate of its regulated ▶▶



Unfamiliar territory

arm, and the company is now getting close to the 70% floor. If its credit rating were to be downgraded, the cost of its planned investment binge would soar. Nearly all of Dominion's business is regulated.

The merger with Dominion is but the latest mega-deal NextEra has pursued this century. It sought to combine with Entergy in 2000, Constellation Energy in 2005, Hawaiian Electric in 2014 and Duke Energy in 2020. Each time it was rebuffed by the company or scuppered by regulators.

This time its target is amenable. But how regulators will respond is not yet clear. Although the Trump administration has blessed big mergers, NextEra and Dominion must also win over state regulators in Virginia and the Carolinas. Voters already blame the AI boom for raising their energy bills—nowhere more so than in Dominion's home territory of Virginia. The state's Democratic governor, Abigail Spanberger, harped on about making data-centre operators "pay their own way" during her campaign last year. Since the merger was announced, Democrats in Washington have begun talking about whether it will raise electricity prices.

NextEra has been labouring to persuade them otherwise. In its press release, it promised that the deal would "drive affordability" thanks to economies of scale, and said that Dominion's customers would receive \$2.25bn-worth of bill credits over the two years following the completion of the transaction. NextEra's renewable energy bona fides, which might help Dominion bring a beleaguered offshore wind development to completion, could also please climate-friendly Democrats. All that will probably be enough to win over regulators, according to Andy DeVries of CreditSights, a research firm, though perhaps not within the 18-month timeline the firms have promised.

If it can get its deal over the line, NextEra will set itself up as the leading power company of the AI boom. First it must convince Americans that the golden age of utilities will be equally golden for them. ■



European trains

Crossing the tracks

BERLIN

Can an Italian company disrupt Germany's broken railway industry?

LUCA CORDERO DI MONTEZEMOLO does not shy away from a challenge. The turbocharged Italian was Ferrari's chairman for more than two decades (and led its racing-car division to victory in 19 Formula One championships). He was also chairman of Fiat, Italy's biggest carmaker, headed Confindustria, the Italian business lobby, and orchestrated Rome's bid for the 2024 Olympics. But his latest ambition might be the boldest yet of his career: he wants to overhaul Germany's crisis-ridden railway industry by bringing Italo, the Italian high-speed rail operator he co-founded, into the market.

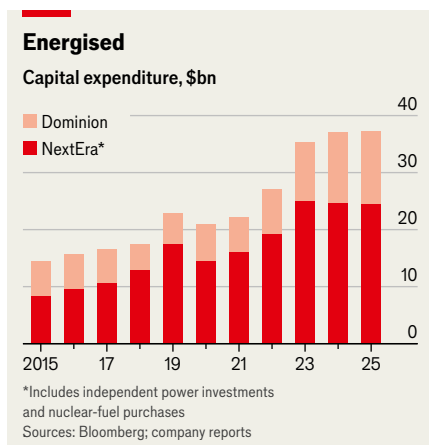
By 2028 Italo wants its new service to link 18 cities in Germany along two main routes: Munich-Cologne-Dortmund and Munich-Berlin-Hamburg. It plans to invest €3.6bn (\$4.2bn) in the venture, which will include the purchase of 30 high-speed trains from Siemens, a German engineering conglomerate, with the option to buy 14 more. Unlike the fiery red carriages it uses in Italy, those deployed in Germany will be a sensibly conservative blue.

Colours aside, the vision is largely the same. "We want to replicate in Germany what we did in Italy," Mr Di Montezemolo tells *The Economist*. Ferrovie dello Stato, Italy's state-owned railway firm, fought tooth and nail to prevent Italo, founded in 2006, from disrupting its business. Nevertheless, Mr Di Montezemolo and his co-founders prevailed in launching their service in 2012. Rail prices in Italy fell by some

40% over the following seven years.

Taking on Deutsche Bahn, however, may be an even bigger challenge. Admittedly, the state-owned German company has gone off the rails in recent years. "We have a dilapidated rail network, broken switches, signal boxes dating back to imperial times, excessive bureaucracy and an oversized administrative apparatus," said Evelyn Palla, its newish chief executive, in March. That is partly the result of decades of underinvestment. In 2004 Deutsche Bahn's annual budget for the building and modernisation of Germany's rail network was cut from €4bn a year to €1.5bn. Last year the government pumped a record €22bn into the company to compensate for its past stinginess, but Ms Palla reckons it will take around a decade to get the business back on track.

Some argue that Germans should not be left waiting for Deutsche Bahn to pick up steam again, and that Italo should be given a chance to disrupt Europe's biggest railway market. Yet Mr Di Montezemolo's plan depends on obtaining access to tracks, which is handled by DB InfraGO, an arm of Deutsche Bahn. Although it is overseen by the Federal Network Agency, a regulator, to ensure it operates fairly, the



We're hiring *The Economist* is seeking a tech-industry writer, to be based in San Francisco. Big ideas are essential; journalism experience is not (though it may help). For more information, visit economist.com/techwriter. The deadline is June 18th.

► tolls it charges rail operators are hefty, and it usually grants access on an annual rather than a multi-year basis. Mr Di Montezemolo says that his company is prepared to pay €250m a year for access to the tracks, but argues that it needs to be granted approval for several years in order to make its investment feasible.

Christian Böttger, a transport expert at the University of Applied Sciences in Berlin, thinks that Italo's case is compelling, given the beneficial effect its entry had on

the Italian market. But he doubts that it will get the necessary political backing for its project. Germany's rail network is already overburdened, he says. Italo's entry would also undermine Deutsche Bahn's high-speed-rail service and could knock the wheels off its integrated business model, in which it makes money from lounges at train stations and the BahnCard, a loyalty programme.

Ten years ago, the European Union adopted the Fourth Railway Package,

which focused on opening domestic markets to competition. Some countries have followed through: in Spain, the high-speed-rail industry is fought over by Renfe, owned by the government, as well as Iryo, which is majority-owned by a subsidiary of Italy's Ferrovie dello Stato, and SNCF, France's national railway company.

But others, including Germany, have dithered. Mr Di Montezemolo hopes to have a decision on track access this month. Without it, his plan may be derailed. ■

BARTLEBY

Almost always look on the bright side of life

Are optimists good for organisations?

IT PAYS TO be an optimist. Upbeat types tend to be in better health. A meta-analysis by Alan Rozanski, a cardiologist, and his co-authors found that optimism was associated with a lower risk of cardiovascular events. They also tend to be resilient. Optimists are likely to see setbacks as temporary and attributable to external circumstances, whereas pessimists regard reverses as a verdict on their own enduring weaknesses.

Optimists are more likely to rise up organisational ladders as well. In a recent paper Nadine Chochoiek of Munich Business School and her co-authors surveyed founders, bosses and employees in the Netherlands, and found that entrepreneurs and managers are as upbeat as each other. Both are more optimistic than employees.

Causality works both ways. Power itself is a source of optimism. It's easier to feel better about the future if you have an ability to shape it. One reason why bosses have a more positive attitude towards AI than workers is surely that they have more control over what will happen. But optimism also propels people onwards and upwards. Optimists are more likely than pessimists to be entrepreneurs. Low expectations of success and a decision to found a business tend not to go together. Daniel Kahneman, a Nobel-prizewinning psychologist, described "delusional optimism" as an engine of capitalism.

Confidence, justified or not, is a big part of why people are chosen for bigger jobs within organisations. The standard psychological test for measuring how optimistic or pessimistic people are is a short questionnaire called the Revised Life Orientation Test, which features statements like "If something can go wrong for me, it will". Would you follow

someone who strongly believes that they are cursed?

Optimism can plainly go too far. In an influential paper published in 2007, Manju Puri and David Robinson of Duke University used the gap between individuals' own longevity expectations and actuarial assumptions as a proxy for people's level of optimism. They found that extreme optimists were more likely to smoke than moderate optimists, and to keep a great share of their personal wealth in illiquid assets. Within organisations, too, excessive optimism often causes trouble. Unrealistic starting expectations make it more likely that projects will miss budgets and deadlines, for instance. Optimism also makes it less likely that failing projects will be canned; decision-makers have a habit of assuming better outcomes than originally planned to justify ploughing on.

A lot depends on the context. "What could possibly go wrong?" sounds much more worrying on the lips of a pilot than a podcaster. A study by Damiano Silipo of the University of Calabria and his co-authors quantified optimism at American

banks by looking at how much money they set aside to cover future loan losses. Optimism prevailed among bankers in the run-up to the 2007-09 financial crisis. Then, suddenly, it didn't.

There are plenty of ideas on how to counter optimism bias. Processes can help—in a "pre-mortem", for example, people deliberately imagine the failure of a proposed initiative and identify the most likely causes. Team composition also matters. A paper by Ulrike Malmendier of the University of California, Berkeley and her co-authors found that overoptimism on the part of the CFO is more predictive than a cocksure CEO when it comes to a preference for debt over equity. But the same paper found that overconfident CEOs tend to hire overconfident CFOs. If it's optimists all the way down, you have a problem.

Yet it is also possible to lean too far in the other direction. In his new book, "The Four Principles", Adrian Gore, founder of the Discovery Group, a big South African financial-services firm, argues that ingrained pessimism is a widespread problem in business.

People are conditioned to look for negative signals, reckons Mr Gore. Explaining why things might go wrong is seen as more sophisticated than believing that things will turn out well. Loss aversion, a strong behavioural bias against giving up what you already have, means that the scales are already tipped against risk-taking. Mr Gore thinks that the trope of learning from failures is overdone; successes teach you more. Performance appraisals ought to focus on putting people in positions that play to their strengths rather than trying to fix their weaknesses. Pessimism has its place, but it is optimism that makes things happen.



Finance & economics



Trump and the economy

Imagining America without the MAGA tax

WASHINGTON, DC

We calculate the drag on growth from fitful presidential policymaking

SINCE DONALD TRUMP took office in January last year, America's economy has continued to be the envy of the world. In 2025, while Britain, France and Japan eked out annual GDP growth of around 1%, and Germany all but stood still, American output grew by 2.1%. In the past 15 months American stockmarkets have broken record after record. And all this even as the president has unleashed seemingly anti-growth policies like mass deportations of migrant workers and chaotic trade wars.

Observers who had predicted economic disaster are left scratching their heads. Perhaps, some now whisper, the policies

are not as destructive as economists had assumed. Others wonder what might have been. For all its strength, America's economy could, on this interpretation, be doing even better. But how much better? Put another way: how big is the "MAGA tax"?

One way to arrive at a figure is to imagine what America's economy would look like in this levy's absence. Mr Trump inherited an economy that was growing strongly. It has since had three boosts, which *The Economist* has roughly quantified.

First, the artificial-intelligence boom. Capital expenditure by just four AI-giants—Alphabet, Amazon, Meta and Mi-

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crosoft—topped \$350bn in 2025 and should hit roughly \$700bn in 2026.

The binge has unleashed a wave of spending on data centres, chips, cooling systems and software. In 2025 real investment in information-processing equipment, software and data centres grew by more than 15%. In gross terms, this surge contributed nearly one percentage point to annualised GDP growth, accounting for about half of the economy's expansion.

This figure, though, overstates AI's true contribution to growth. Roughly two-thirds of data-centre spending is on equipment, much of it imported from Asian manufacturers like South Korea and Taiwan. When American firms buy this, most of the economic activity occurs abroad. To estimate how much the spending contributes to American GDP, we subtracted the rise in imports of real equipment from the surge in AI investment. By our reckoning, around \$50bn of the AI-investment boom in 2025 reflected additional domestic production, adding 0.2 percentage points to annualised GDP growth.

The AI frenzy has also fuelled America's stockmarket, the source of the second boost to growth. Between Mr Trump's election and the end of 2025, the S&P 500 index of large American firms rocketed by 15% in real terms—unusually fast by historical standards. That added \$5trn more to household wealth than would have accrued in a typical year. Americans tend to spend a small share of such windfalls. Still, using a conservative rule of thumb that each dollar of equity wealth raises spending by two cents in the first year, this probably raised consumption by \$100bn in 2025, or 0.5% of total consumer spending. Given shoppers' central role in America's economy, the wealth effect may have added 0.3 percentage points to growth.

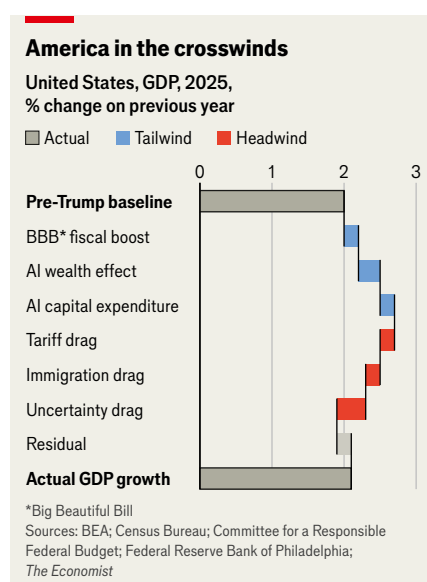
The third boost to America's economy has come from Mr Trump's growth-promoting policies. His administration has eased corporate mergers, ordered federal agencies to cut red tape and loosened constraints on private credit. The tax-cutting bill passed in 2025 injected trillions of dollars' worth of stimulus into the economy. It also probably improved the economy's rate

of long-term growth, by making pre-existing tax cuts permanent, restoring firms' ability to fully expense spending on research and development, and allowing them to depreciate assets more rapidly. All of this encourages investment. On average, the independent forecasts we reviewed—including those by the Congressional Budget Office, Tax Foundation, the Tax Policy Centre and the Yale Budget Lab—estimated that the legislation would add 0.2 percentage points to GDP growth in its first year and 0.4 percentage points in 2026.

Combine all three boosts and America's economy ought to be breaking the decibel-meter. Before the presidential election—and before economists were able to size up Mr Trump's ideas—the consensus forecast was for 2% growth in 2025. Adding in AI investment, soaring stockmarkets and tax cuts might have got America to 2.7%. That is more than a half percentage point faster than reported growth.

Another way to calculate the MAGA tax is to try to capture its economic drag directly. Economists have done this for some of Mr Trump's policies. According to the Peterson Institute, a think-tank, his tariffs reduced real GDP growth by about 0.2 percentage points in 2025, by squeezing household purchasing power and compressing firms' profit margins. The Brookings Institution, another think-tank, estimates that in 2025 the president's deportations and border shutdown turned net migration negative for the first time in at least half a century. This reduced labour supply and, since migrant workers spend money, consumer demand. All this may have slowed growth by 0.2 percentage points.

Such figures are instructive, but they do not capture the full cost of uncertainty stemming from Mr Trump's erratic policy-making. Tariffs are announced, delayed, revised and revived. Immigration agents are deployed, recalled and redeployed elsewhere. Wars are waged. An index of economic-policy uncertainty developed by Scott Baker of Northwestern University



and colleagues rose by over 100 points from before Mr Trump's election to the end of 2025. Swings of that magnitude are typically followed by growth in business investment slowing by five to ten percentage points, as firms postpone capital spending and supply-chain adjustments.

Indeed, strip out the splurge on information-processing gear and software—the categories most closely tied to AI—and the picture looks grim. Over the past four quarters non-residential fixed investment, excluding AI-related categories, has contracted at an annualised rate of roughly 3%. It grew by over 5% per year during the previous decade. Investment in industrial and transport equipment has fallen by more than 2% over the past year. Manufacturing construction is down by 20%. In total, non-AI investment is running about \$130bn below its trend from the last decade. This capex recession is reducing GDP growth by 0.4 percentage points.

Could AI itself explain the weakness? The contraction in non-AI investment is too large and broad to be down to firms

merely reallocating capital from other sectors towards data centres. It spans oil and gas, carmaking and factory construction. Trade-policy uncertainty probably played a large role. In a survey a year ago, the Federal Reserve Bank of Atlanta found that 45% of executives planned to cut capital spending owing to policy uncertainty.

Another potential explanation—that strong demand and heavy government borrowing have pushed up interest rates and crowded out private investment—is also unpersuasive. Credit remains plentiful. Investment-grade corporate borrowing has seldom been so cheap, compared with Treasury yields, since the 1990s. It is therefore a good bet that presidential policy-making is to blame for sour sentiment.

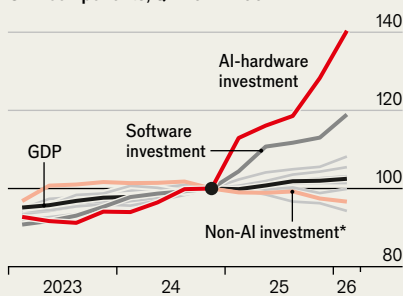
Together, the squeeze from tariffs on real incomes, reduced labour supply and capex-shy companies adds up to 0.8 percentage points. That is in line with the earlier figure we arrived at by comparison with a counterfactual American economy. Looking ahead, there is little sign of relief. Tariffs remain in flux, sustaining high uncertainty for businesses and households. The war in Iran and the closure of the Strait of Hormuz have triggered an energy shock that will further compress real incomes and corporate margins, dampening investment even more.

A natural reaction to such figures is to despair at how much damage bad policies can cause. Another, though, is to marvel at the awesome power of America's economic engine. Despite everything Mr Trump has put in its way, GDP may grow at an annualised rate of 4% in the current quarter, if you believe the latest real-time forecast by the Federal Reserve's Atlanta branch. Without the deadweight of the MAGA tax, in other words, America might be rocketing ahead with annualised growth of nearly 5%. It has notched up such performance in just nine quarters this century, and only five if you exclude the recovery after the covid-19 pandemic. If only the president would let it, it could be doing so again. ■

Artificially enhanced

United States

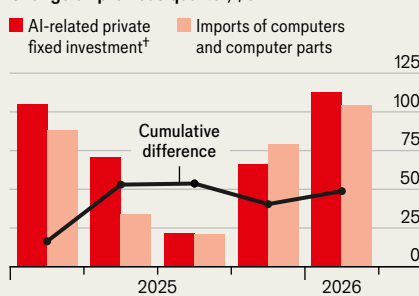
GDP components, Q4 2024=100



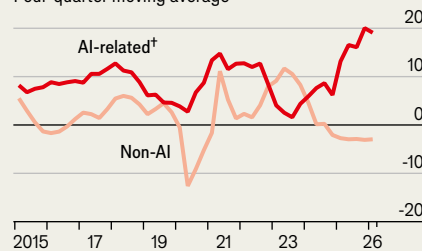
Sources: BEA; Census Bureau

*Private non-residential investment minus investment in information-processing equipment and software

Change on previous quarter, \$bn



Private non-residential fixed investment, % change on previous quarter, annualised Four-quarter moving average



*Information-processing equipment, software and data centres

Specialty insurance

Loss leaders

The underwriters on the hook for war in Iran

THE INSURANCE industry exists to contemplate worst-case scenarios, but the events of the past three months have nevertheless been hard to take in. The Iran war has unleashed havoc in the Middle East. It has also presented two classes of specialty insurer—those offering protection against political violence and marine war, which covers ships in conflict zones—with their biggest shock in decades.

The impact on those who insure against political violence has been most dramatic. Such policies protect businesses from terrorist attacks, sabotage, civil unrest, war and other nasty events. Since Iran started firing missiles at its neighbours, local firms in the Gulf have scrambled to find coverage for exposed infrastructure and property. Their clamour, and the higher risks they now face, has transformed the market. Insuring some assets now costs 40 times the pre-war rate.

Some of this is because previous prices were too low, says Fergus Critchley, head of terrorism and political violence at WTW, an insurance broker. A run of strong performance in the sector had brought in competition, pushing down underwriters' premiums. They had room to fall, in turn, because few foresaw a conflict on anything like the present scale.

Claims are thought to run into the billions of dollars, with many stemming from attacks on energy infrastructure. For a niche industry that generates about \$1.2bn of annual income from premiums, the resulting losses may erase years' worth of revenue. Many insurance contracts also promise reimbursement for indirect losses, such as interruption of business. The Gulf is packed with offices belonging to multinationals, so such payouts will be steep. The market will tighten as a result. Contracts now being signed are dearer than in peacetime and offer narrower coverage.

What if peace is declared? Mr Critchley says that premiums would probably stay at their current, expensive levels for at least three to six months. In part this is because any news of a peace deal would meet significant scepticism. Even if a lasting one is struck, insurance would not be as cheap as before the war. Oleksii Omelianchuk of FortuneGuard, a firm that assesses conflict risks, points out that insurers who did not price in the risk of the present war are unlikely to make the same mistake twice.

Whether war or peace prevails, politi-

cal-violence insurers may soon scale back. They typically negotiate their contracts with reinsurers—to whom they offload some of their risk—each year on January 1st. When this date rolls around in 2027, reinsurers may well reduce the coverage they offer and raise prices, just as insurers are doing already. Oliver Martin, head of political violence at Atrium, an underwriting syndicate, says that could deter insurers from signing bumper policies today, since they may not be able to reinsure them after next year's renegotiation.

Prices have also risen sharply for insurance against marine war. For ships attempting to traverse the Strait of Hormuz, premiums for such protection are now often between 10 and 20 times the pre-war rate, according to WTW. So far the losses sustained by marine-war insurers are nowhere near as steep as in the political-violence market. Stale Hansen, boss of Skuld, one such insurer, says that even the total loss of a ship would not be a "game changer" for the market. Oil tankers are not hugely expensive, often fetching between \$80m and \$120m apiece. For comparison, marine insurers collect premiums totalling around \$40bn per year. And few skippers, after all, have yet risked destruction by attempting to cross the strait.

As a result, marine insurers may do well from the war. But the longer it lasts, the more uncertain their prospects become. Many policies have provisions for "blocking and trapping". These trigger payouts after a vessel has been stranded for a set period—usually six or 12 months—after which it is deemed lost. Should Hormuz stay shut for long enough, such provisions might apply to 2,000 or so ships stuck behind it. After Russia invaded Ukraine, blocking-and-trapping clauses eventually led to hundreds of millions of dollars in insurance losses. And far more boats are now in the Gulf than ever were in the Black Sea, threatening much bigger payouts.

Insurers are therefore eager for a programme to help vessels safely escape. Some of those stranded in the Black Sea by war in Ukraine were eventually able to leave under a UN-brokered scheme. Government support for a similar scheme in Hormuz would help. However, the most recent attempt by President Donald Trump—his short-lived Project Freedom—provoked Iran to launch more missiles and was then promptly called off. Progress in peace talks, meanwhile, seems about as dependable as a fair wind. It may be a long time before skippers and their insurers have room to relax. ■

Correction In a leader last week we wrote that a South Korean presidential adviser had backtracked on a proposal for AI dividends. In fact, the details of the proposal were at first widely misreported. There was no backtracking. Sorry.

Chinese industry

The pterodactyl's downdraft

HONG KONG

Does China's manufacturing success leave space for anyone else?

"WILD GEESE fly in orderly ranks," noted Akamatsu Kaname, a Japanese economist, when contemplating the progress of Japan's textile industry in the 1920s and 1930s. The birds' V-shaped formation has served as a popular analogy for how manufacturing spread in East Asia. As Japan prospered and its wages rose, labour-intensive industries migrated from the leader of the flock to followers like South Korea and Taiwan.

Many expected China to provide such an updraft to poorer countries. Now they have doubts. Despite China's rising wages and sophistication, it "still commands a historically unusual share" of low-end manufacturing, according to a new paper by Shoumitro Chatterjee of Johns Hopkins University and Arvind Subramanian of the Peterson Institute for International Economics. Even as the likes of Germany fret about China's entry into high-spec industries, poorer places wonder when it will vacate lower-end niches they want to fill.

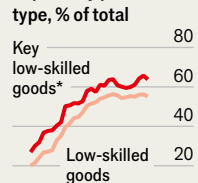
Manufactured goods are sometimes divided into four types: low-, medium- and high-tech, plus those based on natural resources, such as refined petroleum. Between 2010 and 2024, China's share of global exports rose in all four categories, according to a paper by Yu Fei and Guo Kai of the CF40 Institute, a Chinese think-tank. Its "manufacturing appears to be competing with all countries", they wrote.

In some cases, China's share is even bigger than it appears from raw export figures. The country makes valuable parts for labour-intensive goods that may be

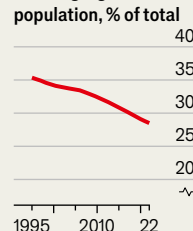
Rags and riches

China's share within group of 30 developing economies

Exports by product type, % of total

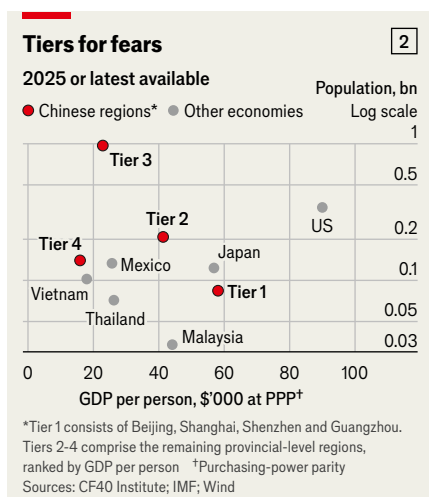


Working-age population, % of total



*Clothing, textiles, leather, footwear, etc

Source: "China's mercantilist squeeze on developing countries", by S. Chatterjee et al., Peterson Institute for International Economics, 2026



Housing in America

Home economics

What economists can learn from the world's biggest DIY store

HOME DEPOT is not an ordinary retailer, but a barometer for America's housing market. Comparable sales at the world's biggest DIY chain fell sharply in 2007, before many on Wall Street cottoned on to the coming housing crash. More recently its revenues surged along with the boom that followed the covid-19 pandemic.

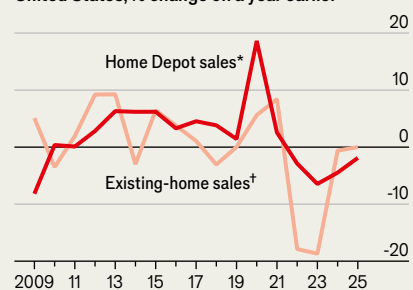
Since then Home Depot has again felt the strain. Though home prices have held up, the home-buying frenzy subsequently ground to a halt. Annual transactions fell by 20% in 2022 and by 30% in 2023. Mortgage rates have fallen from their peak in 2023, but sales of existing homes over the past two years have been the lowest in three decades, according to the National Association of Realtors. Speaking in April, Richard McPhail, Home Depot's finance chief, put it plainly: "We have never seen housing activity this slow for this long."

All this means fewer DIY projects. Favourable currency exchange rates boosted revenue growth in Home Depot's latest fiscal quarter, for which it released results on May 19th. But strip these out and annual sales in America have been falling or flat for three years. Customer transactions last year were 9% below a peak in 2021.

Worse, there is little sign of improvement. Home Depot expects comparable sales to be unchanged in 2026, with a best-case scenario of 2% growth. Its share price has plunged by a quarter from a peak last

Screwed

United States, % change on a year earlier



*Inflation-adjusted comparable-store sales.

Fiscal years ending Jan/Feb

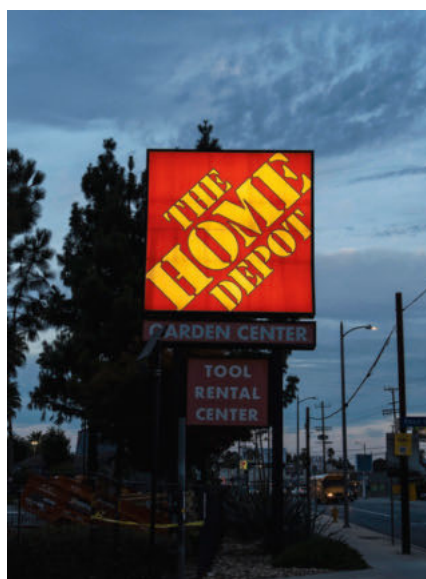
†Volume

Sources: LSEG Workspace; company reports

► stitched or bashed into final form elsewhere. China provides 64% of the value embodied in all the garments, textiles, leather and similar goods exported from 30 low- and middle-income countries for which suitable data are available. This percentage is far higher than China's share of the working-age population for this income group (see chart 1).

What explains China's failure to fly in an orderly rank? One factor is its sheer size. China added "a demographic pterodactyl to the flying geese", noted Lant Pritchett, a development economist, in 2010. Another factor is the staggered nature of its development. Some parts of the country have become well-to-do, even as others remain relatively poor. Ms Yu and Mr Guo point out that China's four richest cities (with a combined population of 84m) have a GDP per person that exceeds Japan's. Its poorest four provinces (population 140m), meanwhile, are closer in income to Vietnam (see chart 2). China, they say, is equivalent to 0.7 Japans, nearly six Malaysias, five Mexicos, four Thailands and 1.4 Vietnams. No wonder it competes with all of them.

But there is more to it than that. Workers in those five countries cannot easily move between them. Chinese ones, on the other hand, are relatively free to migrate from its Vietnam-equivalents to its Malaysias and Japans. Their ability to depart for higher-wage provinces should raise pay in low-end industries, rendering them uncompetitive. China's policymakers once seemed resigned to this evolution. Now they seem determined to resist it. "We must continue to transform and upgrade traditional industries," said President Xi Jinping in 2023, "and avoid simply treating them as 'low-end industries' to be abandoned". One way to offset wage costs is automation, robotics and other technological innovations—so China increasingly strives to make low-end goods with high-tech methods. Wild geese fly in orderly ranks. But not if drones get in the way. ■



Leading indicator flashing red

year. Shoppers still snap up smaller items like gardening tools and paint. But they are not starting large projects such as kitchen and bathroom renovations—because the housing market is gridlocked.

Dearer mortgages are partly to blame. Homeowners with pandemic-era fixed rates of 3% or below are reluctant to sell and abandon them, now that the average rate on a new 30-year mortgage has risen to nearly 6.5%. Such rates also put first-time buyers off. More than half of outstanding mortgages still had rates of 4% or less in the third quarter of 2025.

The more expensive mortgages are, the less buyers can afford to pay for a house. But "sellers are extremely reluctant to realise losses from their homes," says Tarun Ramadorai of the London School of Economics. This has also frozen the housing market in Britain, where transactions remain subdued. In both places, homeowners' loss-aversion has contributed to prices not moving much. Instead, their unwillingness to sell at a loss has caused sale volumes to collapse.

Before the Iran war, America's housing market looked ripe for a recovery. Unemployment was low, existing homeowners were sitting on near-record equity and affordability was improving, in part because of rising incomes. But the closure of the Strait of Hormuz has sent energy prices rocketing, threatening to drive up inflation and, in turn, mortgage rates. And so the housing market remains on ice.

What might prompt a thaw? Some homeowners have little option but to sell up: those undergoing divorces, for instance, or moving for work. Home Depot, however, cannot rely on this trickle of transactions. So it is instead courting builders, who tend to spend more than the DIY crowd and now account for half of its revenue. Over the past few years, the firm has spent over \$30bn acquiring wholesale distributors that cater to professional contractors. But for now, plenty of its tills remain empty—and offer a warning that America's housing market is broken. ■

Expat migration

New oases

BERLIN, DUBAI AND SINGAPORE

Want to escape Dubai? Here's where you might end up

LIFE IN DUBAI used to be about as blissful as white-collar expatriate existence gets. The private schools are good, beaches pretty, flight connections plentiful and booze legal (so long as you are not Emirati or Muslim). Expats face no income tax; it barely rains; Russian oligarchs can mingle with Western bankers, Arab property moguls and Israeli entrepreneurs.

All that remains true nearly three months after America and Israel attacked Iran across the Gulf. But Dubai's advantages must now be weighed against the risk of Iranian missiles and drones raining down. On May 18th one hit an electricity generator at the sole Emirati nuclear power plant.

Although most strikes have been intercepted before they could do real damage to Dubai and other parts of the United Arab Emirates (UAE), many footloose foreign residents have already scattered. A few nabbed seats on the last flights out to America or Europe. Others drove to Muscat, in neighbouring Oman, in search of alternative escape routes. Many hoped to return once the hostilities ceased. As these drag on, however, plenty are looking for a new, more peaceful bolt-hole. Where are they headed? And will they ever return?

The UAE does not disclose detailed statistics on foreign residents but estimates suggest that, before the war, perhaps 3m-4m of the country's 12m people were well-heeled outsiders and their families. More than 240,000 of them were millionaires. Dubai received the lion's share of these arrivals. It now probably accounts for an outsize proportion of the departures. Dominic Volek of Henley & Partners, which advises the footloose affluent, reports that enquiries about other jurisdictions from UAE-based residents have risen by over 40% in recent weeks.

More than 35 countries are now competing for the rich and enterprising, says Jean-François Harvey of Harvey Law Group, a firm of immigration lawyers. Long-standing destinations like New Zealand ("at the end of the world, mostly self-sufficient and outside a nuclear blast radius", in the words of one consultant) and Malta have new competition. The Maldives is launching a permanent-residency-by-investment scheme this year. Argentina is also expected to offer deep-pocketed investors citizenship soon. On April 24th Turkey proposed a 20-year exemption to taxes on foreign income and capital gains

for some foreigners. Since the war began about a dozen clients of Mr Harvey's clients have acquired Turkish citizenship by buying a house there.

Milan is particularly popular. "There is an increase in the movement of people out of Dubai," says Roberto Bonomi, a tax lawyer in Milan at Withers, a British law firm. Diletta Giorgolo of Sotheby's International Realty, a posh estate agent, says that interest in Italy from the Gulf has shot up in recent months, compared with a year earlier. What were initially mostly requests related to temporary rentals are increasingly connected "not only to short-term considerations, but also to longer-term lifestyle and investment planning", she explains.

In contrast to sleepy rivals, Italy's capital of both fashion and finance has networks to help those who want to enlarge their fortunes, rather than merely spend them. In recent years American hedge funds such as Millennium Management have set up shop in the city, allowing wealthy traders and fund managers to take advantage of Italian tax rules. High-earners there pay a relatively modest flat tax of €300,000 (\$349,000) per year on their entire foreign income. Parents can now pick between American, British, Canadian, French and German international schools. The weather is tolerable, too.

EU citizens can move to Milan at their leisure, which is why Dubai's European

transplants like it so much, says Mr Bonomi. For non-Europeans the most common way to secure residency is to invest €250,000 in an Italian startup or €500,000 in a more established firm. Alternatively, you can donate €1m to an Italian charity or park €2m in Italy's government bonds.

Another option is Singapore. The city-state had lost out to Dubai in recent years. Heavy-hitters from India and mainland China, in particular, were drawn by the emirate's glitz, permissive rules and business opportunities. Singapore's stricter social mores and the government's obsession with a squeaky-clean image made it seem stuffy by comparison.

That image now looks like a strength. So are Singapore's efficient government, predictable legal system and excellent wealth managers. Big Singaporean banks, such as OCBC, report an uptick in net wealth inflows from Dubai. Singapore's gold imports from the UAE have quadrupled since January.

Ryan Lin of Bayfront Law, a firm in Singapore, says new client inquiries have leapt by a third in the past two months. His existing clientele, mostly newly rich Chinese people, are increasingly interested in leaving the Middle East. Rich Indians, 3,500 of whom leave the country each year with \$1m or more in the bank, also seem keen to give Singapore another look. Mukesh Ambani, India's richest man, opened a family office there in 2022.

Places like Milan and Singapore are not perfect substitutes for Dubai. Russian plutocrats are frowned on in Italy, as in the rest of Europe, with Vladimir Putin waging war on Ukraine. Other wealthy foreigners may fear that next year's elections could usher in a government that abolishes the flat-tax regime. Even the current, friendly one raised the tax on foreign income from



Get me out of here

▶ an initial €200,000 this year.

Singapore, for its part, taxes income at 24% and charges foreigners an eye-watering surcharge on property sales. It has also tightened money-laundering rules in the wake of a \$3bn scandal in 2023, and might be nervous about letting in a rush of dodgy money from Dubai. A law passed in 2024 lets the police prowl through tax and customs data. In recent years 80% of licence applications by crypto firms in Singapore were declined or withdrawn, according to

the Financial Action Task Force, an anti-money-laundering watchdog. "Some investors liked the Emirates because they didn't ask too many questions," says a private banker in Singapore. Moving to the Asian metropolis might feel like it involves a "proctology exam".

"I think crypto wealth will stay in the Middle East," reckons Mr Lin. So will some other sorts. Wealth managers, themselves a loaded bunch, need to be close to their clients, says one who expects his firm's

pinstriped legions to return to Dubai before long. Many foreign firms that allowed similarly well-off employees to "work from home", meaning Milan or London, in the first months of the war expect them to get back to the office in Dubai.

"Time is a great healer," says another wealth manager. Perhaps. But the longer the wound of war remains open, the likelier expats are to be scarred by the experience. In the meantime, many will prefer to convalesce somewhere less hot. ■

BUTTONWOOD *TIPSy-turvy*

Why don't inflation-scarred investors buy more inflation-protected bonds?

CONVULSING BOND markets mean different things to different people. If you run a government, it is plainly bad that governments' borrowing costs are shooting up—as they are across the world. The yield on Japan's 30-year debt has risen above 4%, its highest ever; America's has topped 5%; Britain's is chugging fitfully towards 6%. Since mortgage rates tend to move in line with these yields, anyone looking for a new one will also feel the pinch. For bond traders, in contrast, the past few weeks have been quite the thrill. A decade ago near-zero interest rates seemed to have permanently shoved their market into the deep freeze. Now it is red hot.

It is not hard to see why. Inflation and yawning government deficits have worried investors for years, and the Iran war is making both problems worse. Bank of America conducts a monthly survey of global fund managers; in the latest, 40% of respondents said an inflationary wave was the biggest tail risk for markets. Despite this (or in explanation of it), 78% expected America's Federal Reserve to either cut rates or leave them unchanged over the next year. They have acted accordingly, slashing holdings of cash and bonds, while raising their equity allocations by more than in any other month of the survey's 25-year history. This is just what you might do if you think money is too loose.

Given all this, it may seem puzzling that one market hasn't moved more. Inflation-linked government bonds ("linkers", or "TIPS" in America) offer the ultimate protection against investors' fears: payments that are guaranteed to rise by the same amount as consumer prices. They therefore offer real yields, as opposed to the nominal ones on ordinary bonds, at which inflation nib-

bles away. The nominal yield minus the real one—for, say, ten-year American Treasuries—is widely thought of as the average inflation rate investors expect over the bonds' remaining life.

Fed officials might draw comfort from this ten-year "break-even". At 2.5% it is roughly in line with their target and only a few tenths of a percentage point higher than it was at the start of this year. But that would be a mistake. The TIPS market is not subdued because investors are relaxed about inflation. Rather, it offers a cautionary tale about the limits of market efficiency—and therefore about using prices to forecast the future.

As Exhibit A, consider how poorly TIPS-derived break-evens have sometimes performed at this in the past. During the panic that accompanied the first covid-19 lockdowns, in 2020, America's ten-year break-even hit 0.5% and the two-year equivalent fell to -1%. In fact, barely a year later, the biggest inflationary surge in decades took off, and in 2022 the annual rate peaked above 9%.

No trader has perfect foresight in the

teeth of a crisis. Exhibit B, however, is that even one who did would struggle to act on it when it comes to inflation break-evens. Compared with the market for normal Treasuries, that for TIPS is small, illiquid and lacking in arbitrage opportunities. They account for just 7% of American government bonds and are favourites of pension schemes, which hold them in order to guarantee retirement payouts and so might not sell at any price. In theory, arbitrageurs should smooth out any distortions this causes using derivative contracts. In practice, trading inflation derivatives throws up so many complications that opportunistic traders often cannot find a broker willing to take them on.

Exhibit C is the unique function of ordinary Treasury bonds. Investors do not only, or even mainly, use them to bet on the future path of interest rates and inflation, but as a safe harbour when other markets are turbulent. They also trust that, during a big stockmarket crash, interest rates would fall and Treasury prices would rise, cushioning their losses. Erratic policymaking in America has tested both beliefs recently. But not to destruction: people still hold Treasuries with an eagerness that does not apply to TIPS. So Treasury yields are tugged down and those of TIPS are not, meaning the break-even—the gap between the two—is depressed.

The effect is especially strong during a panic, when the safety of Treasuries is most appealing. Almost certainly, this helps explain why break-evens fell so low in 2020. If you want to guarantee your purchasing power no matter how far consumer prices rise, linkers are just the thing. If you want to know how much investors fear inflation, you may need to look elsewhere.



FREE EXCHANGE

Amoral calculus

Economists prefer to treat morality as an outside constraint. This isn't always possible



WHY IS BUYING heroin easier than hiring a hitman? Both are illegal. Yet find yourself in an unfamiliar town and ask around where you might get your fix and—possibly after being met by a few disgusted glances and wide-eyed stares—you are eventually given a phone number or directions to some sketchy area. Even if someone bothers to report you to the police, the constabulary is too busy to follow up on a single user. Start asking where you can get the services of a contract killer, however, and if you receive any leads at all, you are most likely to find yourself talking to an undercover policeman.

This is one of the questions asked in “Moral Economics”, a new book by Alvin Roth, a Nobel-prizewinning economist at Stanford University. Mr Roth’s Nobel was for his work on market design, particularly in those markets where two or more parties must be paired up in the absence of price signals (think university places or marriage). His studies led him to the trade in human organs—he set up a kidney exchange in which donors who are not a match for their respective intended recipients can swap—and, more broadly, to the murky world of “repugnant transactions”. These Mr Roth defines as ones where, like scoring heroin but not securing a hitman, buyers and sellers would happily transact but someone else not directly harmed by the exchange wishes to prevent it.

Mr Roth’s subject is a good one. Morality clearly shapes markets far beyond those for heroin and hitmen. Like all of his discipline, he makes moral economics about trade-offs: are the harms of allowing an activity greater than the harms of disallowing it? After all, he writes, people “can’t generally expect to reach a consensus” about the principles at stake. The clear-eyed evidence he gathers is a useful place from which to argue about when prohibition works. Yet despite his attempts to “take moral and ethical concerns seriously”, the book too often treats other people’s principles as an irrational bias to be managed rather than a reasoned position to be argued against. That is a missed opportunity.

Some of his frustration is understandable. What different societies consider repugnant can seem arbitrary. Horsemeat, a delicacy in parts of Europe, is off the table in California following a referendum in 1998. Many European countries forbid surrogacy on

the ground that it violates the dignity of women to make the body into a tradable commodity, yet they permit some forms of sex work. In America it is the other way around: prostitution is illegal everywhere except for a few counties in Nevada whereas surrogacy is widely available. California allows paid sex—so long as it is being used to make pornography. Americans can be paid to donate blood but could not partake in “challenge trials” for covid-19 vaccines, tested on donors who expose themselves to the disease. British blood donors can count on at best a biscuit but are allowed to offer their life for a fee to see if a jab works.

Such inconsistencies infuriate ethicists trying to find a coherent set of moral intuitions. For Mr Roth, they are a godsend. They provide an opportunity for some field experiments to see what happens when markets are legal or not and what kind of black markets come with them. Some rules of thumb emerge: bans never work perfectly but they do reduce the frequency of the activity. There are usually ways around restrictions for the suitably motivated. A forum for Kazakhs in America suggests trying one or two Uzbek restaurants in Philadelphia for your horsemeat fix, or heading to Quebec. Horse lovers in California can content themselves that fewer are being eaten locally.

Mr Roth suggests, reasonably, that policy should weigh the costs and benefits of banning an activity. He cites a paper by Scott Cunningham of Baylor University, in Texas, and Manisha Shah of the University of California, Berkeley, that took advantage of an accidental legalisation of indoor prostitution in Rhode Island after state lawmakers did not fully think through the implications of a bill. The researchers found that the result was fewer sexual assaults and sexually transmitted infections. This argues for policy experiments with licensed sex work. Where evidence is scarcer or more ambiguous, as with prohibitions on drugs that are both addictive and lethal, he suspends judgment.

Too often, though, Mr Roth retreats to the position that there is not much point in discussing the principles because there will be no agreement. That makes morality an exogenous fact of life, something generated outside the economic model. It is a constraint, like geography, that must be recognised by economists. Sometimes, on Mr Roth’s view, it is useful: in contrast to this newspaper, he seems sceptical about legalising many drugs. Sometimes it is deadly: hundreds of thousands die every year because they cannot get the organ donations they need.

An accounting for taste

Mr Roth’s reluctance to engage in arguments about principles is a shame. For one thing, he has nothing to be ashamed of. The picture that emerges from the book is of a deeply moral person, who believes in bodily autonomy, in not subordinating individual lives to a collective and in not accepting unnecessary deaths to spare some people from feeling squeamish.

Perhaps more important, it lets Mr Roth skirt an uncomfortable question about the role of repugnance. He is right that it often serves to exercise social control: dietary restrictions let an in-group that shuns pork, or beef, or meat, ostracise the out-group that does not. But it is also a tool of social cohesion. Emile Durkheim, a French sociologist at the turn of the 20th century, argued that societies divide the world into the sacred, covered by ritual and taboo, and the profane, the ordinary. One that treats everything as tradable might struggle to sustain the tacit co-operation on which markets, repugnant or otherwise, ultimately rely. ■



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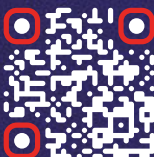
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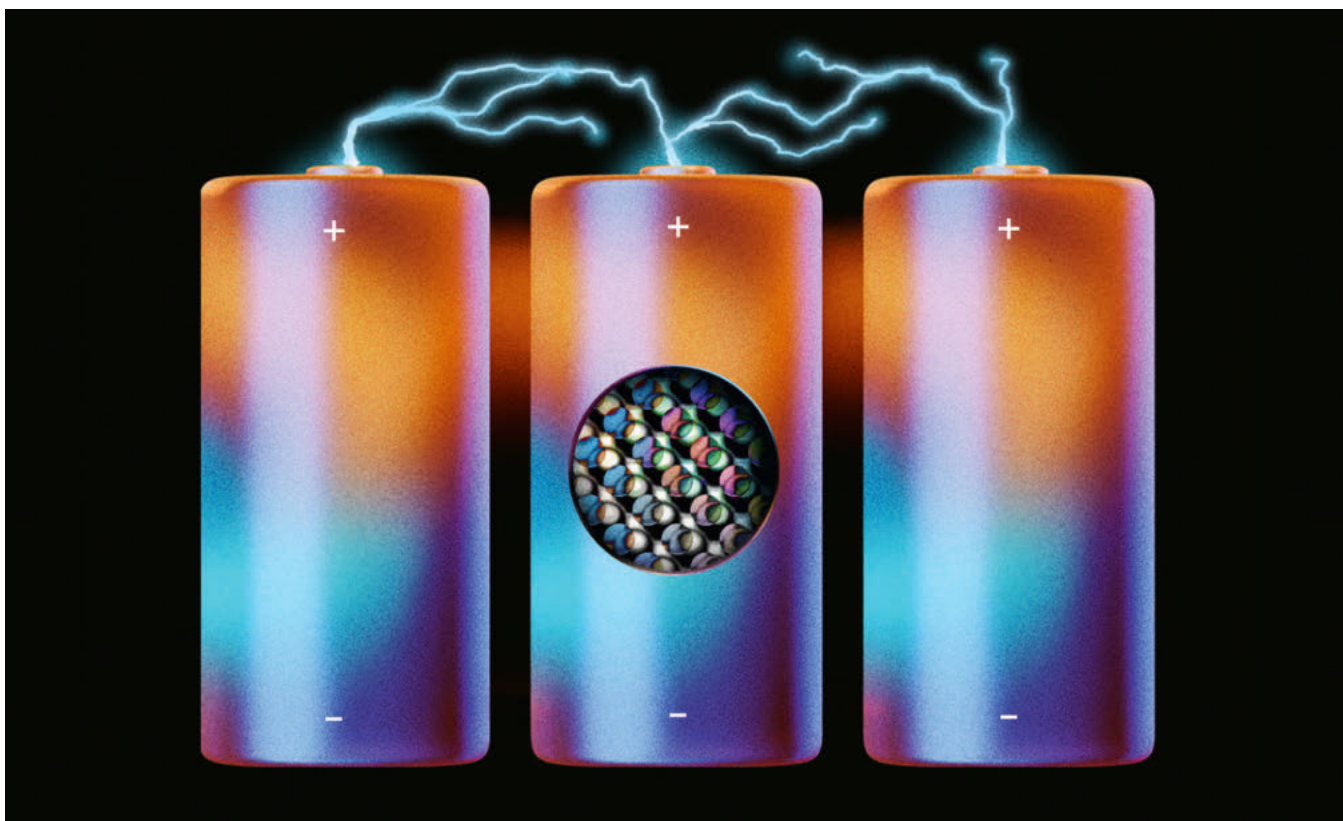
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Science & technology



Technology

Charging ahead

Advances in materials science could at last bring solid-state batteries to market

LIKE ANY champion who spends too long at the top, the lithium-ion battery is stagnating. Over decades as the battery of choice in everything from smartphones to electric cars and drones, its design has been tweaked countless times to improve its energy density and performance. But, some scientists say, those improvements are approaching their theoretical limits. Even the best models are prone to dying out in the cold, rapidly losing capacity or—as is the case for those in household devices—spontaneously catching fire.

At the same time, demand for batteries has never been greater. 30% of cars sold in 2026 are expected to be electric vehicles (EVs) which rely on them for power. Last year American homes and businesses installed a record number of big batteries. According to Wood Mackenzie, a consultancy, by the end of the decade installations could rise by almost 40%. Worthy challengers are desperately needed.

Advances in materials science are at last bringing some within reach. Battery-builders are modifying existing materials and creating novel combinations to design batteries that store more energy while being safer and more stable than anything on the market today. The lithium-ion battery's crown may be up for grabs.

Solid-state batteries are among the most exciting alternatives. When a conventional lithium-ion battery is charged, lithium ions migrate from the cathode to the anode; when it is discharged, they return. The medium the ions shuttle through is called the electrolyte, usually a flamma-

ble, organic solvent soaked into all of a battery's components. In solid-state batteries, however, the anode, cathode and electrolyte are compressed together as slabs. This means more conductive materials can be packed into the same space, allowing for energy densities as high as 500 watt-hours per kilogram (Wh/kg), compared with about 300Wh/kg for liquid electrolytes. They are also less likely to combust.

Although solid-state batteries have been studied for decades, researchers have thus far been able to make only tiny versions for use in such devices as medical implants. The most significant barrier to scaling them up is brittleness. When cells are charged and discharged, the ions repeatedly embed themselves in the electrode material. That causes the battery to expand and contract, creating voids between the components that can lead to cracking and deformation. This slows down the ions and degrades the battery's performance.

In January researchers at the Shenzhen Institutes of Advanced Technology, part of the Chinese Academy of Sciences, took a big step towards overcoming the brittleness problem. They created a high-performing electrolyte material by alternately stacking layers of ceramic 1-100nm thick with similarly thin sheets of polymer. The stack was then placed perpendicularly to the surface of the electrodes, like a layer ►►

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▶ cake sitting on its side. On its own, the ceramic is a good conductor but prone to cracking. The polymer, for its part, is flexible but a poor conductor. The combination allowed ions to flow as smoothly as the best existing solid-state electrolytes, but with a much lower tendency to crack.

There are other hurdles to overcome. As batteries charge and discharge, wiry crystals known as dendrites can grow on the electrodes' surface, leading to cracking and, eventually, short circuits. Scientists have long believed that these form when excess lithium ions from the cathode accumulate on the surface of the anode (rather than being absorbed). Stronger electrode materials, which would resist the cracking, are an obvious solution. In a paper published in March, however, a team led by researchers at the Massachusetts Institute of Technology concluded this understanding was flawed. Instead, they said, dendrites grow when chemical reactions change the electrode's properties, causing them to weaken. That suggests scientists should be looking for electrodes with greater chemical stability, not just strength.

Materials science can also make solid-state batteries faster. In conventional polymer electrolytes, ions can move only as fast as the surrounding polymer segments allow. A group at Oak Ridge National Laboratory in Tennessee, part of America's Department of Energy, found a way of decoupling the two sets of movements. They achieved this by adding chemical compounds called zwitterions to polymer segments that would ordinarily be poor conductors. Although zwitterions are neutral molecules, they have charged regions that can give ions a boost. The team's results showed that this configuration could make ions travel through the electrolyte as much as 10bn times faster. Future tests will show how it performs in a cell.

Braving the elements

One noteworthy advantage of solid-state electrolytes is that they would open the door to materials other than lithium. Sodium-ion batteries, which replace the lithium in the cathode with sodium, are especially attractive. Sodium is not only cheaper and stabler than lithium, it is 1,000 times more abundant in Earth's crust. Unfortunately sodium atoms are bigger and heavier than those of lithium, meaning they are unlikely to embed in conventional graphite electrodes. At present, the result is a heavier battery that can store less energy. Although better electrodes can improve matters—for example hard carbon, which is capable of absorbing sodium ions into its spongelike structure, outperforms graphite—no suitable liquid electrolytes have yet been found.

A solid electrolyte would be easier to work with. For one, the decreased risk of

dendrite formation in solid-state batteries would allow anodes to be made out of highly reactive sodium metal. That would allow them to store more energy per kilogram than would be possible at present. Whereas a battery with a hard carbon anode has an energy density of around 175Wh/kg, sodium metal anodes could enable densities closer to 500Wh/kg.

To boost a solid-state sodium-ion battery's capacity yet further, researchers are experimenting with removing the anode altogether. That would create space for a thicker cathode that can be packed with more sodium, in turn boosting how much energy a battery could store. Removing the anode need not be fatal to the battery's operation. While it charges, the sodium ions would move from the cathode to another battery component known as the current collector, where they would accumulate until discharge occurs. In effect, an anode is created as the battery operates.

The heady pace of progress is the product of a truly global competition to produce the best solid-state design, says Shirley Meng, a materials scientist at the University of Chicago. The contest could also revolutionise the way batteries are manufactured. For now batteries with liquid electrolytes are built by submerging electrodes in vats of solvents and using enormous amounts of energy to dry them off. Solid-state batteries made in this way develop micropores on their surfaces, increasing the odds of malfunction. Thicker electrodes are also trickier to make because they dry unevenly.

So-called dry electrode manufacturing—in which dry powders are pressed together to form solid batteries—is, therefore, being taken increasingly seriously. Trials have shown that it cuts energy use by about half and manufacturing costs by about a fifth, while boosting the overall performance of the batteries. Many companies, including Tesla, a maker of batteries and EVs, and LG Energy Solution, a South Korean battery maker, are competing to be first to perfect it.

Distinguishing hype from reality is not easy. But recent developments mean that ambitious promises could be fulfilled. China's Contemporary Amperex Technology, the world's largest battery manufacturer, has said it will produce solid-state batteries by 2027 and plans to launch the first sodium-ion EV by the middle of this year. Samsung, a South Korean electronics company, has said it will mass produce solid-state batteries by 2027 while Toyota, a Japanese carmaker, has made a similar pledge. Ford Motors, an American car manufacturer, launched a battery-making unit this month, and plans to deliver large-scale batteries for data centres and industrial businesses by next year. In the battery-making business, these are electrifying times. ■

Virology

Cruise control

The hantavirus outbreak offers epidemiologists valuable data

UNCOVERING HOW a virus spreads among a population involves some tricky detective work. The schedules of all those infected have to be carefully examined in the days or weeks around the time they fell ill, in order to work out who infected whom—and, most important, where and how. The more isolated the population, the more accurately such records can be gathered. To epidemiologists, therefore, cruise ships with onboard outbreaks are the equivalent of floating Petri dishes bursting with valuable information.

The *MV Hondius* is the latest example. As of May 20th at least 11 cases of hantavirus were confirmed among its 147 passengers and crew; three of those infected have died. The outbreak was caused by the Andes strain of hantavirus, a bug carried by South American rodents that causes about 100-150 known human infections in Argentina and Chile each year. Human-to-human transmission can occur but such secondary cases are rarer still. The *Hondius* case study is, therefore, a valuable addition to this body of knowledge.

At the moment, the original source of the onboard virus is thought to be exposure to rodent droppings or urine prior to departure. The data—including genomic sequencing of the virus—suggest that the first four cases in the outbreak may have originated that way, possibly from the same source, says Thomas Hofmann from ▶▶



Stacking the deck in our favour

▶ the European Centre for Disease Prevention and Control. But there is rarely complete certainty about how an infection was picked up. People cannot recall every single interaction they have with others. It is even harder to know who used the same toilet or touched the same serving utensils at a lunch buffet. Better records of passenger interactions are needed to get to the bottom of the outbreak.

The disease detectives on board were tasked with working out what sorts of interactions had occurred among the known cases, as well as between that cohort and the healthy passengers, in order to help manage the outbreak. In time, the data will also help researchers assess the odds of a particular type of contact leading to transmission of the virus (such as sharing a dinner table with someone who was already infected, or giving them a hug), which will help manage future outbreaks.

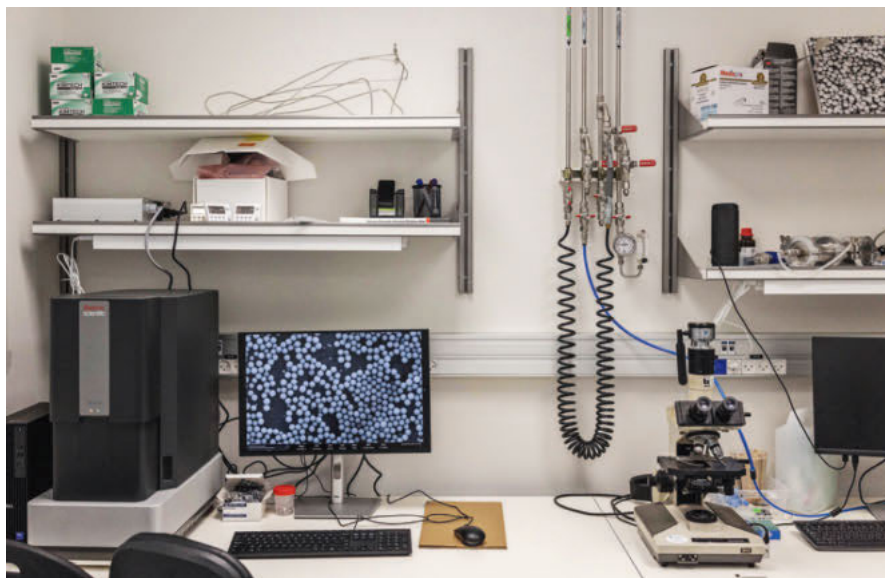
Although the pieces of the puzzle are still being put together, the results so far are consistent with what was already known about the Andes strain, says Dr Hofmann. Crucially, it is not a virus that transmits easily among people, such as those that cause covid-19 and the flu. If it did, he says, there would be far more cases on the *Hondius*, where passengers spent a lot of time in communal indoor lounges.

That being said, widespread transmission between people cannot be ruled out. In an outbreak that began in Argentina in 2018 one person unleashed a chain of transmission that ultimately infected 33 others. That outbreak has led researchers to think some people may be hantavirus “superspreaders”. For reasons that are unclear, they may be shedding and dispersing exceptionally high quantities of the virus.

Even so, the studies of the Andes strain (to which the *Hondius* data will soon be added) show that the virus does not have what it takes to be a pandemic threat. Genomic sequencing of samples taken over the years shows that it changes very little as it circulates in rodents. What is more, samples from the ship outbreak do not show the emergence of any adaptations that could make it better at transmitting between humans.

As the typical incubation period is around three weeks and transmission generally occurs when people already have symptoms, there is usually plenty of time to find and isolate close contacts. Such measures cannot contain viruses with short incubation periods (like influenza) or that are spread by asymptomatic carriers (like covid and influenza).

It remains a disease to be taken seriously: mortality can be as high as 30% even when the disease is recognised early and intensive care is available. But, thankfully, becoming infected from a rodent is rare—and from a human rarer still. ■



Climate modification

A load of (very tiny) balls

CHICAGO

Private money is bringing new ideas—and new concerns—to solar-geoengineering research

IN FEBRUARY 2024 an article in the *Wall Street Journal* revealed that Stardust Solutions, an Israeli startup, was developing tiny particles which, if lofted into the stratosphere in sufficient number, might be used to cool the Earth. The idea of putting stuff into the stratosphere to lower temperatures was not new; it is the most discussed of the various sunlight-blocking technologies gathered under the rubric of solar geoengineering. Stardust’s novelty was that it claimed to be developing special particles which might do particularly well. What it was that made them so super, though, remained a secret.

That secrecy ended last week. On May 14th the company published preprints of papers which it is submitting to peer-reviewed journals describing its particles in detail. Most of the company’s data, it turns out, concerns spheres less than a thousandth of a millimetre across and made entirely of amorphous silica (the same stuff that opals are made of) with a specially treated surface. A second, similarly sized version has a surface shell of amorphous silica but a core of calcium carbonate.

In an interview at a meeting on geoengineering hosted by the Climate Systems Engineering initiative (CSEi) at the University of Chicago a few days after the preprints were uploaded, Yanai Yedvab, Stardust’s boss, stressed the degree to which the particles were part of an “end to end” approach to geoengineering technology

which the company hoped to provide to any governments which might, eventually, decide that geoengineering is a good idea. The reception was decidedly mixed.

Although the use of calcium carbonate and silica to these ends has been explored before, it has never been done in such depth. The lion’s share of research on stratospheric geoengineering assumes instead that the cooling would be done by mimicking the cooling effect of large volcanic eruptions: in other words, injecting a sulphur-rich chemical high into the atmosphere, where the sulphur ends up in tiny reflective droplets.

There is a problem with this approach which has nothing to do with climate. Breathing in sulphate particles is bad for people. And particles high up in the stratosphere will inevitably drift down. The amount of sulphate that would be used would, admittedly, be small relative to the total quantities emitted by industry, and it would not be concentrated close to where people live. Nevertheless, when Dr Yedvab says “Dispersing millions of tonnes of toxic materials above the heads of their children for decades is something [people] wouldn’t feel that comfortable about,” it is hard to disagree.

Amorphous silica is not in itself a health concern. But that does not mean that the Stardust particles will necessarily be safer than sulphates after a few years of weathering. On this and other matters—▶

such as the idea that the particles could be chemically tagged—the scientists gathered in Chicago seemed unconvinced. “I think the work they’ve done on the particles is really terrific,” says David Keith, CSEI’s faculty director and a longtime leader in geoeengineering research. “What I don’t buy is that it’s definitely safer or definitely a good thing.”

It might be tempting to see this as a reflexive resistance to incomers. But there is something deeper at play. Almost all geoeengineering researchers, as well as the charities which fund a significant part of their work, say they do so disinterestedly, equally happy to discover good news or bad. (The Degrees Initiative, a British charity which funds such research and takes that stance, is chaired by a member of *The Economist’s* editorial staff.) Stardust, by contrast, is a commercial undertaking. If its technology is never used, its investors will lose out. That means it has a vested interest in geoeengineering going ahead, and in perpetuating a dim view of sulphates.

“It’s not that there’s anything fundamentally wrong with for-profit companies,” said Dakota Gruener, who runs Reflective, a philanthropically funded solar-geoeengineering research outfit, in an on-stage discussion with Dr Yedwab at the Chicago meeting. “But when you have investors who are saying what we bet upon is that this will be deployed, it makes it hard to have that same level of objectivity.”

Objectivity tends to engender a sense of trust. So does transparency—a norm in the field that was challenged by Stardust’s long silence. It does not help the case for trust that its original funding came through Awz Ventures, a technology investor based in America, Canada and Israel that has various links to intelligence and security services. Awz also has an advisory board led by Stephen Harper, a former prime minister of Canada noted for his friendliness to the fossil-fuel industry.

Perhaps more important than the source of its funds, though, is their quantity. In October 2025 the original investment of \$15m through Awz was joined by a second round of investors bringing in \$60m. (One of those investors was the venture-capital arm of Exor, a shareholder in *The Economist’s* parent company.) A survey of research funding published by SRM360, a non-profit which provides information about solar geoeengineering, suggests that this makes Stardust easily the most generously funded research outfit in the field.

Solar geoeengineering has long been controversial. Those studying it have nonetheless made real progress in understanding its potential, and its scope for misapplication. Stardust’s ideas may in the long run be seen as a fruitful contribution to that progress. In the short run, it may well add to the controversy. ■

Well Informed

How well do anabolic steroids work?

Very. But beware the side-effects

MOST SPORTING bodies prohibit performance-enhancing drugs. The Enhanced Games revels in the possibilities they offer. The competition, which kicks off on May 24th, allows athletes to use all sorts of substances, provided they are licensed and that they are administered under a doctor’s supervision.

Many athletes competing in the games have been cagey about their plans. But in April Mitchell Hooper, a Canadian who has twice won the World’s Strongest Man competition, revealed his in detail. Aside from Adderall, a stimulant, Mr Hooper’s “stack” consists of various anabolic-androgenic steroids (AAS), a class of compounds that are chemical cousins of testosterone, the chief male sex hormone.

That makes sense: AAS are some of the most potent performance-enhancing drugs known, especially for boosting strength and power. They will probably be doing most of the heavy lifting for every athlete at the games. And they are popular with non-athletes, too: one meta-analysis from 2014 estimated that 6% of men have used them at least once. But just how effective are they?

In short, very—at least when it comes to packing on muscle. In a much-cited study done in 1996, for instance, young men given high-ish levels of testosterone and told to do no exercise saw a 19%

improvement in their lower-body strength after ten weeks. That was about the same as participants given placebo drugs but who hit the weight room three times a week. Those who combined steroids with training saw a 38% increase. Several other randomised control trials conducted since then have reported similar results.

Steroids also boost levels of oxygen-carrying red blood cells, which might help with endurance sports. Gym wisdom holds that they improve recovery too, allowing athletes to train harder—though that has yet to be rigorously proved in humans.

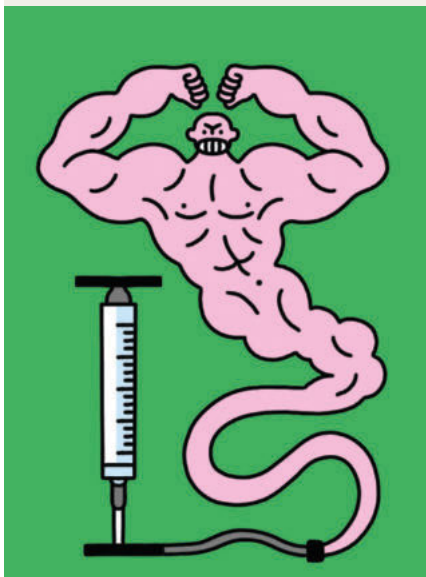
What this means for sporting performance is harder to quantify. Since steroids are banned in most sports, controlled trials on elite athletes are usually a non-starter. But in 1997 *Clinical Chemistry* published a remarkable paper based on documents from the East German state-sponsored doping programme that began in the 1960s. One chart shows turinabol, a steroid taken orally, improving the shot-put distance of a female athlete by around 15% in just 11 weeks.

All this extra power comes with side-effects. For one thing, taking high levels of steroids raises the risk of heart disease. Artificial AAS also suppress production of the natural sort, causing infertility and, in men, testicular shrinkage. Usually, the body will resume production when you stop taking the drugs—but sometimes it does not.

Cosmetic side-effects are another worry. In men genetically predisposed to baldness, steroids will accelerate hair loss. And as testosterone is the chemical from which the body synthesises oestrogen, some AAS can lead to unnaturally high levels of that hormone—causing some men to grow breasts.

In women, steroids can enlarge the clitoris; encourage the growth of body hair and beards; and deepen the voice. Some of the doped East German athletes (most of whom took steroids unwittingly) suffered lifelong health complications, and in 2006 won payouts from the company that made the drugs.

Steroids clearly work. But powerful drugs have powerful downsides—a lesson that competitors in the Enhanced Games would do well to bear in mind.



Culture



The Judgment of Paris

Red, white and cru

PARIS

Fifty years ago, a blind tasting revolutionised the wine world. Just don't mention it to the French

"MAYDAY!" COMES from the French *m'aider* (help me). But no one in the stiff-lipped wine world expected a disaster on that sunny May day in 1976. The great names and noses of French wine had been invited by Steven Spurrier, an English wine merchant, to judge a blind tasting. Spurrier hoped to garner publicity for his shop and wine school in Paris. To mark America's bicentennial, he had tracked down a few bottles from young, unknown producers in Napa Valley, California, and put them alongside celebrated ones from Bordeaux and Burgundy. Spurrier, like the judges, expected the French to easily win the taste-off. France was "the undisputed mistress of the vine", according to the "World Atlas of Wine" (1971), which devoted 73 pages to it and just eight to California.

But 50 years ago, on May 24th, the unthinkable happened. Red and white wines from Napa made in the French style—a bottle of Stag's Leap and Chateau

Montelena from 1973—won the contest, ranking above acclaimed names such as Château Mouton Rothschild and Château Haut-Brion. Unlike the mythical Judgment of Paris, which led to the Trojan war, golden grapes did not provoke an actual conflict—but they did prompt some fighting talk. One French wine official later described the tasting as "our Waterloo". The French first blamed the judges. They also, as usual, blamed the English. "Perfidious Albion" is what one winemaker called

Spurrier to his face; others lobbed less literary insults behind his back. (In 2008 Spurrier was played memorably by Alan Rickman in "Bottle Shock", Hollywood's depiction of the tasting.)

The Judgment of Paris was the shot heard round the wine world. Napa's victory had two Nebuchadnezzar-size effects. First, it sparked demand for Californian wine at home and abroad. The volume of exported bottles of American wine more than quintupled between 1975 and 1980. Second, it inspired confidence in "new world" producers, including Argentina and Australia. "There is no more influential event in the wine world, ever," claims Juan Muñoz-Oca, estate director at Stag's Leap. Today the two winning bottles can be found at the Smithsonian's National Museum of American History, where George Washington's uniform and the original star-spangled banner are also on display.

Much can be discerned from how the two countries are marking the tasting's 50th anniversary. Stag's Leap and Chateau Montelena are hosting events around the world and pouring tastes of the acclaimed vintages. (The 1973 Stag's Leap and Montelena were released with a price tag of \$6 and \$6.50, respectively, around \$35-38 in today's dollars. Now a single bottle costs thousands of dollars, if you can find one.) And following Ralph Waldo Emerson's ob- ➤

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► servation that “music and wine are one,” in July a new one-act opera based on the tasting will premiere in Napa.

In France there will be no songs or spectacle. Ask winemakers there how often the Judgment of Paris is discussed, and they will claim almost never. (Though they will also point out that the French wines, on average, scored higher than Napa’s. And that they aged better, too.)

The competitive element of the tasting obscures the fact that, in the decades following the Judgment, both American and French wines boomed, enjoying new markets and newly minted middle-class wine drinkers. Fifty years on, the scene is less convivial. Wine consumption is falling, with Gen Z and baby-boomers alike preferring to drink “less but better”. After being associated with heart health, now wine is on the heart and mind of health officials, who say no amount of alcohol is safe. Wine is being “put in the same category as prostitution and drugs”, complains Stéphanie de Bouard-Rivoal of Château Angelus, a top-tier winery in Saint Émilion.

Today winemakers in Bordeaux talk openly about *la crise* (the crisis). “Pretty bad” is how Bo Barrett, who runs Chateau Montelena, describes the situation in Napa. Both regions are seeing vines ripped out, as the supply of grapes outstrips demand. France and the EU have committed around €250m (\$290m) to vine-uprooting.

Of all the world’s wine regions, Burgundy has fared best. At about a third of Bordeaux’s size, it is smaller, less corporate and more coveted by connoisseurs. Its wines—reds made from Pinot Noir and whites from Chardonnay grapes—boast elegance and finesse. (Bordeaux’s reds are based on Cabernet Sauvignon and Merlot.) Burgundy’s rarity, exclusivity and sometimes astronomical prices have prompted even millionaires to complain they can no longer afford to drink it. On Liv-ex, a wine exchange, Burgundy’s share of sales has risen to around 23%, more than triple what it was ten years ago (see chart).

Both Bordeaux and Napa, in contrast, are experiencing “economic crises” and are “at a turning-point”, says William Kelley, editor-in-chief of the *Wine Advocate*. Yet their suffering has different causes. Bordeaux is “paying for the mischiefs of the past”, says Omri Ram of Château Lafleur, namely that it produced a “bubble” of too much wine that was “sold for the wrong reasons to the wrong clients”. Chinese buyers had thirsted after Bordeaux’s prestige labels—until Xi Jinping denounced lavish gifts, and sales sputtered. That left winemakers, who had been raising prices, vulnerable to a collapse in demand.

The *en primeur* system governing how Bordeaux’s wines are sold has also stoked *la crise*. It facilitates pre-orders for new vintages, which hands cash to winemakers for

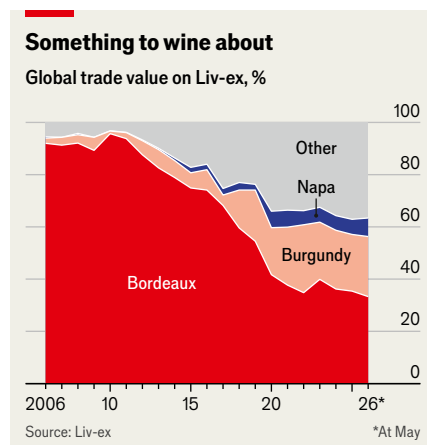
cases delivered years later. With the price of older vintages now comparable to, or below, new releases, buyers are less keen to snap up the young stuff, which requires ageing and storage. It has created a doom loop, especially for the mid- and lower-tier wines. Last month wineries hosted their annual *en primeur* tastings, and the mood was tense. Some warn *en primeur* may cease to exist.

Napa’s winemakers are feeling pinched for different reasons. In recent decades they doubled down on the American market, selling to baby-boomers at high prices rather than prioritising a global footprint. But consumers are ageing and buying less. In addition, Napa’s most important international market, Canada, has dried up: sales of American wine there fell by nearly 80% in 2025, after most provinces banned American alcohol in retaliation for tariffs.

The cost of doing business in California (like France) is hangover-inducing, as is the red tape. “It’s hard for small producers to stay in business, so they’re gobbled up by larger wine companies,” says Mr Muñoz-Oca. Stag’s Leap is a case in point: Marchesi Antinori, an Italian firm, acquired sole ownership of it in 2023.

Palates have also changed. Napa underwent a “Cabernetisation”, as it chased the high scores and price tags of big red wines. Consumers, however, have shifted away from that “monster truck” style of red wine, says Mr Kelley. Cuisine has become lighter, and people want more balanced reds with lower alcohol levels.

Both Napa and Bordeaux are trying to navigate their new reality. Consumers are opening bottles younger, so “we try to make them approachable,” says Blandine de Brier Manoncourt, co-owner of Château Figeac in Bordeaux. Some, including Château Angelus, have launched lower-tier wines to appeal to younger buyers. Its 2023 Tempo d’Angelus sells for around €22 a bottle, about a twentieth the price of its flagship wine. Today the competition is not between America and France. It is between wine and everything else. ■



Christianity in North Korea

The Kim catechism

Korean Messiah. By Jonathan Cheng.

Knopf; 768 pages; \$36

PEOPLE HAVE been predicting the collapse of North Korea’s communist dictatorship since 1948, when Kim Il Sung seized power with Josef Stalin’s backing. And yet, almost 80 years later, the regime is still there, and the Kim family are still in charge of it, having survived the Korean war, the fall of the Soviet Union, decades of national self-isolation and two dynastic transitions. Totalitarian repression, as well as assistance from powers such as China and Russia, help explain the regime’s resilience. “Korean Messiah” by Jonathan Cheng, a journalist at the *Wall Street Journal*, sheds light on a less appreciated force: faith.

In keeping with communist principles, North Korea is officially atheist. Nonetheless, in Mr Cheng’s telling, its system owes as much to Jesus, Mary and Joseph as it does to Marx and Engels. He argues convincingly that Kim Il Sung appropriated the rituals and dogma of religion when forging his political ideology. For example, all North Koreans are taught a set of decrees called the “Ten Principles for the Monolithic Ideology”. They are “so similar in language to the biblical Ten Commandments, and...fulfilled so similar a function in the lives of its population” that they “might as well have been etched on a pair of stone tablets”, Mr Cheng writes.

Pyongyang had long been a hotbed of Christian thought. In the late 19th and early 20th centuries the city became known as “the Jerusalem of the East” among Western missionaries in Asia. Kim’s ancestors were among the first converts; his parents were fervent Presbyterians. As a child, he accompanied his father to chapel, where he sang hymns and learned to play the organ. “This upbringing gave the young man a front-row seat to the power of faith—its power to mobilise, to offer succour, and, perhaps most importantly, to inspire awe, devotion, fear and zeal.”

Kim’s exposure to Christianity also gave him an acute sense of the threat it posed to his hopes for absolute rule. After taking power in North Korea, Kim, by then an ardent Marxist and hardened anti-American crusader, came to see Christianity as a rival faith. Portraying himself as the true messiah, he brutally suppressed Christians, killing thousands of believers and sending more fleeing to the South. ►►

BACK STORY

Reading between the sheets

The copious sex in "Rivals", a TV bonkbuster, is more than mere titillation

EVERYONE IS A sex detective now. A train entering a tunnel, a jutting skyscraper: in the post-Freudian age, readers and viewers are primed to spot innuendo and phallic symbolism. Along with death, sex is said to be art's bedrock subject. But occasionally—and more interestingly—the imagery works in the opposite way. Instead of other things standing in for sex, sex is a metaphor for other themes. Take "Rivals".

The TV bonkbuster, set in the mid-1980s in the fictional British county of Rutshire, is back for a second season. Based on Dame Jilly Cooper's novels, it once again follows the antics and adulteries of alpha men and their wives (and other people's wives), all driven and riven by class, money and ambition. Again there is copious rumpy-pumpy—still the most meaningful part of the drama.

In the first season's finale, Tony, a villainous media baron played deliciously by David Tennant, was thwacked on the head with a gold statuette. Fans shouldn't fret: he is soon well enough to keep his end up (and get it away). A few new faces join the cast, including "Muffy, short for Caroline" and some walking appendages. This time the '80s soundtrack includes "The Chicken Song", a novelty tune which, for many British viewers, will feel like a gruesome flashback to a humiliating bender. "What the fuck is this?" asks Cameron (Nafessa Williams), a badass American TV producer, quite understandably.

The heart of the show is still the slightly yucky yearning between Rupert (Alex Hassell, pictured right), a priapic toff and politician, and moony Taggie (Bella Maclean, pictured left), who long ago forgave him for groping her. That, and the sex. Unfolding in and around country piles, "Rivals" resembles a mur-

der mystery, only the question is not who will get bumped off, but who will next shag whom. Now the characters are at it in the stables and jacuzzi and on a piano (again), with a splash of skinny-dipping instead of nude tennis.

Mindless titillation? Au contraire. Much as a cigar in art is rarely just a cigar—least of all when Tony is smoking it—a bonk in "Rivals" is never just a bonk. For starters, sex is sometimes about power and its abuses. "I can't," says Cameron when rebuffing Tony's advances. "You can if you want to be promoted," he snarls.

Next, sex is about freedom: from the constraints of class, social expectations, corporate decorum or dismal marriages. Some of Rutshire's denizens grab their freedom with both hands. "I don't wait around to be kissed by somebody. If I want to kiss someone, I do it," dauntless Cameron declares. Others are more hesitant. "We can't all just do what we want," a timid figure laments.

Sex is history. Along with the music of the 1980s—and its knitwear, tobacco habit, convertibles, game-show formats

and boozy lunches—"Rivals" resurrects the decade's sexual prejudices. There are double standards: were he a woman, Rupert would be "tarred and feathered" for his philandering, his ex-wife complains. Homophobia is rife, compelling people to live painful double lives. "Do you even really exist if no one sees you're there?" wonders a bereft gay man.

W.B. Yeats, a poet, reckoned the title couple's smutty exchanges in "Lady Chatterley's Lover" are "a forlorn poetry uniting their solitudes". Likewise in "Rivals", sex is often about loneliness. It can be a salve: Declan (Aidan Turner), a thrusting TV presenter and Taggie's dad, knows that his passionate wife plays around for "relief from the loneliness" his workaholic causes. Yet if sex can fulfil a need for intimacy, the compulsive, Casanovan kind may suggest an unreachable isolation. Rupert's conquests, he is told, are a doomed bid to "fill that emptiness" inside him.

Sex is a matter of luck, the arc of one life (or character) intersecting with a partner's at the right moment. And, sometimes, sex is about honesty. Beneath the illicit hanky-panky, "Rivals" is an ode to marriage at its frankest. When they aren't lambasting or betraying each other, Declan and his wife candidly share their fantasies and feelings.

The ostensible plot of the new series involves a struggle over a broadcasting franchise. But compared with the saucy bits, that is trivial. The real, profound action is in the bedroom, not the boardroom. (Metaphorically speaking—in practice they blur, the characters going in for al desko trysts as well as the al fresco sort.) The final moral of "Rivals" is surprising. Maybe everything does come down to sex—except for sex itself, which is often about something else.



► Eventually the Great Leader's teachings coalesced into a formal doctrine, known as Juche (self-reliance), which supplanted Marxism-Leninism as the foundation of the regime. Ideologues insisted that Kim's dogma be treated as infallible. Mr Cheng quotes one North Korean escapee's description of encountering Christianity after fleeing the country: "When they talked about God, it kept reminding me of Kim Il Sung...When they mentioned Jesus, it reminded me of Kim Jong Il." The state

enforces cult-like devotion to the ruling family. Jim Jones of Jonestown infamy was an admirer of Kim Il Sung. Jones's acolytes met with North Korean officials more than a dozen times.

In arguing that Christianity holds the key to understanding the regime, Mr Cheng underplays other influences, such as Korea's long history of monarchy, Stalinism and communist ideology itself. But seeing North Korea as a quasi-religious project is illuminating. In addition to brute

force, the Kim dynasty relies on indoctrination and inquisition to stay in power. If its subjects harbour doubts, they are not allowed to express them, on pain of the gulag or worse. The regime persecutes any belief system that could rival its own. North Koreans who watch television shows smuggled in from rich, democratic South Korea—which reveal how much better life is without a god-king—can face the death penalty. The greatest threat the Kim cult faces is not America, but apostasy. ■

History

Zionism's Jewish opponents

Here Where We Live Is Our Country.

By Molly Crabapple. *One World*; 480 pages; \$32. Bloomsbury Circus; £25

AT THE DAWN of the second world war, the majority of the world's Jews, 9.5m of them, lived in Europe—and had for centuries. Though there were brief periods of acceptance, for the most part Jews were barely tolerated and subject to sporadic violence and restrictive laws governing where they lived and worked. Throughout the 19th century nationalist aspirations grew, and Jews debated where they belonged. Zionists had one answer: they had to create their own state. Bundists had another. It is their story Molly Crabapple, a Puerto Rican-Jewish writer and artist from New York, resurrects.

The General Jewish Labour Bund in Lithuania, Poland and Russia was founded in Vilnius in 1897. Bundists were socialists who believed in workers' revolutions, but also recognised that Jews had particular needs and interests. They formed defence organisations to protect Jews from pogroms and promoted Yiddish culture. By 1906 the Bund had attracted 40,000 supporters, making it one of the largest socialist parties in the Russian empire.

Bundists bitterly opposed Zionism, believing, as Ms Crabapple explains, that "the establishment of Israel would lead to perpetual war with its neighbours and the people it had dispossessed." Instead they believed in *do'ikayt*, Yiddish for "hereness", meaning "the diaspora was home" and "Jews had the right to live in freedom and dignity wherever it was they stood."

It may strike a contemporary reader as odd, but before the Holocaust most Jews were not Zionists. The Orthodox believed only God could call Jews back to the Holy Land; liberals believed Zionism was a concession to antisemitism; and many simply found the idea of nearly 10m people emigrating impractical.

Ms Crabapple's interest in the Bund is both political and personal. Her great-grandfather, Sam Rothbort, was born in 1882 in what today is Belarus. The Bund created believers like Rothbort who were committed enough to withstand ostracism, beatings, imprisonment and worse. Their bitterest battles were internecine, against Soviet-aligned communists and Zionists, neither of whom had much patience for competition. But their willingness to fight on behalf of Jews and along-

side non-Jewish socialists earned them the tragic honour of becoming "as of 1938...the most popular Jewish party in Poland".

The Holocaust changed everything. After the war, "Bundist survivors tried to keep their spirits up," but "the movement they built had been decimated, as had the Polish world of their birth," Ms Crabapple writes. The Zionists, whatever their arguments' merits, at least had a state to offer survivors. The Bund had nothing but dashed utopian hopes, and the party ceased operations in 1949. The only reason this book exists is because the Bund's archivist smuggled two train-cars' worth of papers from Berlin to Paris in 1933.

This impassioned history of a century-old failed political movement briefly made it on to the *New York Times* bestseller list. Not everyone is happy about it. The editor of *Commentary*, a right-leaning Jewish magazine in America, has called Ms Crabapple "a sixth-rate middle-aged anti-Zionist screed-meister" and "an enemy to her people". But though anti-Zionism sometimes covers for antisemitism, they are not identical, as Ms Crabapple's book shows.

In America and elsewhere, the share of people with negative views of Israel has risen markedly during the course of its brutal war in Gaza (however initially justified) and attacks on Iran. According to a recent survey, today just 37% of Jewish Americans identify as Zionists, though 71% profess an "emotional connection" to Israel and 88% say it has the right to exist as a Jewish and democratic state (whatever they mean by that). Jews in the diaspora who want to celebrate their heritage without tying themselves to Israel can look to the Bundist concept of "hereness" this book celebrates—and to the long and honourable Jewish history of fighting for others alongside themselves. ■



Forgotten history

Literary mishaps

Foul copy

Rogues, Widows and Orphans.

By Rebecca Lee. *Profile*; 304 pages; £20. To be published in America by the University of Chicago Press in September; \$29.95

WHAT A DIFFERENCE a word makes. In 1631 printers omitted three letters from the seventh commandment. The bungled injunction—"Thou shalt commit adultery"—cost them their licence and a £300 fine (worth around £54,000, or \$72,000 today). The "Wicked Bible" is a warning of the perils of misprints.

A new book offers an entertaining look at "all the ways words can go wrong". Rebecca Lee, who works at Penguin Random House, explores publishers' "glitches and gremlins", as well as censorship, plagiarism, hoaxes, failures and feuds. She chronicles how the written word can be "divisive, damaging, disruptive and unwelcome".

Take Winston Churchill's history of the second world war, which called France's army "the poop of the French nation" (he had written "prop"). Editors—or "correctors", as their 17th-century counterparts were called—emerge as crucial props against blunders. Ms Lee also offers many examples of atrocious writing. "Bad" style can ascend to a kind of genius. Amanda McKittrick Ros was an Irish writer of prolix, overblown 19th-century novels. Her euphemisms alone astonish: trousers become "the southern necessity". Hearing about the Nobel prize, she asked her publisher: "Do you think I should make a 'dart' for it?"

At times, this inky history feels rushed. Chapters scoot through censorship's most famous instances and skip across rejections and vicious reviews. But Ms Lee's judgments are sound: "Freedom to read", she observes, "remains a right constantly under siege."

Inevitably, the book closes with AI as literary threat and promise. The technology offers a "flood of free or cheap content" and plunders past creativity to deliver "rear-view mirrors of storytelling". Error-strewn, vice-prone, human-authored books have already weathered almost six centuries of debacles. But this next chapter could be the most dramatic yet.

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Storytelling and video games

Ready Reader One

Gamified novels—known as LitRPG—are a winning format

ALL CARL wants to do is smoke his cigarette, but Princess Donut, his ex-girlfriend's cat, has escaped. He reluctantly goes after the creature when—"slam"—his building, and the city beyond it, is flattened like "a massive tin can that had been crushed by a giant cosmic boot". Carl is one of the few survivors left on Earth. Along with Princess Donut, he is forced into a dungeon to fight various monsters; the battles will be televised throughout the universe. As they "level up" through the floors and try to make their escape, they earn XP (experience points).

It sounds like the premise of a video game. But it is the introduction to "Dungeon Crawler Carl", a book series by Matt Dinniman, which has sold over 2.7m copies in print. The first instalment has been on the *New York Times* paperback bestseller list for 19 weeks and counting. The eighth, "A Parade of Horribles", was released on May 12th. The audiobook is in first place on Audible's bestseller list in America; all the other books are in the top 20. A TV adaptation and a graphic novel of Carl's misadventures are in the works.

Gamified novels—known as literary role-playing games, or LitRPG—are going mainstream. Such novels borrow the tropes of video and tabletop games. (Unlike choose-your-own-adventure tales, however, the reader does not make narrative choices.) Characters face challenges and grow stronger. They go on quests to obtain rewards. In Mr Dinniman's novels Carl gets tougher as he punches goblins and defeats a "Krakaren", a monster that is a combination of a "cosmic octopus" and "your average, suburban, anti-vax, let-me-talk-to-your-manager mom". Throughout the book the reader is updated on his character stats, health bar, XP and special skills.

Video-game vernacular provides a useful shorthand. Minor figures are called NPCs: non-playable characters. By "just saying, 'He levelled up', I can communicate a tremendous amount of information that if I was writing traditional fantasy, I would have to demonstrate," says Seth Ring, who has published nearly 50 LitRPG titles.

The core readers of LitRPG are gamers in their 30s who appreciate such elements. But its biggest audience is audiophiles, from truckers to stay-at-home mothers. The novels often have only one perspective, and are usually narrated in the first person, which makes them easy to follow.

Fans gush about the voice actors almost as much as they do about the authors.

Though a relatively new player in Western publishing, LitRPG has been around for a while. (A Russian publisher claims to have coined the term in 2013.) Versions of it have long been popular in Asia. *Xuanhuan* novels, a type of Chinese fantasy, took off in the early 2000s. Such tales drew on Daoist ideas of "cultivation" and self-improvement and followed heroes as they progressed through different realms. Japanese manga and anime often feature *isekai* (another world), where characters are transported into video-game-like settings with magic and sword fights. The idea is also a trope in Korean webtoons.

Western writers have borrowed these ideas and created an "unholy mish-mash" to forge their own version of LitRPG, says Andrei Nadir, an author who writes under the pen name Actus. Authors including Mr Dinniman argue LitRPG is less a genre and more a style of writing: stories might be about aliens and post-apocalyptic survival or they might feature steampunk cowboys and orcs who enjoy cooking.

In the early- to mid-20th century pulp fiction such as Raymond Chandler's "The Big Sleep" and Dashiell Hammett's "The Maltese Falcon" hooked readers with a punchy style. LitRPG follows in that tradition, reckons Rhett Bruno, head of Aethon, a publishing house. It is "filling a big void",

he insists, between ponderous fantasy epics such as "Game of Thrones" and smutty romantasy series like "A Court of Thorns and Roses". LitRPG offers "fun, light, rompy-type action fantasy". Chapters are short, action-packed and usually end with a cliffhanger. The format delivers regular dopamine hits to readers.

Writers of LitRPG are thinking strategically on and off the page. Many self-publish their work online, chapter by chapter; some are prolific, posting new material daily. Readers often interact with their favourite writers and leave comments on chapters, which then shape the stories. Mr Nadir, for instance, aims for 5,000 words a day and completes a book about every three months. Yet it is not unusual for him to rewrite an already published chapter after receiving feedback from fans.

Authors are gaining more than XP. Loyal followers will pay for early access to chapters. One writer, with the cryptic pen name Zogarth, makes nearly \$90,000 every month from subscribers on Patreon, making him one of the most successful authors on the platform. Writers have ample leverage if they choose to negotiate with publishing houses. Mr Dinniman sold the print rights to "Dungeon Crawler Carl" to Ace Books and the audio rights to Audible, but retained the digital ones. "We found something that might change the rules of publishing," declares Seth Fishman, his agent. (In fact, bespoke rights arrangements are not new, but they are increasingly common among digital-first authors.)

To be sure, LitRPG is not going to win any prestigious awards. Quantity has been trouncing quality. But readers looking for escapist thrills are often forgiving. "We are trying to write really, really good stories," Mr Ring says. "We are not trying to create the next great American novel." ■



Winning fans' hearts

Economic & financial indicators

	Gross domestic product				Consumer prices				Unemployment		Current-account balance		Budget balance		Interest rates		Currency units		
	% change on year ago latest		quarter* 2026†		% change on year ago latest		2026†		rate		% of GDP, 2026†		% of GDP, 2026†		10-yr gov't bonds latest, %		change on year ago, bp		per \$ May 21st
United States	2.7	Q1	2.0	2.2	3.8	Apr	3.7	4.3	Apr	-3.4	-6.5	4.6		9.0	-				
China	5.0	Q1	5.3	4.9	1.2	Apr	1.6	5.2	Apr†§	3.2	-5.8	1.4	§	-14.0	6.80	6.2			
Japan	0.6	Q1	2.1	0.5	1.4	Mar	2.3	2.7	Mar	3.3	-1.8	2.8		124	159	-9.1			
Britain	1.1	Q1	2.5	0.8	2.8	Apr	3.5	5.0	Feb††	-3.1	-5.1	5.1		48.0	0.74	nil			
Canada	0.7	Q4	-0.6	1.2	2.8	Apr	2.6	6.9	Apr	-0.4	-2.2	3.6		29.0	1.37	1.5			
Euro area	0.8	Q1	0.6	1.0	3.0	Apr	2.6	6.2	Mar	2.5	-3.3	3.1		48.0	0.86	3.5			
Austria	0.6	Q4	0.1*	0.8	3.5	Apr	2.6	5.6	Mar	1.0	-4.1	3.3		34.0	0.86	3.5			
Belgium	0.7	Q1	0.8	0.9	4.2	Apr	3.2	6.3	Mar	-3.2	-4.9	3.7		60.0	0.86	3.5			
France	1.1	Q1	nil	0.8	2.5	Apr	2.2	7.7	Mar	-0.6	-5.3	3.9		62.0	0.86	3.5			
Germany	0.3	Q1	1.1	0.6	2.9	Apr	2.7	4.0	Mar	4.2	-3.7	3.1		48.0	0.86	3.5			
Greece	2.5	Q4	3.2	2.0	4.6	Apr	3.4	9.0	Mar	-5.5	0.8	3.8		38.0	0.86	3.5			
Italy	0.7	Q1	0.7	0.8	2.8	Apr	2.1	5.2	Mar	0.9	-3.0	3.8		21.0	0.86	3.5			
Netherlands	1.2	Q1	0.2	1.1	2.5	Apr	3.1	3.9	Apr	11.7	-2.6	3.2		39.0	0.86	3.5			
Spain	2.7	Q1	2.5	2.2	3.5	Apr	2.9	10.3	Mar	2.3	-2.4	3.6		36.0	0.86	3.5			
Czech Republic	2.5	Q4	2.9	2.2	2.5	Apr	2.2	3.0	Q1*	0.3	-2.6	5.0		74.0	20.9	5.5			
Denmark	5.8	Q1	7.8	1.8	1.3	Apr	2.0	3.0	Mar	12.4	1.2	2.8		39.0	6.43	2.8			
Norway	2.2	Q4	-1.3	1.5	3.4	Apr	2.9	4.8	Feb††	11.8	8.9	4.6		42.0	9.26	10.8			
Poland	3.4	Q1	2.0	3.4	3.2	Apr	3.2	6.1	Mar§	-1.3	-7.0	5.9		43.0	3.65	2.7			
Russia	-0.2	Q1	na	1.5	5.6	Apr	5.7	2.2	Mar§	2.6	-2.4	14.8		-98.0	71.0	13.2			
Sweden	2.0	Q1	-0.8	2.3	-0.1	Apr	1.8	8.7	Apr§	5.7	-2.0	2.9		46.0	9.34	3.2			
Switzerland	0.7	Q4	0.6	1.0	0.6	Apr	0.4	3.0	Apr	5.2	0.1	0.6		28.0	0.79	5.1			
Turkey	3.4	Q4	1.5	3.0	32.4	Apr	30.1	8.1	Mar§	-3.0	-3.6	33.1		210	45.6	-14.8			
Australia	2.6	Q4	3.2	2.1	4.6	Mar	4.3	4.5	Apr	-1.9	-1.6	4.5		65.0	1.40	11.4			
Hong Kong	5.9	Q1	12.2	3.2	1.7	Mar	2.0	3.7	Apr††	10.2	-3.6	3.3		23.0	7.83	-0.1			
India	7.8	Q4	7.1	6.5	3.5	Apr	4.8	6.7	Apr	-1.3	-4.5	7.1		85.0	96.6	-11.4			
Indonesia	5.6	Q1	6.0	5.2	2.4	Apr	3.5	4.7	Feb§	-0.9	-3.5	6.8		-5.0	17,632	-6.9			
Malaysia	5.4	Q1	1.7	4.7	1.9	Apr	2.2	2.9	Mar§	2.6	-4.1	3.6		3.0	3.97	8.3			
Pakistan	4.8	2026**	na	3.0	10.9	Apr	7.8	6.9	2025	-1.1	-4.7	12.9	†††	56.0	279	1.2			
Philippines	2.8	Q1	3.6	2.7	7.2	Apr	6.8	5.8	Q1§	-4.0	-6.3	7.7		152	61.5	-9.4			
Singapore	4.6	Q1	-1.3	2.9	1.8	Mar	2.8	2.1	Q1	15.3	0.8	2.2		-26.0	1.28	1.6			
South Korea	3.6	Q1	6.9	2.9	2.6	Apr	2.9	2.9	Apr§	8.2	-2.4	4.2		148	1,498	-7.0			
Taiwan	13.7	Q1	11.9	8.1	1.7	Apr	1.6	3.4	Mar	28.5	0.8	1.6		7.0	31.6	-4.6			
Thailand	2.8	Q1	2.7	1.8	2.9	Apr	2.9	1.0	Mar§	1.0	-5.5	2.4		53.0	32.6	1.0			
Argentina	2.1	Q4	2.5	3.0	32.4	Apr	31.6	7.5	Q4§	-0.2	0.3	na		na	1,397	-18.2			
Brazil	1.8	Q4	0.6	1.8	4.4	Apr	4.5	6.1	Mar§††	-2.6	-7.3	14.3		21.0	5.01	13.0			
Chile	-0.5	Q1	-1.1	1.4	4.0	Apr	3.7	8.9	Mar§††	-2.2	-2.5	5.6		-17.0	898	4.9			
Colombia	2.2	Q1	2.4	2.7	5.7	Apr	5.7	8.8	Mar§	-2.5	-6.6	14.0		175	3,730	11.9			
Mexico	0.1	Q1	-3.2	1.0	4.4	Apr	4.1	2.8	Mar	-0.3	-3.8	9.4		-5.0	17.3	11.3			
Peru	3.2	Q4	2.5	2.5	4.0	Apr	4.0	5.0	Apr§	1.8	-2.4	6.3		-9.0	3.43	7.6			
Egypt	5.3	Q4	0.8	4.5	14.9	Apr	13.5	6.0	Q1§	-4.9	-7.5	25.8		101	53.4	-6.6			
Israel	2.0	Q1	-3.3	3.1	1.9	Apr	2.2	2.9	Apr	1.8	-4.5	4.0		-48.0	2.90	21.4			
Saudi Arabia	4.5	2025	na	-2.0	1.7	Apr	2.5	3.5	Q4	-1.4	-4.0	na		na	3.75	nil			
South Africa	0.8	Q4	1.5	1.6	3.8	Apr	3.8	32.7	Q1§	-1.8	-4.4	8.7		-172	16.5	8.8			

Source: Haver Analytics *% change on previous quarter, annual rate †The Economist Intelligence Unit estimate/forecast §Not seasonally adjusted *New series **Year ending June ††Latest 3 months †††3-month moving average
 §\$5-year yield †††Dollar-denominated bonds Note: Euro-area consumer prices are harmonised

Markets

In local currency	Index		% change on:	
	May 20th	one week	Dec 31st 2025	
United States S&P 500	7,433.0	-0.2	8.6	
United States NAS Comp	26,270.4	-0.5	13.0	
China Shanghai Comp	4,162.2	-1.9	4.9	
China Shenzhen Comp	2,869.2	-2.7	13.4	
Japan Nikkei 225	59,804.4	-5.5	18.8	
Japan Topix	3,791.7	-3.3	11.2	
Britain FTSE 100	10,432.3	1.0	5.0	
Canada S&P TSX	34,161.8	0.4	7.7	
Euro area EURO STOXX 50	5,976.1	2.0	3.2	
France CAC 40	8,117.4	1.4	-0.4	
Germany DAX*	24,737.2	2.5	1.0	
Italy FTSE/MIB	49,181.7	-0.6	9.4	
Netherlands AEX	1,033.7	2.3	8.7	
Spain IBEX 35	18,051.7	2.2	4.3	
Poland WIG	133,684.9	1.0	14.0	
Russia RTS, \$ terms	1,172.3	1.3	5.8	
Switzerland SMI	13,399.3	1.4	1.0	
Turkey BIST	14,012.0	-4.0	24.4	
Australia All Ord.	8,717.0	-1.8	-3.3	
Hong Kong Hang Seng	25,651.1	-2.8	0.1	
India BSE	75,318.4	1.0	-11.6	
Indonesia IDX	6,318.5	-6.0	-26.9	
Malaysia KLSE	1,717.7	-1.6	2.2	

	Index		% change on:	
	May 20th	one week	Dec 31st 2025	
Pakistan KSE	164,831.4	-1.6	-5.3	
Singapore STI	5,044.9	0.8	8.6	
South Korea KOSPI	7,209.0	-8.1	71.1	
Taiwan TWI	40,020.8	-3.3	38.2	
Thailand SET	1,528.4	0.7	21.3	
Argentina MERV	2,788,517.0	1.8	-8.6	
Brazil BVSP*	177,355.8	0.1	10.1	
Mexico IPC	68,893.9	-1.8	7.1	
Egypt EGX 30	51,936.8	-2.8	24.2	
Israel TA-125	4,325.9	-1.4	18.1	
Saudi Arabia Tadawul	10,985.6	-0.3	4.7	
South Africa JSE AS	114,634.1	-2.3	-1.0	
World, dev'd MSCI	4,766.7	-0.2	7.6	
Emerging markets MSCI	1,637.3	-4.0	16.6	
US corporate bonds, spread over Treasuries				
				Dec 31st 2025
Basis points				latest
Investment grade				87
High-yield				331

Sources: LSEG Workspace; Moscow Exchange; Standard & Poor's Global Fixed Income Research *Total return index

Commodities

The Economist commodity-price index		% change on	
2020=100		May 19th*	month
Dollar Index			
All items	158.2	154.9	1.5
Food	150.2	148.0	2.2
Industrials			
All	164.8	160.5	1.0
Non-food agriculturals	149.8	150.1	1.6
Metals	168.7	163.2	0.9
Sterling Index			
All items	150.3	148.6	2.4
Euro Index			
All items	154.1	152.6	2.9
Gold			
\$ per oz	4,664.8	4,498.2	-5.3
Brent			
\$ per barrel	108.0	111.5	13.3

Sources: CME Group; LME; LSEG Workspace; NOREXCO; NZ Wool Services; S&P Global Commodity Insights; Thompson Lloyd & Ewart; USDA *Provisional

For historical indicators data, visit economist.com/economic-and-financial-indicators

OBITUARY

Edith Eger

The “Ballerina of Auschwitz” died on April 27th, aged 98



“DANCE!” THAT was what he said to her on her first night in the camp. Edith Eger could dance, of course: she had been learning ballet since she was five years old. In her leotard and tights, she had practised her stretches. Down she would go: spine straight, tummy taut, breathe in, breathe out...She knew how to use her breath to control her body. She knew an entire routine to “The Blue Danube”. She thought she was imagining it when she heard that being played in the camp on her first night. Then he had appeared in her barracks. “Little dancer,” he said. “Dance for me.” That was what he would call her: “My little dancer.” They would call him the “Angel of Death”.

Later, after she moved to America, she didn’t want to call that place anything at all. She didn’t want anyone’s pity. She just wanted to be a Yankee doodle American. Besides, she didn’t even know how to pronounce that word in English: “Auschwitz”. How do you say that without an accent? So she never said it. Though later, when other people found out about her past, they would use its name to her. It’s pretty bad, they’d say of some problem or other, but “it’s not *Auschwitz*”. She felt that was wrong: there was no hierarchy of suffering. She didn’t want people to see her suffering and think theirs was less significant. She wanted them to look at her and think: if she can do it, so can I!

She started to speak out. First, in therapy herself, then as she trained to become a psychologist, then finally as a bestselling author. Always, her message was the same: you cannot change or forget the past but you can choose how you respond to it. You can be depressed or you can be happy. That choice at least is yours. Though she, at first, had had no choice. In her new life as a wife and mother in Baltimore she could not control how she reacted to things. She felt anxious when she heard a siren. She felt dizzy when she heard heavy footsteps. When a bus driver shouted at her she huddled on the floor of the bus, crying and shaking. It wasn’t that bad. It wasn’t *Auschwitz*. And it was always Auschwitz.

What do you do with your past? She was only 17 when the war ended but she already had so much past; she had already seen so much. Like that boy in the camp. He had been tied to a tree then SS soldiers had shot at him. They shot his foot, his arm, his hands,

an ear. A little boy, used as target practice. Then there was the girl who tried to escape. The soldiers had shot her then hung her body in the middle of the camp as an example. And there was the pregnant woman: when she went into labour, the SS tied her legs together. She had never seen agony like hers.

She would remember good things, too, like her home in what was then Hungary. It had been full of music: one sister was a violinist, another a pianist—and she was the dancer, then a gymnast. At 16 she was suddenly chosen for the Olympic training team. Then, equally suddenly, she was kicked off it because of her “background”. Her “background” had become her foreground: she now wore a yellow star. Then, one night, soldiers came and took her family away. Yet the music continued: as they walked into Auschwitz beneath the “Arbeit macht frei” sign, the camp orchestra started up. Her father turned to her, delighted. “You see,” he said. “It can’t be a terrible place.” She would never see him again.

Memory was what mattered. That was one of the last things her mother said to her. As the train drew closer to Auschwitz she had turned to Edith. All that they had had been taken but, she said, “no one can take away from you what you’ve put in your mind.” When they got off the trucks, they took her mother, too. A gap-toothed soldier was sorting women into two queues, asking each: are you sick? His voice was almost kind. The ill, the old—and her mother—he sent to the left. She tried to follow but he held her shoulder. “She’s just going to take a shower.” Later, she asked another prisoner where her mother was. The prisoner nodded at a smoking chimney. “Your mother is burning in there. You better start talking about her in the past tense.”

The gap-toothed man came back that night. She learned that he was Josef Mengele, the camp doctor; the “Angel of Death”. She learned he liked to walk through each evening, looking for inmates to entertain him. He was a refined killer and a lover of the arts. There was a camp orchestra: prisoners were played in to jaunty marches; beaten to Viennese waltzes. Mengele stopped at her barracks. He wanted a dancer and she found herself being pushed forward. The camp orchestra struck up “The Blue Danube”. “Dance!” he said, but she could not move. Then, she remembered her mother’s words. Her body was trapped but her mind was her own. So she freed herself with it. She imagined that she was not dancing in Auschwitz for her mother’s killer but on stage in the Budapest opera house. She danced for love. She danced for life.

Though when the war ended and the death march began she nearly died. The Germans made them march for days, until she was so hungry that she ate grass, so weak that she could not stand. She lay on a pile of corpses. She heard someone shout “The Americans have arrived!” but could not move. She weighed about 30kg. A hand—she would learn that it belonged to an American GI—was held out to her; in his palm were red, green and yellow beads. “Food,” he said. She would learn that these were called M&M’s.

Later she would learn other things, too. She would learn that of the more than 15,000 deportees of her hometown, 70 survived. She would learn that what made her fall to the floor shaking was called “post-traumatic stress disorder”—though she didn’t like that word “disorder”. Her reactions were not disordered, but natural. And she would learn how to pronounce that word: “Auschwitz”—though she would, to the end of her days, say it with an accent. But that no longer mattered. You cannot, she had learnt, change your past. But you can choose how you feel about it. You can choose happiness. You can choose to love. You can choose to dance. ■

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